MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR, KEY CODE ERD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.421 failures per million operating hours for class I or 1.456 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ERD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3 5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirement for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/301-1 for class I and M28787/301-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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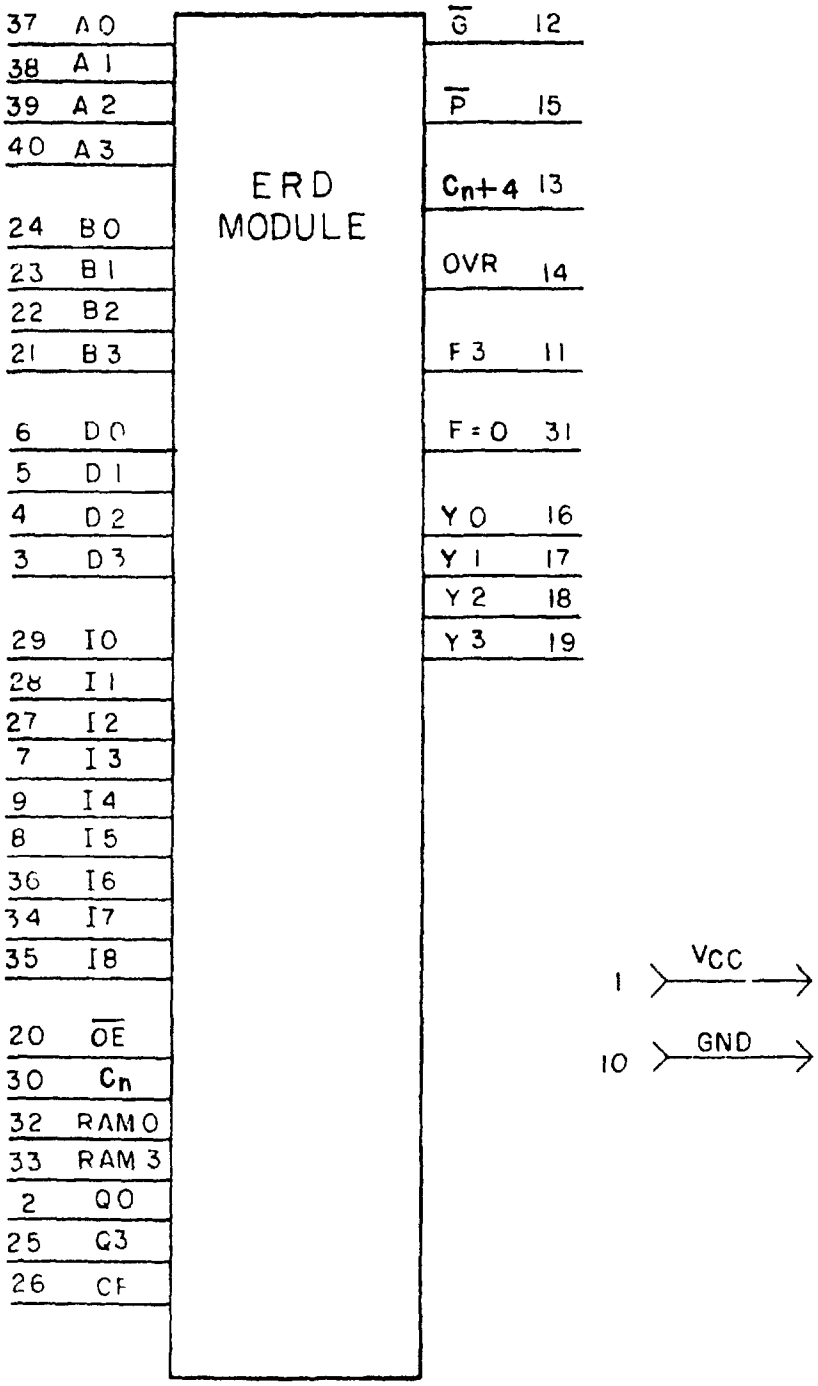


FIGURE 1 Module function diagram, microprocessor

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} voltage stress	IV	S	← Must pass remaining tests →										
C1-C28	Input breakdown current	V	S	0 0	1 0	0 0	1 0	0 0	1 0	0.0	1 0	0.0	1.0	mA
D1-D28	Input clamp diode voltage	VI	S	-0 4	-1 5	-0 4	-1.5	-0 4	-1 5	-0 4	-1 5	-0.4	-1 5	V
E1-E13	Output short circuit current	VII	X	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F15	Input current, high level	VIII	X	0	20	0	20	0	20	0	20	0	20	µA
F16-F23	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	0	40	µA
F24-F27	Input current, high level	VIII	X	0	100	0	100	0	100	0	100	0	100	µA
F28	Input current, high level	VIII	X	0	200	0	200	0	200	0	200	0	200	µA
G1-G15	Input current, low level	IX	X	-0 00	-0 40	-0 00	-0 40	-0 00	-0.40	-0 00	-0.40	-0 00	-0.40	mA
G16-G23	Input current, low level	IX	X	-0 00	-0 80	-0 00	-0 80	-0 00	-0 80	-0 00	-0 80	-0 00	-0.80	mA
G24-G27	Input current, low level	IX	X	-0 00	-0 88	-0 00	-0 88	-0 00	-0.88	-0.00	-0.88	-0.00	-0.88	mA
G28	Input current, low level	IX	X	-0 00	-3 98	-0 00	-3 98	-0.00	-3 98	-0.00	-3 98	-0.00	-3 98	mA
H1-H13	Output voltage, high level	X	X	2 4	4 5	2 4	4 5	2 4	4 5	2 4	4 5	2 4	4 5	V
J1-J14	Output voltage, low level	XI	X	0 0	0 5	0 0	0 5	0 0	0 5	0 0	0 5	0 0	0 5	V
K1-K4	High impedance leakage current, high level	XII	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
K5-K8	High impedance leakage current, high level	XII	X	-1	100	-1	100	-1	100	-1	100	-1	100	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements-Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L4	High impedance leakage current, low level	XIII	X	-50	1	-50	1	-50	1	-50	1	-50	1	µA
L5-L8	High impedance leakage current, low level	XIII	X	-800	1	-800	1	-800	1	-800	1	-800	1	µA
"	Output leakage current	XIV	X	0	250	0	250	0	250	0	250	0	250	µA
N	Power supply current	XV	X	25	250	26	265	26	265	28	280	28	280	mA
P1-P10	Dynamic functional test	XVI	X	← See table XVI →										
Q1-Q13	Transition time, t_{TLH}	XVII	S	1	15	1	15	1	20	1	15	1	25	ns
R1-R14	Transition time, t_{THL}	XVIII	S	1	15	1	15	1	20	1	15	1	25	ns
S1, S2, S10, S11	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	7	75	7	75	8	80	7	75	8	85	ns
S3, S12, S46, S47, S56, S57	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	7	70	7	70	7	75	7	70	8	80	ns
S4, S7, S13, S26, S27, S55	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	59	5	59	6	64	5	59	7	70	ns
S5, S14,	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	8	85	8	85	9	91	8	85	9	97	ns
S6, S15	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	7	76	7	76	8	83	7	76	9	90	ns
S8, S9, S17, S18,	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	9	90	9	90	9	95	9	90	10	100	ns
S16	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	62	5	59	6	64	5	59	7	70	ns
S19, S23	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	31	3	31	3	35	3	31	4	40	ns
S20	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	39	3	39	4	44	3	39	5	50	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S21	Propagation delay time, t_{PLH}, C_L - 50 pF	XIX	S	3	37	3	37	4	43	3	37	5	50	ns
S22,S43,S48	Propagation delay time, t_{PLH}, C_L - 50 pF	XIX	S	4	41	4	41	4	45	4	41	5	50	ns
S24	Propagation delay time, t_{PLH}, C_L - 50 pF	XIX	S	5	55	5	55	5	58	5	55	6	62	ns
S25	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	45	4	45	5	52	4	45	6	60	ns
S28	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	36	3	36	4	40	3	36	4	45	ns
S29	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	34	3	34	3	39	3	34	4	45	ns
S30	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	51	5	51	5	54	5	51	5	57	ns
S31, S32	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	53	5	53	5	59	5	53	6	65	ns
S33	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	2	27	2	27	3	31	2	27	3	35	ns
S34	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	2	27	2	24	2	29	2	24	3	35	ns
S35	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	2	20	2	20	2	22	2	20	2	25	ns
S36	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	46	4	46	4	49	4	46	5	52	ns
S37,S58	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	2	26	2	26	3	30	2	26	3	35	ns

See footnotes at end of table

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						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S38, S39	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	45	4	45	5	50	4	45	5	55	ns
S40,S41,S49, S50, S51	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	50	5	50	5	55	5	50	6	60	ns
S42	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	46	4	46	5	50	4	46	5	55	ns
S44,S53	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	6	65	6	65	6	68	6	65	7	72	ns
S45	Propagation delay time, t_{PLH} C_L = 50 pF	XIX	S	5	57	5	57	6	63	5	57	7	70	ns
S52,S54	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	42	4	42	4	48	4	42	5	55	ns
S59	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	35	3	35	4	42	3	35	5	50	ns
S60,S61,S62	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	52	5	52	5	58	5	52	6	65	ns
S63,S64	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	4	41	4	41	4	48	4	41	5	55	ns
S65	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	7	70	7	70	7	76	7	70	8	82	ns
S66	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	5	57	5	57	6	66	5	57	7	75	ns
S67,S69	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	3	30	3	30	3	32	3	30	3	35	ns

See footnotes at end of table.

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TABLE I Module electrical requirements- Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S68,S70	Propagation delay time, t_{PLH}, C_L = 50 pF	XIX	S	7	71	7	71	7	78	7	71	8	85	ns
T1,T2,T10, T11	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	7	75	7	75	8	80	7	75	8	85	ns
T3,T12,T46, T47,T56,T57	Propagation delay time, t_{PHL}, C_L = 50 pF	XY	S	7	70	7	70	7	75	7	70	8	80	ns
T4,T7,T13, T16,T26,T27, T55	Propagation delay time, t_{PHL}, C_L = 50 pf	XY	S	5	59	5	59	6	64	5	59	7	70	ns
T5, T14	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	8	85	8	85	9	91	8	85	9	97	ns
T6, T15	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	7	76	7	76	8	83	7	76	9	90	ns
T8,T9,T17, T18	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	9	90	9	90	9	95	9	90	10	100	ns
T19, T23	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	31	3	31	3	35	3	31	4	40	ns
T20	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	39	3	39	4	44	3	39	5	50	ns
T21	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	37	3	37	4	43	3	37	5	50	ns
T22,T43,T48	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	41	4	41	4	45	4	41	5	50	ns
T24	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	55	5	55	5	58	5	55	6	62	ns

See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued

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						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T25	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	45	4	45	5	52	4	45	6	60	ns
T28	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	36	3	36	4	40	3	36	4	45	ns
T29	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	34	3	34	3	39	3	34	4	45	ns
T30	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	51	5	51	5	54	5	51	5	57	ns
T31, T32	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	53	5	53	5	59	5	53	6	65	ns
T33	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	2	27	2	27	3	31	2	27	3	35	ns
T34	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	2	29	2	26	2	31	2	26	3	37	ns
T35	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	2	22	2	22	2	24	2	22	2	27	ns
T36	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	46	4	46	4	49	4	46	5	52	ns
T37,T58	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	2	26	2	26	3	30	2	26	3	35	ns
T38,T39	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	45	4	45	5	50	4	45	5	55	ns
T40,T41,T49, T50,T51	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	50	5	50	5	55	5	50	6	60	ns

See footnotes at end of table.

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						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T42	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	46	4	46	5	50	4	46	5	55	ns
T44, T53	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	6	65	6	65	6	68	6	65	7	72	ns
T45	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	57	5	57	6	63	5	57	7	70	ns
T52, T54	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	42	4	42	4	48	4	42	5	55	ns
T59	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	35	3	35	4	42	3	35	5	50	ns
T60,T61,T62	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	52	5	52	5	58	5	52	6	65	ns
T63	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	45	4	41	4	48	4	41	5	55	ns
T64	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	4	43	4	41	4	48	4	41	5	55	ns
T65	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	7	70	7	70	7	76	7	70	8	82	ns
T66	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	5	63	5	57	6	66	5	57	7	75	ns
T67	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	33	3	33	3	35	3	33	3	38	ns
T68,T70	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	7	71	7	71	7	78	7	71	8	85	ns
T69	Propagation delay time, t_{PHL}, C_L = 50 pF	XX	S	3	30	3	30	3	32	3	30	3	35	ns
U1-U4, U9-U12	Disable recovery time	XXI	S	6	66	6	66	7	71	6	66	7	76	ns

See footnotes at end of table.

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* TABLE I Module electrical requirements- Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U5-U8, U13-U16	Enable recovery time	XXI	S	6	60	6	60	6	65	6	60	7	70	ns
V1-V4, V9, V10, V12	Disable recovery time	XXII	S	9	90	9	90	9	95	9	90	10	100	ns
V11	Disable recovery time	XXII	S	9	105	9	105	9	110	9	105	10	115	ns
V5-V8, V13-V16	Enable recovery time	XXII	S	5	50	5	50	5	55	5	50	6	60	ns
W1-W10	Output cross talk, low level	XXIII	Q	0	600	0	600	0	600	0	600	0	600	mV
X1-X9	Output cross talk, high level	XXIV	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XXV	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply filter capacitance	XXVI	S	3.30	5.40	3.30	5.40	3.30	5.40	3.30	5.40	3.30	5.40	µF
AA1-AA24	Input capacitance	XXVII	Q	2	30	2	30	2	30	2	30	2	30	pF
BB1-BB4	Input/output capacitance	XXVIII	Q	2	30	2	30	2	30	2	30	2	30	pF
CC1-CC4, CC6-CC10	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF
CC5	Output capacitance	XXIX	Q	2	35	2	35	2	35	2	35	2	35	pF
DD1,DD3, DD5,DD7	Hold time, t_{HH}	XXX	S	900	1100	900	1100	900	1100	900	1100	900	1100	ns
DD2,DD4, DD6,DD8	Hold time, t_{HL}	XXX	S	900	1100	900	1100	900	1100	900	1100	900	1100	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through CC shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module function. The 2901A bipolar four-bit slice microprocessor module is designed as a high-speed cascadable element intended for use in CPU's, peripheral controllers, programmable microprocessors, and numerous other applications. The microinstruction flexibility of it will allow efficient emulation of any digital computing machine. The module consists of a 16-word by a four-bit two-port RAM, a high-speed ALU and the associated shifting, decoding and multiplexing circuitry. The nine-bit microinstruction work is organized into three groups of three bits each, and selects the ALU source operands, and the ALU function, and the ALU destination register (see table II). The microprocessor is cascadable with full look-ahead or with ripple carry, has three-state outputs, and provides various status flag outputs from the ALU.

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* 6.4.1 Module use. MIL-M-28787/301, 304, 305, and 308 modules are a series of designs implementing the 2900 family as building blocks to emulate microprogrammed computers or controllers.

a. MIL-M-28787/301 (ERD) - Microprocessor.
b. MIL-M-28787/304 (ZJU) - Microprogram controller.
c. MIL-M-28787/305 (ZJX) - Interrupt controller.
d. MIL-M-28787/308 (MJC) - Program control unit.

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

6.6 Changes for previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II Microinstruction decode

ALU source operand control

Micro code				ALU source operands	
I2	I1	I0	Octal Code	R	S
L	L	L	0	A	Q
L	L	H	1	A	B
L	H	L	2	0	Q
L	H	H	3	0	B
H	L	L	4	0	A
H	L	H	5	D	A
H	H	L	6	D	Q
H	H	H	7	D	0

ALU function control.

Micro code				ALU function	Symbol
I5	I4	I3	Octal code		
L	L	L	0	R Plus S	R + S
L	L	H	1	S Minus R	S - R
L	H	L	2	R Minus S	R - S
L	H	H	3	R OR S	R V S
H	L	L	4	R AND S	R ∧ S
H	L	H	5	$\overline{R}$ AND S	$\overline{R}$ ∧ S
H	H	L	6	R EX-OR S	R ∀ S
H	H	H	7	R EX-NOR S	$\overline{R \forall S}$

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TABLE II Microinstruction decode - Continued

ALU destination control. 1/-3/

Microcode				RAM function		Q-reg function		Y	RAM shifter		Q shifter	
I8	I7	I6	Octal Code	Shift	Load	Shift	Load	Output	RAM 0	RAM 3	Q0	Q3
L	L	L	0	X	NONE	NONE	F→ Q	F	X	X	X	X
L	L	H	1	X	NONE	X	NONE	F	X	X	X	X
L	H	L	2	NONE	F→ B	X	NONE	A	X	X	X	X
L	H	H	3	NONE	F→ B	X	NONE	F	X	X	X	X
H	L	L	4	DOWN	F/2→ B	DOWN	Q/2→ Q	F	F0	IN3	Q0	IN3
H	L	H	5	DOWN	F/2→ B	X	NONE	F	F0	IN3	Q0	X
H	H	L	6	UP	2F→ B	UP	2Q→ Q	F	IN0	F3	IN0	Q3
H	H	H	7	UP	2F→ B	X	NONE	F	IN0	F3	X	Q3

1/ X = Don't care Electrically, the shift pin is a TTL input internally connected to a three-state output which is in the high-impedance state

2/ B = Register addressed by B inputs.

3/ UP is toward MSB, DOWN is toward LSB

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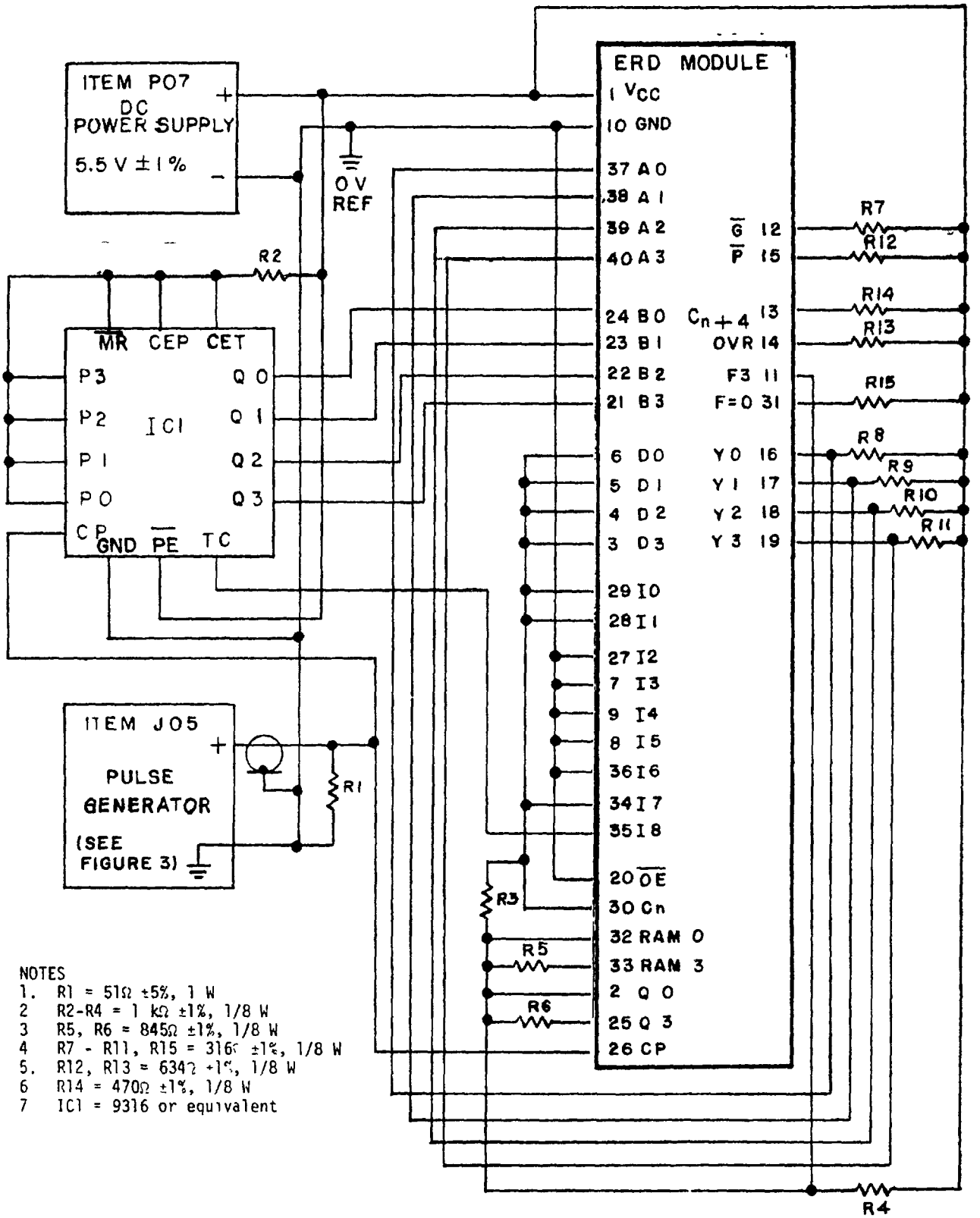
TABLE II. Microinstruction decode - Continued.

Source operand and function matrix. 1/

I_{210} OCTAL / ALU Source / ALU Function / I5 I4 I3	0	1	2	3	4	5	6	7
	A,Q	A,B	0,Q	0,B	0,A	D,A	D,Q	D,0
0	C_n = L	A+Q	A+B	Q	B	A	D+A	D+Q	D
	R Plus S C_n = H	A+Q+1	A+B+1	Q+1	B+1	A+1	D+A+1	D+Q+1	D+1
1	C_n = L	Q-A-1	B-A-1	Q-1	B-1	A-1	A-D-1	Q-D-1	-D-1
	S Minus R C_n = H	Q-A	B-A	Q	B	A	A-D	Q-D	-D
2	C_n = L	A-Q-1	A-B-1	-Q-1	-B-1	-A-1	D-A-1	D-Q-1	D-1
	R Minus S C_n = H	A-Q	A-B	-Q	-B	-A	D-A	D-Q	D
3	R OR S	AVQ	AVB	Q	B	A	DVA	DVQ	D
4	R AND S	AΛQ	AΛB	0	0	0	DΛA	DΛQ	0
5	$\overline{R}$ AND S	$\overline{A}$ΛQ	$\overline{A}$ΛB	Q	B	A	$\overline{D}$ΛA	$\overline{D}$ΛQ	0
6	R EX-OR S	AVQ	AVB	Q	B	A	DVA	DVQ	D
7	R EX-NOR S	$\overline{A \forall Q}$	$\overline{A \forall B}$	$\overline{Q}$	$\overline{B}$	$\overline{A}$	$\overline{D \forall A}$	$\overline{D \forall Q}$	$\overline{D}$

1/ + = Plus, - = Minus, V = OR, Λ = AND; ∀ = EX-OR

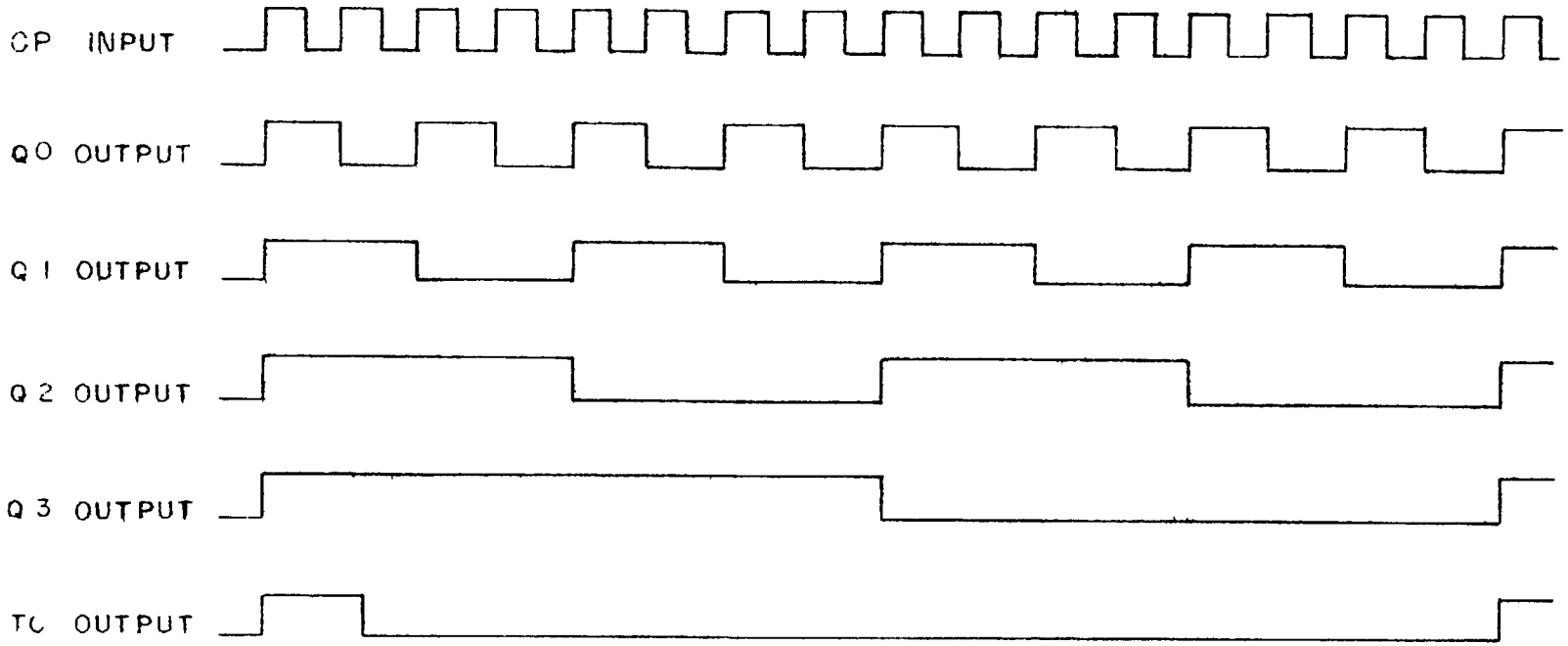
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NOTES
1. R1 = 51Ω ±5%, 1 W
2 R2-R4 = 1 kΩ ±1%, 1/8 W
3 R5, R6 = 845Ω ±1%, 1/8 W
4 R7 - R11, R15 = 316Ω ±1%, 1/8 W
5. R12, R13 = 634Ω ±1%, 1/8 W
6 R14 = 470Ω ±1%, 1/8 W
7 ICI = 9316 or equivalent

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTES
1. CP = 100 kHz ±20% square wave, duty cycle = 50 ±20%
2. Pulse width of the 18 input pulse = 10 to 11 µs.
3. Waveform input voltage = 5.5 V $\geq V_{IH} \geq$ 2.4 V
   = 0.5 V $\leq V_{IL} \leq$ 0.5 V
4. Item J05 setup
   PA = 3.0 V ±10%
   PW = 5.0 µs ±10%
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.
   50% duty cycle

FIGURE 3 Operating temperature, component temperature, and life test circuit waveforms for figure 2.

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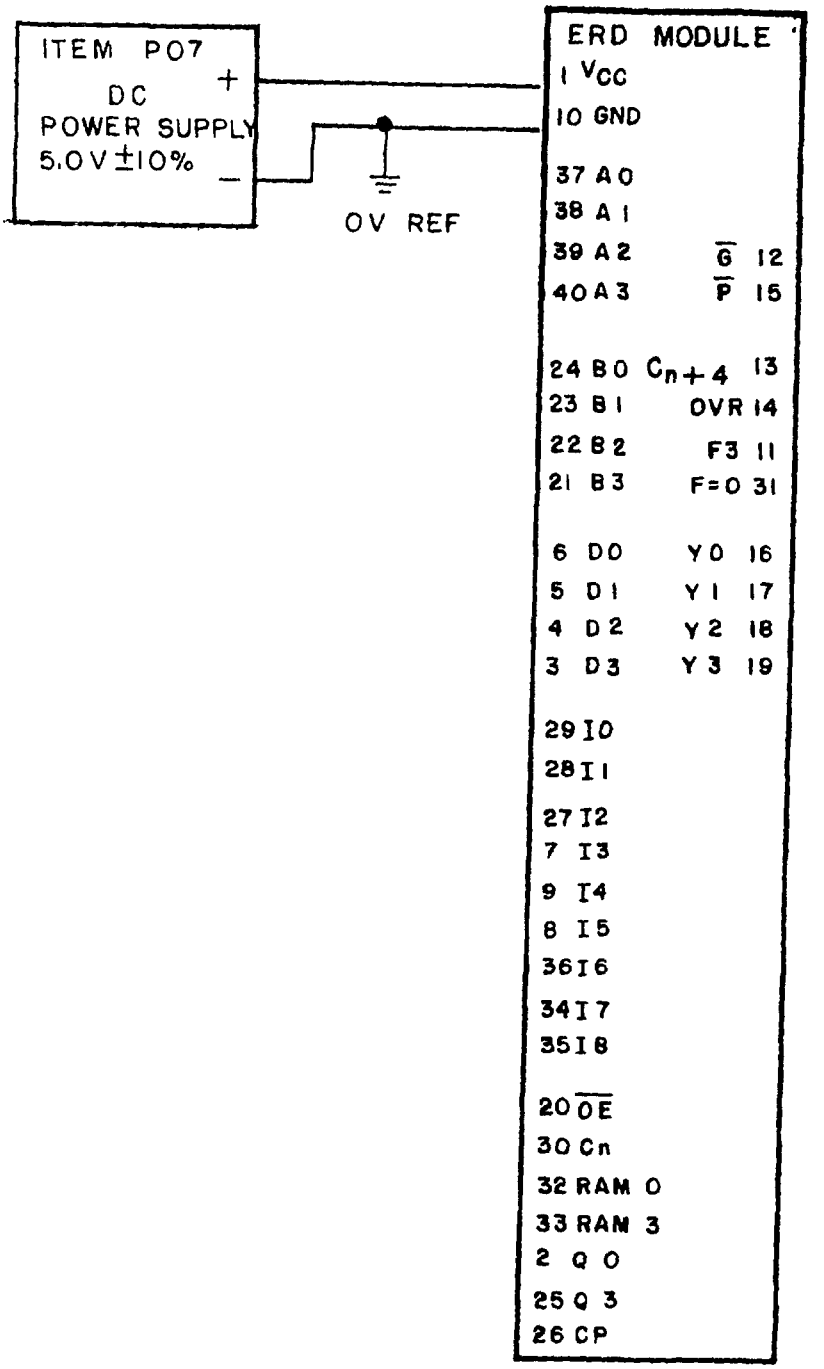


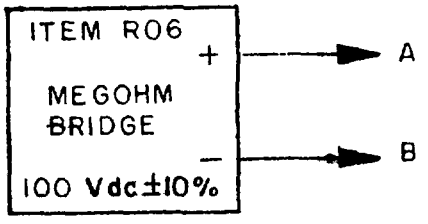
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

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WARNING·

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE· Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

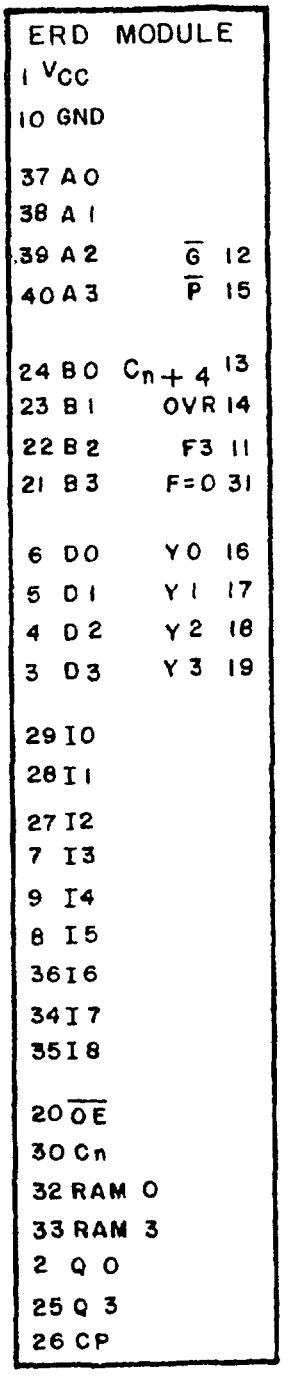


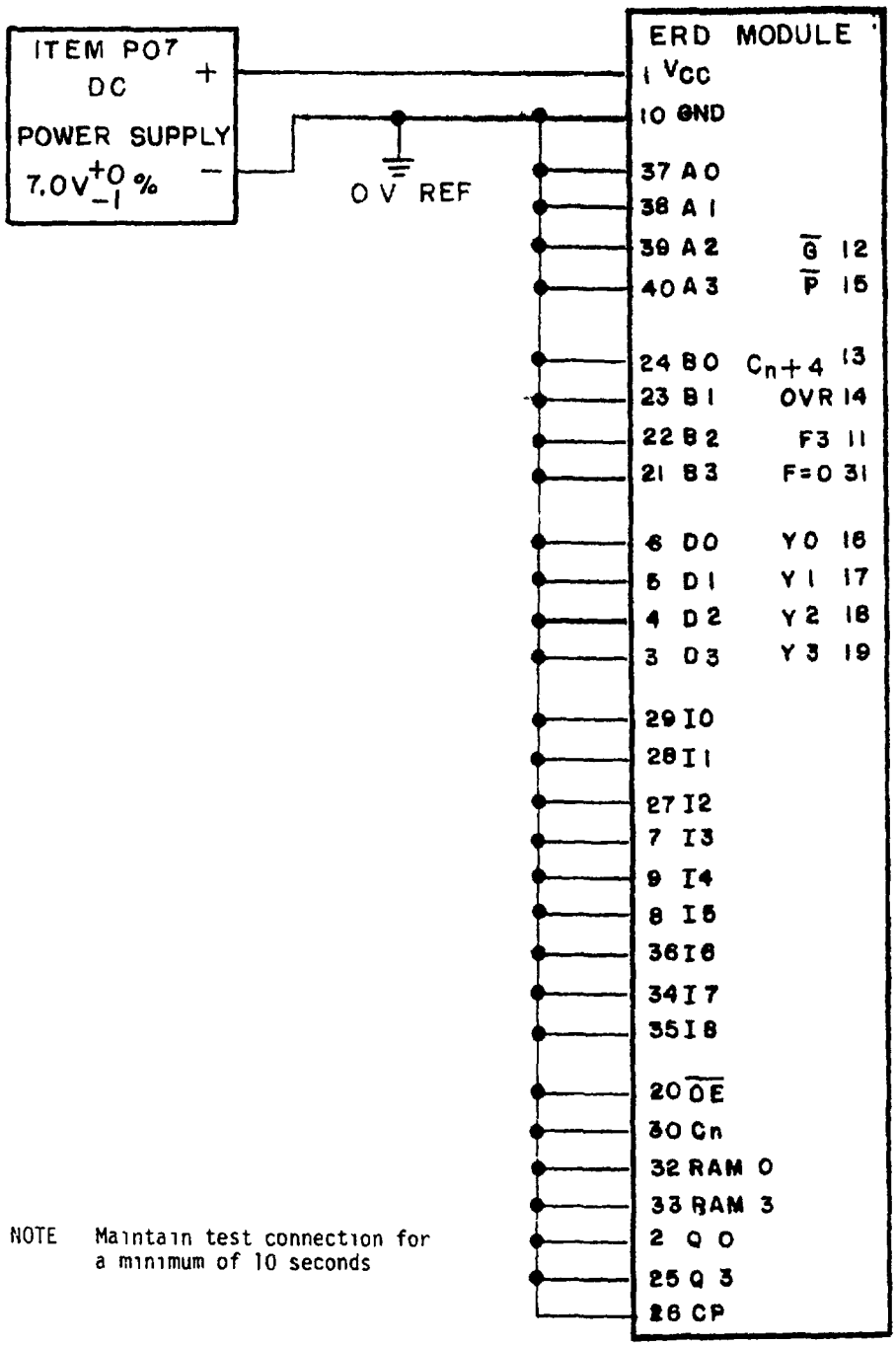
FIGURE 5 Isolation test circuit for table III.

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TABLE IV. V_{CC} voltage stress test

Test condition	Parameter		Initial limits 25° ±5°C ambient		Unit	Test figure
			Min	Max		
B	V_{CC} voltage stress	← Must pass remaining tests →				6

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NOTE Maintain test connection for a minimum of 10 seconds

FIGURE 6. V_{CC} voltage stress test circuit for table IV.

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TABLE V Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A0 Input breakdown current	37	38,39,40,24,23,22,21,6,5,4,3, 27,28,29,7,9,35,34,36,20,30, 32,33,2,25,26,8	0.0	1.0	mA	7
C2	A1	38	37,39,40,24,23,22,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C3	A2	39	37,38,40,24,23,22,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C4	A3	40	37,38,39,24,23,22,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C5	B0	24	37,38,39,40,23,22,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C6	B1	23	37,38,39,40,24,22,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C7	B2	22	37,38,39,40,24,23,21,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C8	B3	21	37,38,39,40,24,23,22,6,5,4,3, 27,28,29,7,9,8,35,34,36,20,30, 32,33,2,25,26				
C9	D0	6	37,38,39,40,24,22,5,4,3,27, 28,29,7,9,8,35,34,36,20,30,32, 33,2,25,26,23,21				

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TABLE V Input breakdown current test - Continued

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C10	D1 Input breakdown current	5	37,38,39,40,24,23,22,21,6, 4,3,27,28,29,7,9,8,35,34, 36,20,30,32,33,2,25,26	0 0	1.0	mA	7
C11	D2	4	37,38,39,40,24,23,22,21,6, 5,3,27,28,29,7,9,8,35,34, 36,20,30,32,33,2,25,26				
C12	D3	3	37,38,39,40,24,23,22,21,6, 5,4,27,28,29,7,9,8,35,34, 36,20,30,32,33,2,25,26				
C13	I0	27	37,38,39,40,24,23,22,21,6, 5,4,3,28,29,7,9,8,35,34, 36,20,30,32,33,2,25,26				
C14	I1	28	37,38,39,40,24,23,22,21,6, 5,4,3,27,29,7,9,8,35,34,36, 20,30,32,33,2,25,26				
C15	I2	29	37,38,39,40,24,23,22,21,6, 5,4,3,27,28,7,9,8,35,34, 36,20,30,32,33,2,25,26				
C16	I3	7	37,38,39,40,24,23,22,21,6 5,4,3,27,28,29,9,8,35,34, 36,20,30,32,33,2,25,26				
C17	I4	9	37,38,39,40,24,23,22,21,6, 5,4,3,27,28,29,7,8,35,34, 36,20,30,32,33,2,25,26				
C18	I5	8	37,38,39,40,24,23,22,21,6, 5,4,3,27,28,29,7,9,35,34, 36,20,30,32,33,2,25,26				
C19	I6	35	37,38,39,40,24,23,22,21,6, 5,4,3,27,28,29,7,9,8,34, 36,20,30,32,33,2,25,26				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	I7 Input breakdown current	34	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,36,20,30,32,33,2,25,26	0.0	1.0	mA	7
C21	I8	36	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,20,30,32,33,2,25,26				
C22	$\overline{OE}$	20	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,36,30,32,33,2,25,26				
C23	C_n	30	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,36,20,32,33,2,25,26				
C24	RAM 0	32	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,36,20,30,33,2,25,26				
C25	RAM 3	33	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,20,30,32,2,25,26,34,36				
C26	Q0	2	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,20,30,32,33,25,26,34,36				
C27	Q3	25	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,20,30,32,33,2,26,36				
C28	CP	26	37,38,39,40,24,23,22,21,6,5,4,3,27,28,29,7,9,8,35,34,20,30,32,33,2,25,36				

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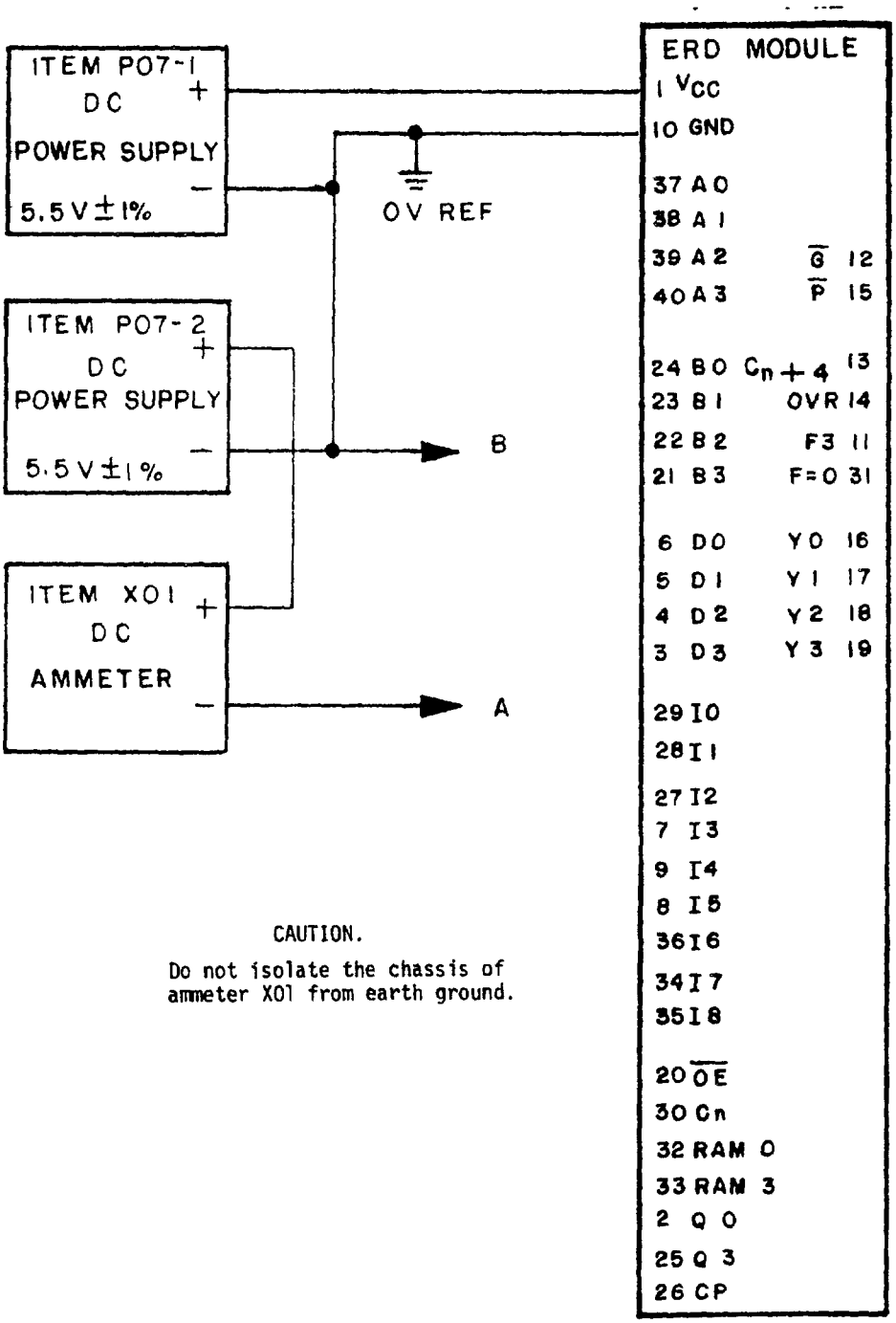


FIGURE 7 Input breakdown current test circuit for table V.

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TABLE VI Input clamp diode voltage test.

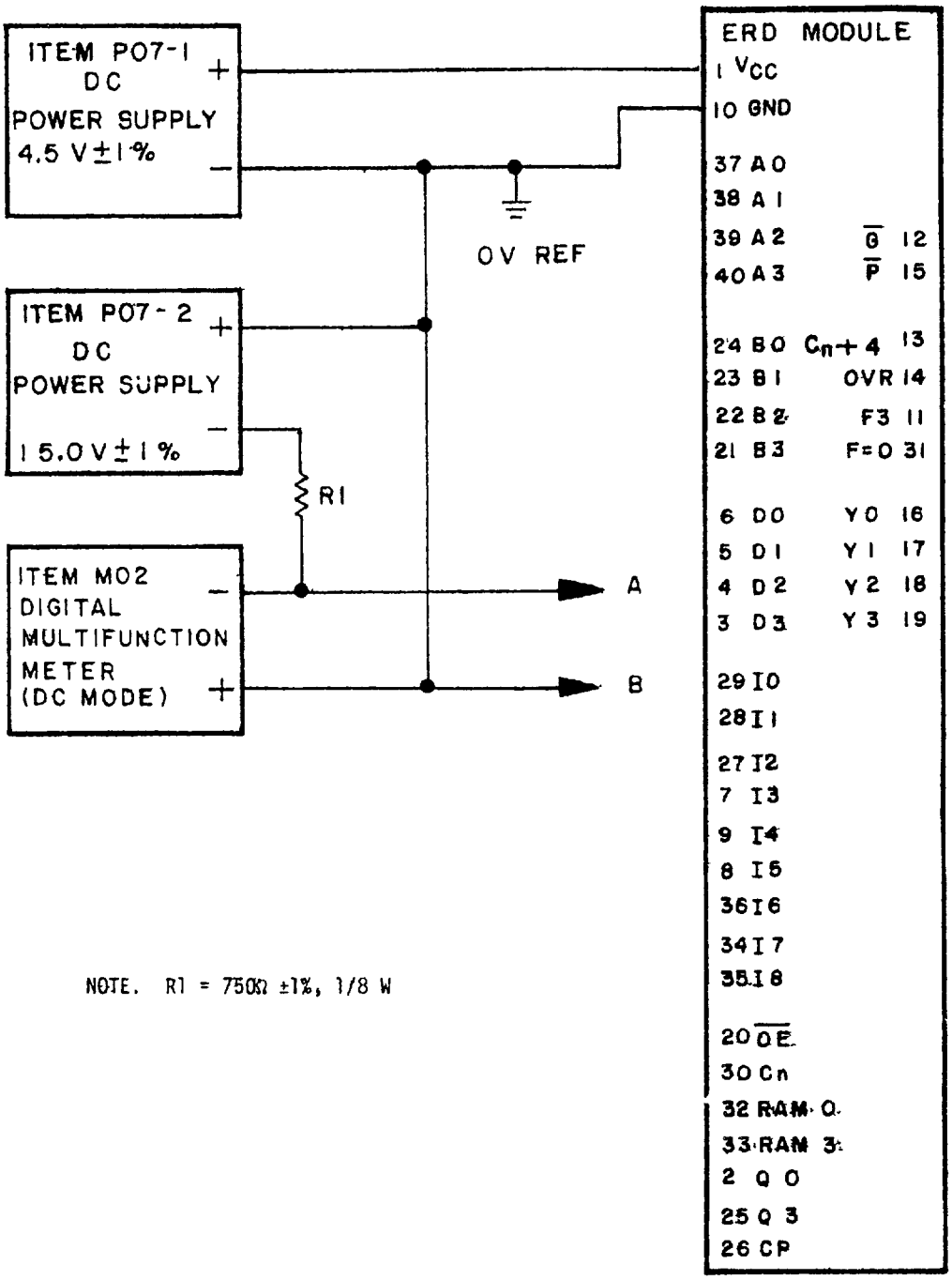
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	A3 Input clamp diode voltage	40	Open	-0 4	-1.5	V	8
D2	A2	39					
D3	A1	38					
D4	A0	37					
D5	I6	36					
D6	I8	35					
D7	I7	34					
D8	I0	29					
D9	I1	28					
D10	I2	27					
D11	CP	26					
D12	B0	24					
D13	B1	23					
D14	B2	22					
D15	B3	21					
D16	D3	3					
D17	D2	4					
D18	D1	5					
D19	D0	6					
D20	$\overline{QE}$	20					

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TABLE VI Input clamp diode voltage test - Continued

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D21	I3	Input clamp diode voltage	7	Open	-0 4	-1 5	V	8
D22	I5		8	Open				
D23	I4		9	Open				
D24	RAM 3		33	34,35,36				
D25	RAM 0		32					
D26	Q3		25					
D27	Q0		2					
D28	C_n		30					

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NOTE. R1 = 750Ω ±1%, 1/8 W

FIGURE 8 Input clamp diode voltage test circuit for table VI.

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TABLE VII Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	RAM 0 Output short circuit current	8,34	3,4,5, 6,7,9, 27,28, 29,35, 36	Open	32	-30	-85	mA	9,10
E2	RAM 3 Output short circuit current	8	3,4,5, 6,7,9, 27,28, 29,34, 35,36	Open	33	-30	-85	mA	
	Setup	8,34, 35,36	3,4,5, 6,7,9, 27,28, 29	26	Open	---	---	---	
E3	Q3 Output short circuit current	8	3,4,5, 6,7,9, 27,28, 29,34, 35,36, 26	Open	25	-30	-85	mA	
E4	Q0 Output short circuit current	34,8	35,36, 27,28, 29,3,4, 5,6,7, 9,26	Open	2	-30	-85	mA	
E5	Y0 Output short circuit current	7	34,35, 36,27, 28,29 3,4,5, 6,8,9	Open	16	-30	-85	mA	

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TABLE VII. Output short circuit current test - Continued

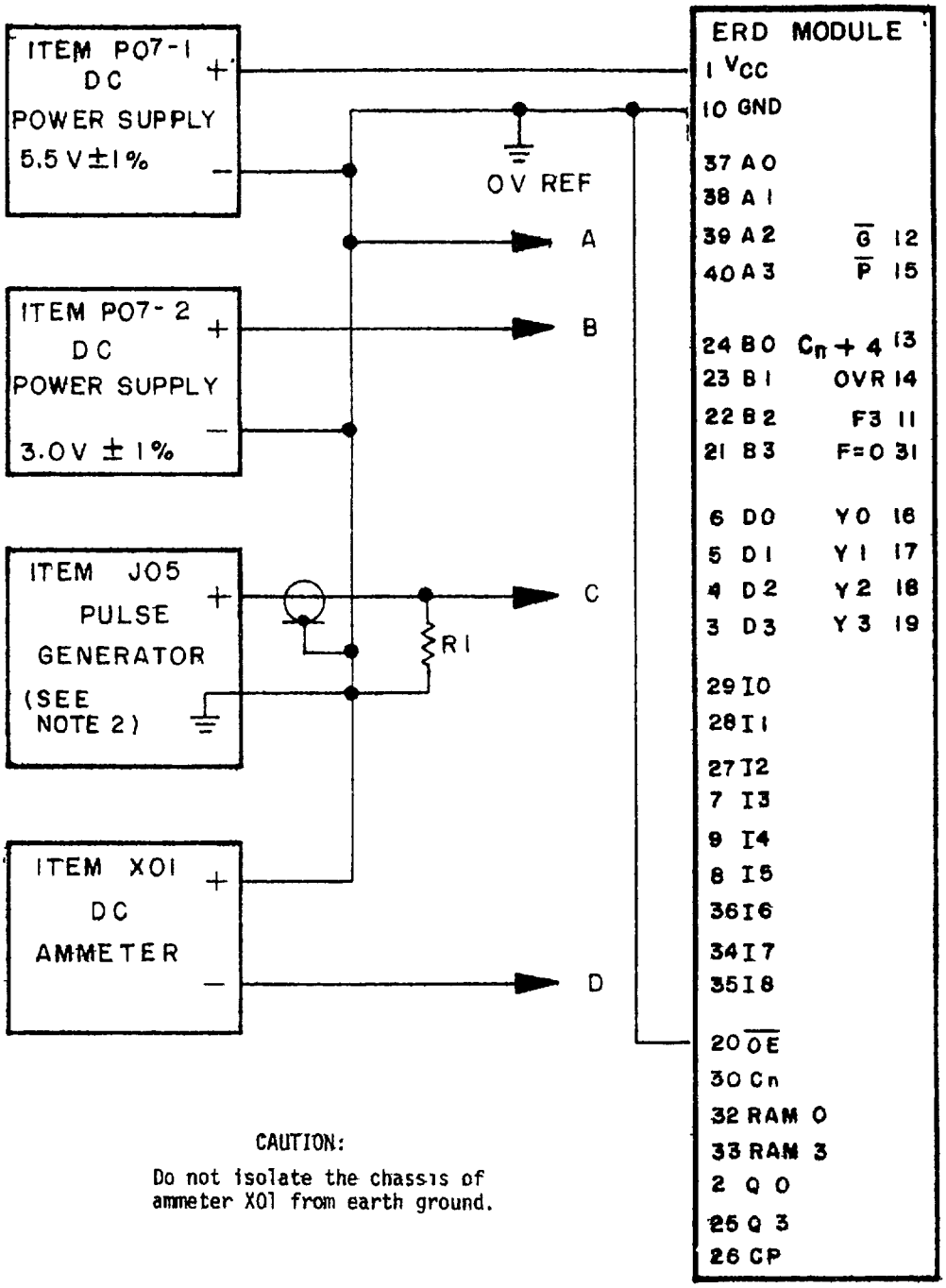
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E6	Y1 Output short circuit current	7	34,35, 36,27, 28,29,3, 4,5,6,8, 9	Open	17	-30	-85	mA	9,10
E7	Y2 Output short circuit current	7	34,35, 36,27, 28,29,3, 4,5,6,8, 9	Open	18	-30	-85	mA	
E8	Y3 Output short circuit current	7	34,35, 36,27, 28,29,3, 4,5,6,8, 9	Open	19	-30	-85	mA	
	Setup	36,21, 22,23, 24,8,3, 4,5,6	34,35, 27,28, 29,7,9	26	Open	---	---	---	
E9	$\overline{G}$ Output short circuit current	7,8,9, 28,29, 37,38, 39,40, 21,22, 23,24	27,26	Open	12	-30	-85	mA	
E10	F3 Output short circuit current	35,4,5, 6,8	36,34, 29,28, 27,3,7, 9	Open	11	-30	-85	mA	

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TABLE VII Output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25°, ±5°C ambient		Units	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
	Setup	35,21, 22,23, 24,4,5, 6,8	34,36, 27,28, 29,3,7, 9	26	Open	---	---	---	9,10
E11	C_{n+4} Output short circuit current	40,39, 38,37, 27,28, 21,22, 23,24, 7,8,9	29,26	Open	13	-30	-85	mA	
E12	OVR Output short circuit current	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	29,26	Open	14	-30	-85	mA	
E13	$\overline{P}$ Output short circuit current	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	29,26	Open	15	-30	-85	mA	

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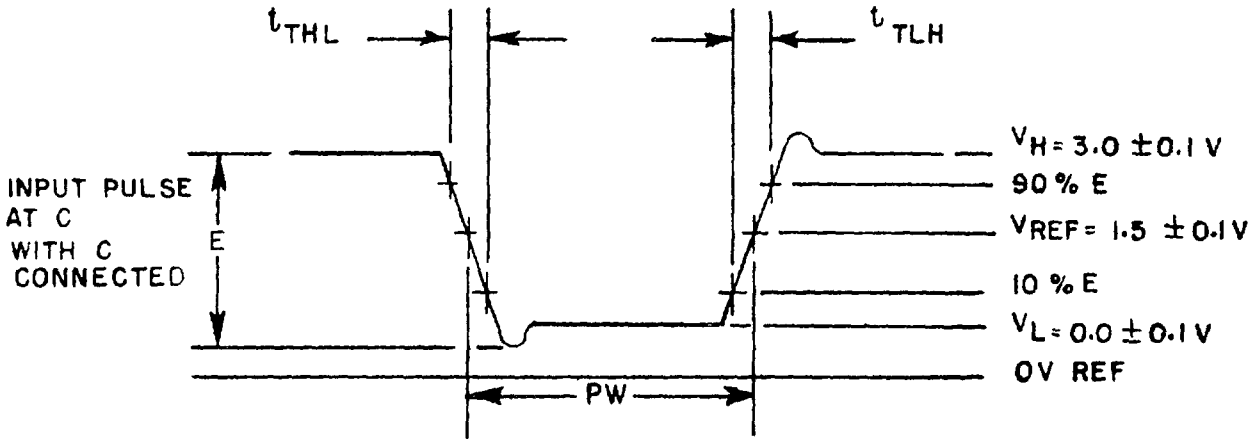
CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Item J05 is used in single pulse mode. When point C is referenced, apply single pulse as specified on figure 10.

FIGURE 9. Output short circuit current test circuit for table VII.

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NOTE. Input pulse parameters
$t_{TLH} = t_{THL} = 10 \pm 5$ ns.
PW = 500 ±50 ns.

FIGURE 10. Output short circuit current test circuit waveform for table VII.

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TABLE VIII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	A3 Input current, high level	Open	40	Open	0	20	µA	11
F2	A2		39					
F3	A1		38					
F4	A0		37					
F5	I6		36					
F6	I8		35					
F7	I0		29					
F8	I1		28					
F9	I2		27					
F10	CP		26					
F11	B0		24					
F12	B1		23					
F13	B2		22					
F14	B3		21					
F15	$\overline{OE}$		20					
F16	D0		3			40		
F17	D1		4			40		
F18	D2		5			40		

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TABLE VIII Input current (high-level) test - Continued

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F19	D3	Input current, high level	Open	6	Open	0	40	µA	11
F20	I7			34					
F21	I3			7					
F22	I5			8					
F23	I4			9					
F24	RAM 3		34,35,36	33			100		
F25	RAM 0		34,35,36	32					
F26	Q3		34,35,36	25					
F27	Q0		34,35,36	2					
F28	C_n		34,35,36	30			200		

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TABLE IX Input current (low-level) test

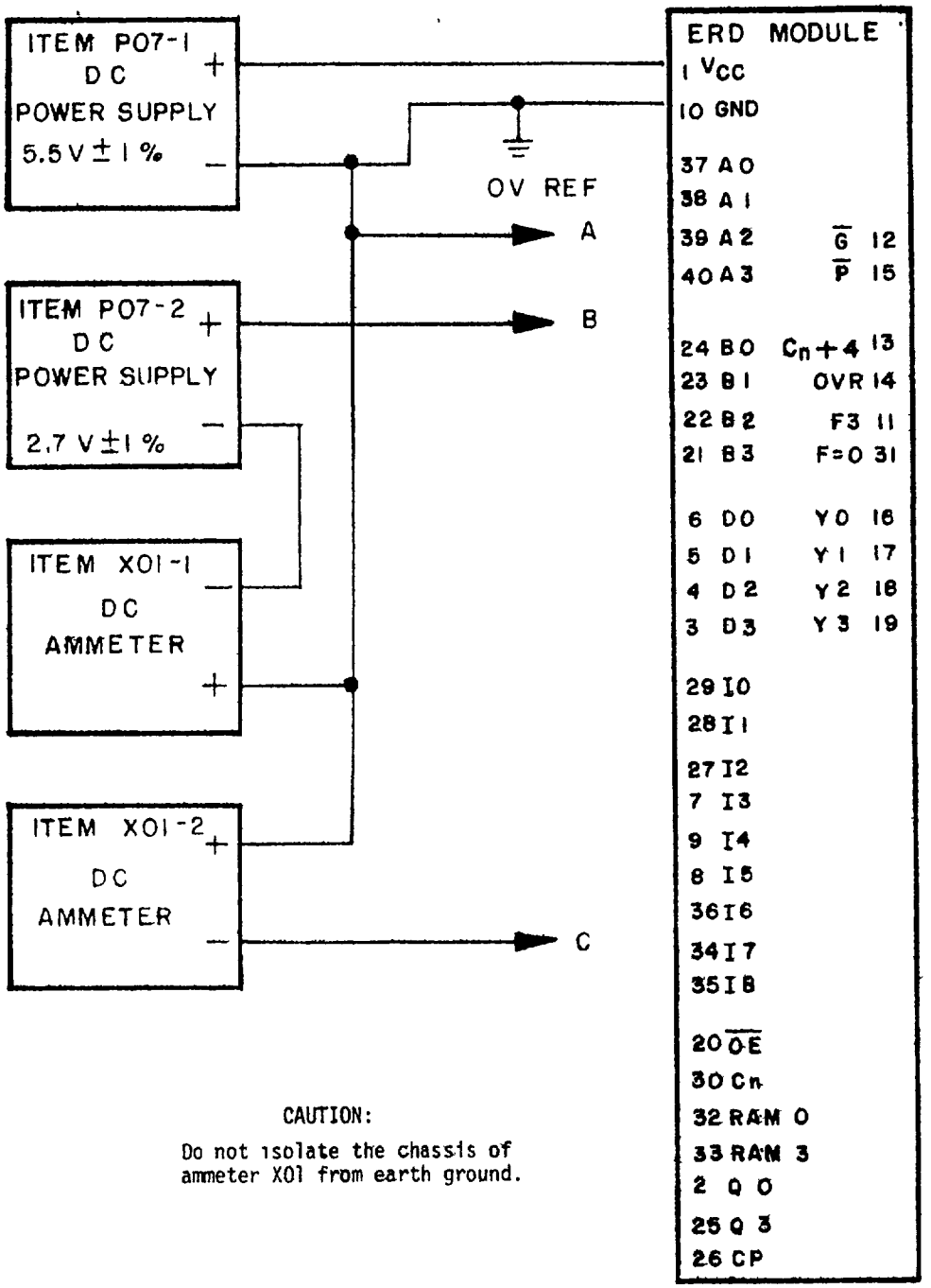
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	A3 Input current, low level	Open	Open	40	-0.00	-0 40	mA	11
G2	A2			39				
G3	A1			38				
G4	A0			37				
G5	I6			36				
G6	I8			35				
G7	I0			29				
G8	I1			28				
G9	I2			27				
G10	CP			26				
G11	B0			24				
G12	B1			23				
G13	B2			22				
G14	B3			21				
G15	$\overline{OE}$			20				
G16	D3			3		-0.80		
G17	D2			4				
G18	D1			5				
G19	D0			6				

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TABLE IX. Input current (low-level) test - Continued

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G20	I7	Input current, low level	Open	Open	34	-0.00	-0.80	mA	11
G21	I3				7				
G22	I5				8				
G23	I4				9				
G24	RAM 3		34,35,36		33		-0.88		
G25	RAM 0		34,35,36		32				
G26	Q3		34,35,36		25				
G27	Q0		34,35,36		2				
G28	C_n		34,35,36		30		-3.98		

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Only one ammeter XO1 is required for test conditions F and G

FIGURE 11. Input current (high-level and low-level) test circuit for tables VIII and IX.

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TABLE X Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	RAM 0 Output voltage, high level	36,35, 29,28, 27,3, 4,5,6, 7,9	34,8	32	Open	2.4	4.5	V	12,13
H2	RAM 3 Output voltage, high level	36,35, 34,29, 28,27, 3,4,5, 6,7,9	8	33	Open	2 4	4.5	V	
	Setup	29,28, 27,3, 4,5,6, 7,9	36,35, 34,8, 20	Open	26	---	---	---	
H3	QO Output voltage, high level	36,35, 29,28, 27,26, 3,4,5, 6,7,9	34,8	2	Open	2 4	4.5	V	
H4	Q3 Output voltage, high level	36,35, 34,29, 28,27, 26,3, 4,5,6, 7,9	8	25	Open	2.4	4.5	V	
H5	YO Output voltage, high level	36,35, 34,29, 28,27, 3,4,5, 6,8,9	7,20	16	Open	2.4	4.5	V	

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TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H6	Y1 Output voltage, high level	36,35, 34,29, 28,27, 3,4,5, 6,8,9	7,20	17	Open	2 4	4.5	V	12,13
H7	Y2 Output voltage, high level	36,35, 34,29, 28,27, 3,4,5, 6,8,9	7,20	18	Open	2 4	4 5	V	
H8	Y3 Output voltage, high level	36,35, 34,29, 28,27, 3,4,5, 6,8,9	7,20	19	Open	2.4	4 5	V	
	Setup	36,34, 29,28, 27,7,9	35,24, 23,22, 21,3,4, 5,6,8, 20	Open	26	---	---	---	
H9	$\overline{G}$ Output voltage, high level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	12	Open	2.4	4.5	V	
H10	F3 Output voltage, high level	36,34, 29,28, 27,3,7, 9	35,4,5 6,8	11	Open	2 4	4.5	V	

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TABLE X Output voltage (high-level) test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
	Setup	36,34, 29,28, 27,3,7, 9	35,24, 23,22, 21,4,5, 6,8,20	Open	26	---	---	---	12, 13
H11	C_{n+4} Output voltage, high level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	13	Open	2.4	4.5	V	
H12	OVR Output voltage, high level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	14	Open	2.4	4.5	V	
	Setup	36,34, 29,28, 27,7,9	35,24, 23,22, 21,3,4, 5,6,8, 20	Open	26	---	---	---	
H13	$\overline{P}$ Output voltage, high level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	15	Open	2.4	4.5	V	

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TABLE XI Output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	RAM 0 Output voltage, low level	36,35, 29,28, 27,7,9	34,3, 4,5,6, 8	32	Open	0 0	0 5	V	12,13
J2	RAM 3 Output voltage, low level	36,35, 34,29, 28,27, 7,9	3,4,5, 6,8	33	Open	0.0	0 5	V	
	Setup	29,28, 27.7,9	36,35, 34,3,4, 5,6,8, 20	Open	26	---	---	---	
J3	Q0 Output voltage, low level	36,35, 29,28, 27,7,9	34,3,4, 4,5,6, 8	2	Open	0 0	0 5	V	
J4	Q3 Output voltage, low level	36,35, 34,29, 28,27, 7,9	3,4,5, 6,8	25	Open	0.0	0 5	V	
J5	Y0 Output voltage, low level	36,29, 28,27, 7,9	35,34, 3,4,5, 6,8,20	16	Open	0.0	0.5	V	
J6	Y1 Output voltage, low level	36,29, 28,27, 7,9	35,34, 3,4,5 6,8,20	17	Open	0.0	0.5	V	
J7	Y2 Output voltage, low level	36,29, 28,27 7,9	35,34, 3,4,5, 6,8,20	18	Open	0 0	0 5	V	

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TABLE YI Output voltage (low-level) test - Continued.

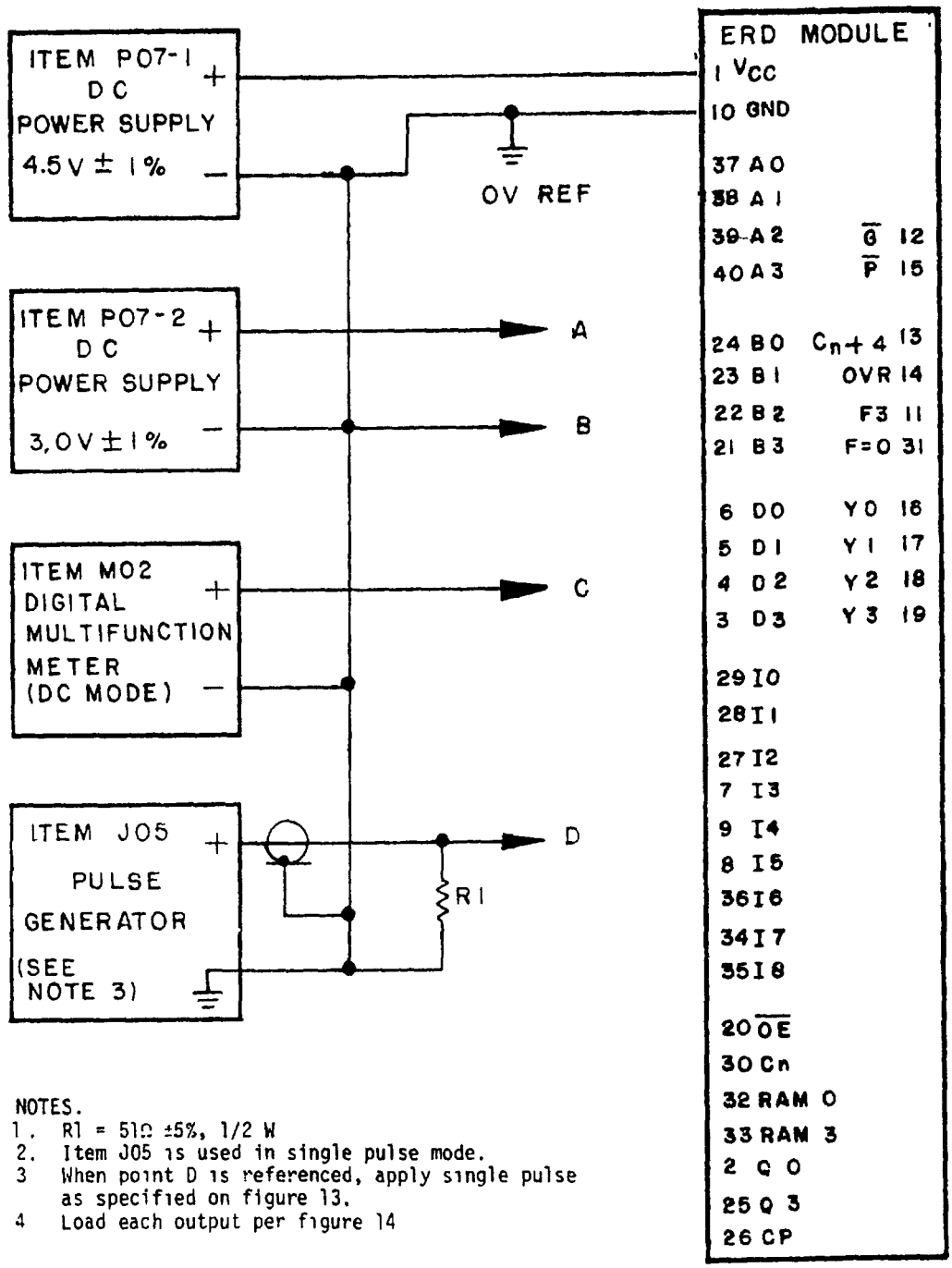
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J8	Y3 Output voltage, low level	36,29, 28,27, 7,9	35,34, 3,4,5, 6,8,20	19	Open	0 0	0.5	V	12,13
	Setup	36,34, 29,28, 27,3,4, 5,6,7, 9	35,8, 20,24, 23,22, 21	Open	26	---	---	---	
J9	$\overline{G}$ Output voltage, low level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	12	Open	0.0	0.5	V	
J10	F3 Output voltage, low level	36,34, 29,28, 27,7,9, 26	35,3,4, 5,6,8	11	Open	0.0	0.5	V	
	Setup	36,34, 29,28, 3,4,5, 6,7,9, 27	35,24, 23,22, 21,8, 20	Open	26	---	---	---	
J11	$\overline{P}$ Output voltage, low level	29,26	40,39, 38,37, 28,27, 24,23, 22,21 7,8,9	15	Open	0.0	0.5	V	

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TABLE XI. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
	Setup	36,34, 29,28, 27,7,9	35,24, 23,22, 21,3,4, 5,6,8	Open	26	---	---	---	12, 13
J12	C_{n+4} Output voltage, low level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	13	Open	0 0	0 5	V	
J13	OVR Output voltage, low level	29,26	40,39, 38,37, 28,27, 24,23, 22,21, 7,8,9	14	Open	0.0	0.5	V	
J14	F=0 Output voltage, low level	36,34, 29,28, 27,3,7, 9	35,4, 5,6,8	31	Open	0.0	0 5	V	

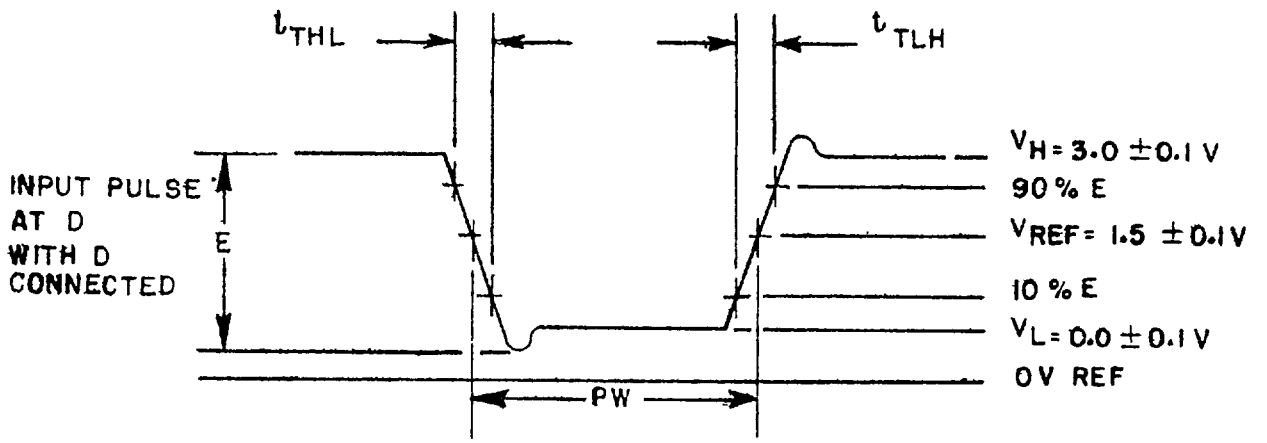
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
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NOTES.
1. R1 = 510 ±5%, 1/2 W
2. Item J05 is used in single pulse mode.
3. When point D is referenced, apply single pulse as specified on figure 13.
4. Load each output per figure 14

FIGURE 12. Output voltage (high-level and low-level) test circuit for tables X and XI.

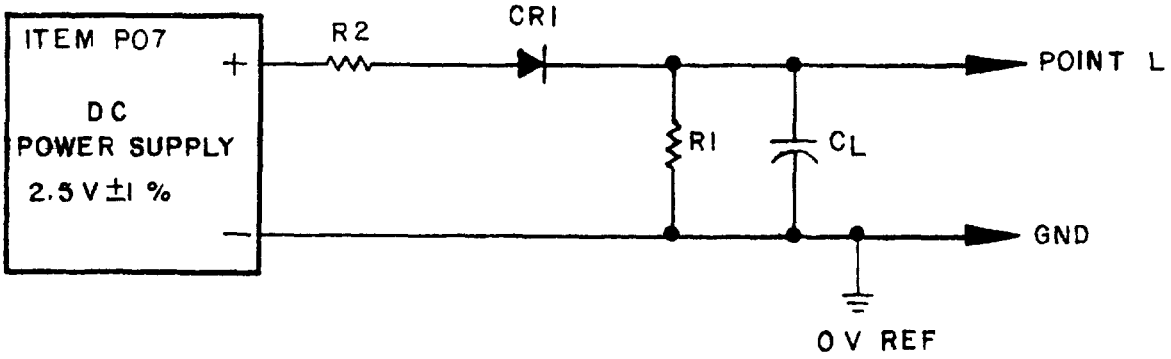
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
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NOTE Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 5$ ns.
PW = 500 ±50 ns.

FIGURE 13. Output voltage (high-level and low-level) test circuit waveform for tables X and XI.

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NOTES:
1 Load for Y0, Y1, Y2, Y3, and $\overline{G}$ outputs
CR1 = 1N4150 or equivalent.
C_L = 50 pF ±5% (including probe and parasitics) for test conditions: Q, R, S, and T.
R1 = 1.50 kΩ ±1%, 1/8 W.
R2 = Select value for a current flow of 16 mA ±5% out of point L with L held at 0.5 V.

2 Load for C_{n+4} output
CR1 = 1N4150 or equivalent
C_L = 50 pF ±5% (including probe and parasitics) for test conditions Q, R, S, and T
R1 = 2.43 kΩ ±1%, 1/8 W
R2 = Select value for a current flow of 10 mA ±5% out of point L with L held at 0.5 V.

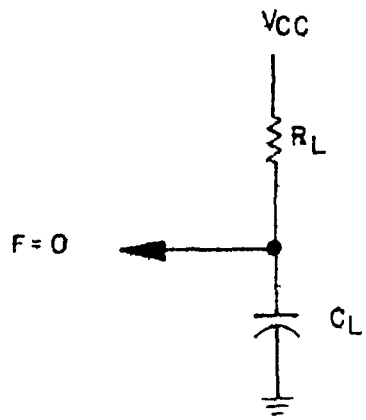
3 Load for OVR and $\overline{P}$ outputs:
CR1 = 1N4150 or equivalent.
C_L = 50 pF ±5% (including probe and parasitics).
R1 = 3.01 kΩ ±1%, 1/8 W
R2 = Select value for a current flow of 8.0 mA ±5% out of point L with L held at 0.5 V.

FIGURE 14 ERD logic load circuit.

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4. Load for F3, RAM 0, RAM 3, Q0, and Q3 outputs.
   CR1 = 1N4150 or equivalent.
   C_L = 50 pF ±5% (including probe and parasitics) for test conditions: Q, R, S, and T
   R1 = 4.02 kΩ ±1%, 1/8 W.
   R2 = Select value for a current flow of 6.0 mA ±5% out of point L with L held at 0.5 V.

5. Load for F = 0 output:



R_L = 255Ω ±1%, 1/8 W.
C_L = 50 pF ±5% (including probe and parasitics) for test conditions. Q, R, S, and T

6 Capacitor C_L is not required for dc electrical test parameters.

FIGURE 14 Logic load circuit - Continued.

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TABLE XII High impedance leakage current (high-level) test.

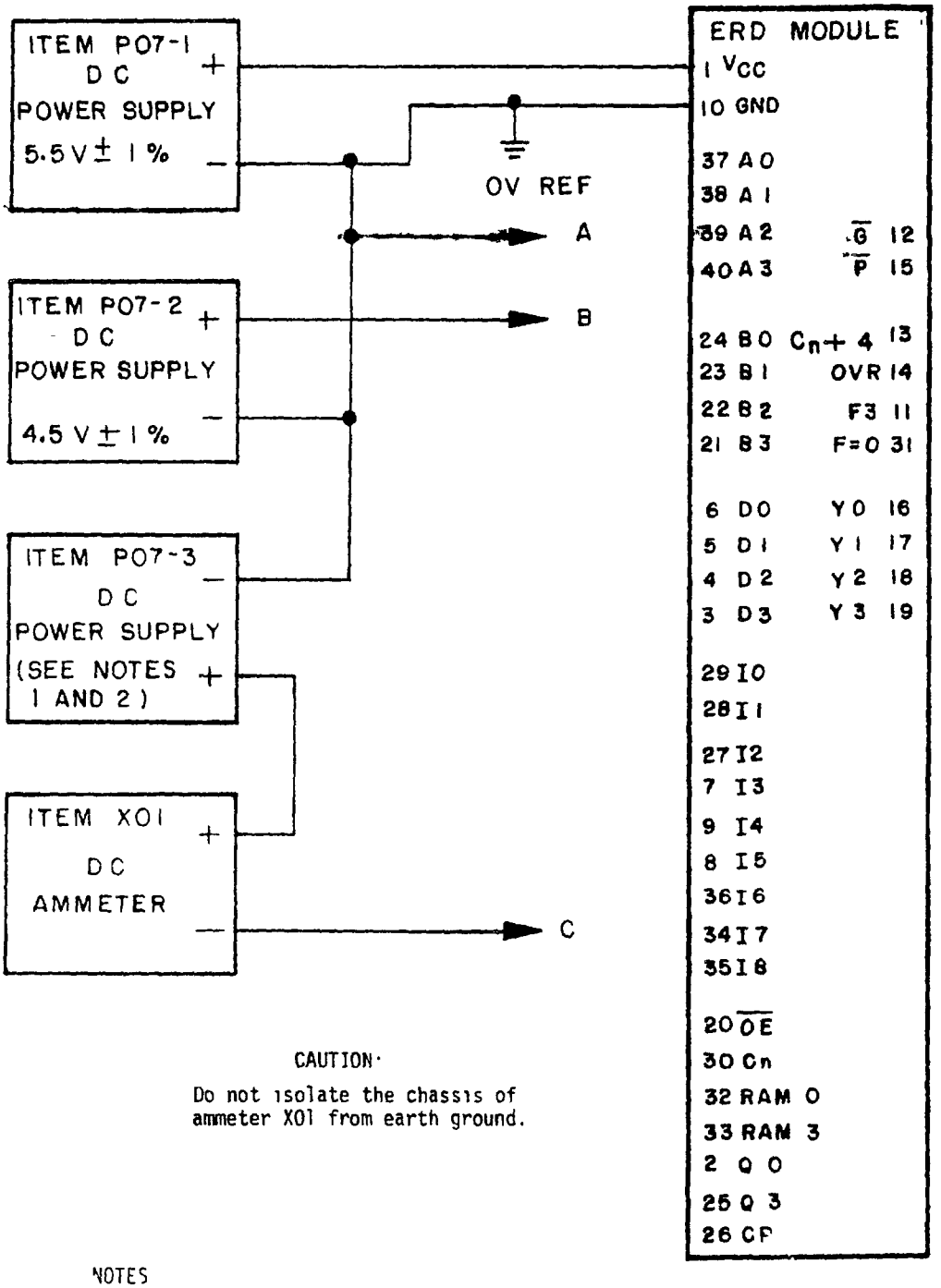
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Y0 High impedance leakage current, high level	Open	20	16	-1	50	µA	15
K2	Y1			17				
K3	Y2			18				
K4	Y3			19				
K5	RAM 3	36,35,34	Open	33		100		
K6	RAM 0			32				
K7	Q3			25				
K8	Q0			2				

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TABLE XIII High impedance leakage current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	IO High impedance leakage current, low level	Open	20	16	-50	1	µA	15
L2	Y1			17				
L3	Y2			18				
L4	Y3			19				
L5	RAM 3	36,35,34	Open	33	-800			
L6	RAM 0			32				
L7	Q3			25				
L8	Q0			2				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES
1. For test conditions K1 - K8 adjust item PO7-3 to 2.4 V ±1%
2. For test conditions L1 - L8 adjust item PO7-3 to 0.5 V ±1%

FIGURE 15 High impedance leakage current (high-level and low-level) test circuit for tables XII and XIII.

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TABLE XIV Output leakage current test

Test condition	Parameter	Connect			Initial limits 23° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M	F=0 Output leakage current	35,3,4,5,6, 8	36,34,29, 28,27,7,9	31	0	250	µA	16

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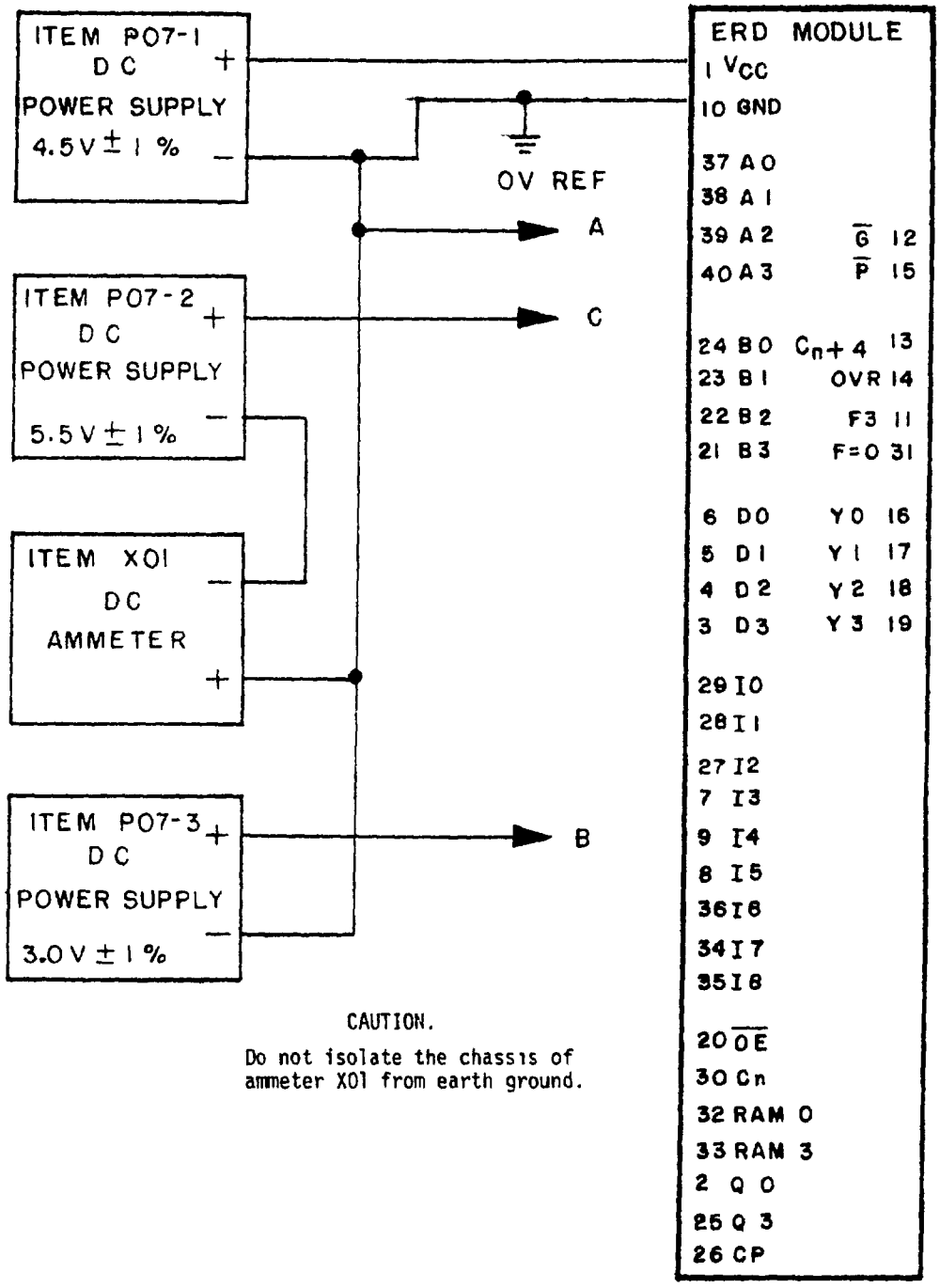


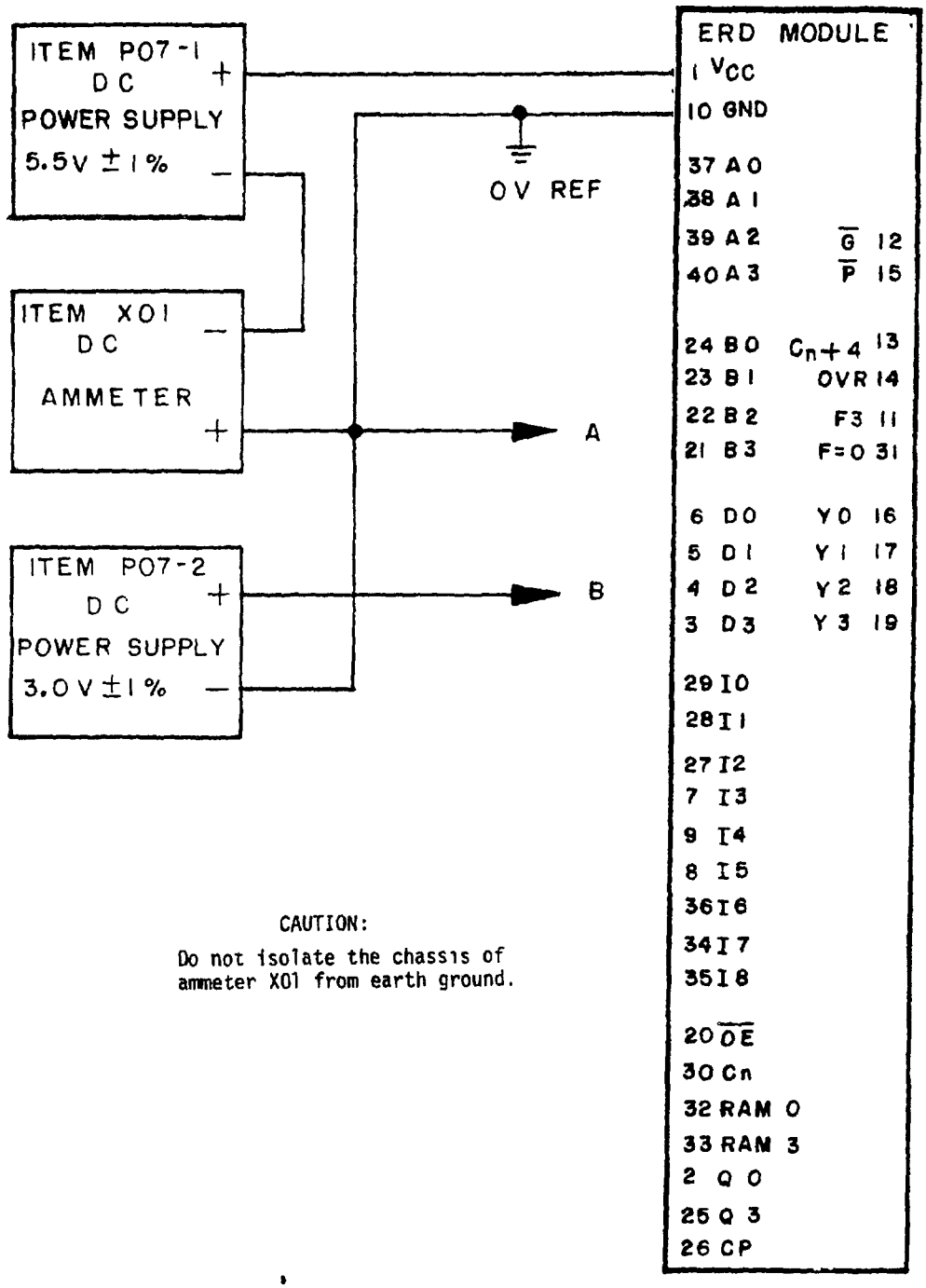
FIGURE 16. Output leakage current test circuit for table XIV.

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TABLE XV. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N	Power supply current	40,39,38,37, 36,35,24,23, 22,21,7,8,9,26	34,29,28,27, 3,4,5,6,30	25	250	mA	17

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 17 Power supply current test circuit for table XV.

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* TABLE XVI. Dynamic functional test. 17/, 18/

Test condition	Test description	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
P1	V_{CC} = 4.5V Dynamic functional test, minimum clock pulse width, low level	1/ - 12/		ns	18, 19, 20, 21, 22, 23
P2	V_{CC} = 5 5V Dynamic functional test, minimum clock pulse width, low level	1/-3/, 5/-12/		ns	18, 19, 20, 21, 22, 23
P3	V_{CC} = 4 5V Dynamic functional test, minimum clock pulse width, high level	13/		ns	24
P4	V_{CC} = 5.5V Dynamic functional test, minimum clock pulse width, high level	13/		ns	24
P5	V_{CC} = 4 5V Dynamic functional test, read-modify-write cycle time check	14/		ns	25
P6	V_{CC} = 5.5V Dynamic functional test, read-modify-write cycle time check	14/		ns	25
P7	V_{CC} = 4.5V Dynamic functional test, maximum shift frequency of register check	15/		MHz	26, 27
P8	V_{CC} = 5 5V Dynamic functional test, maximum shift frequency of register check	15/		MHz	26, 27
P9	V_{CC} = 4.5V Dynamic functional test, catch all	16/			28
P10	V_{CC} = 5 5V Dynamic functional test, catch all	16/			28

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1/ Test conditions P1, P2, pattern FPAT 1 and FPAT 11 are tests in which the A-address of the 2-port RAM is tested for galloping "ones" in a field of "zeroes". During these tests, the B-address is the same as the A-address and OP code 337 is used for a write operation, while OP code 134 is used for a read operation. The four shift operation pins Q0, Q3, RAM 0, and RAM 3 are "don't cares". A-address (source), B-address (destination) and Data (I = X37) minimum setup times relative to the clock rising edge are verified. See figure 18.

2/ Test conditions P1, P2, FPAT 2 and FPAT 12 are tests in which the B-address of the 2-port RAM is tested for galloping "ones" in a field of "zeroes". During these tests, the A-address is the inverse of the B-address and OP code 337 is used for a write operation while OP code 133 is used for a read operation. The four shift operation pins, Q0, Q3, RAM 0, and RAM 3 are "don't cares". A-address (source), B-address (source), Data (I = X37), and I0 to I5 minimum setup times relative to the clock rising edge are verified. See figure 19.

3/ Test conditions P1, P2, pattern FPAT 3 and FPAT 13 are tests in which the A-address of the 2-port RAM is tested for galloping "ones" in a field of "zeroes". During these tests, the B-address is the inverse of the A-address and OP code 337 is used for a write operation while OP code 233 is used for a read operation. The four shift operation pins, Q0, Q3, RAM 0, and RAM 3 are "don't cares". A-address (source), B-address (source), Data (I = X37), and I6 I8 minimum setup times relative to the clock rising edge are verified. See figure 20.

4/ Test condition P1, pattern FPAT 4 and FPAT 14, vectors 1 through 19 are tests in which the A and B-address are at word locations preloaded with known values. The Q register is also preloaded with known values. Then, with the ALU destination OP code = 1 (No OP) and the ALU function code = 6 (exclusive OR), the source code is cycled through 0-7. The function code is then modified to 7 (exclusive NOR) and the source code sequence is cycled through once more. The A-address (source), B-address (source) and I0-I5 minimum setup times relative to the rising clock edge are verified. See figure 21.

5/ Test condition P1, P2, FPAT 4 and FPAT 14, vectors 19 through 299 are tests in which the memory is pre-loaded with content equal to the address. Then, with A-address = B-address, a destination OP code of 1 (No OP), and a source OP code of 1 (A & B Port selected), the ALU function code is cycled through the sequence of 7, 5, 4, 0, 1, 3, 2, 6 for every set of A & B address. The whole sequence is then repeated with A-address equal to the inverse of the B-address. The A-address (source), B-address (source), and I0 to I5 minimum setup times relative to the clock rising edge are verified. See figure 21.

6/ Test conditions P1, P2, FPAT 5 and FPAT 15 are tests in which the memory is preloaded with content equal to address. With OP code 105, (adds D input to the A-port memory). The pattern cycles through every possible D input added to every word in memory with carry in being both one and zero. A-address (source), B-address (source), C_n, and Data (arith. mode) minimum setup times relative to the clock rising edge are verified. See figure 22.

7/ Test conditions P1 and P2, patterns FPAT 6 and FPAT 16 are tests in which the Q register is first loaded with all zeroes. Then with C_n = 0 and with OP code 006 which loads the Q register with the sum of data input and the Q register contents on every clock cycle, the device is clocked through all possible data inputs. The C_n input is then changed to a ONE, and with OP code 016, which loads the Q register with the difference of Q-D, the device is clocked through all possible data inputs again. This checks both the add and substract modes of the ALU, the internal-carry-look ahead circuitry and the Q register operation. The A-address (source), B-address (source), C_n, and Data (arith. mode) minimum setup times are verified. See figure 22.

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8/ Test conditions P1 and P2, patterns FPAT 7 and FPAT 17 are tests in which a unique string of data is shifted into the appropriate shift inputs of the Q register. OP codes used in this group of tests are 432 for shift left, 532 for no shift, 632 for shift right and 732 for no shift. The A-address (source), B-address (source), I8 I6, Q3 and Q0 minimum setup time relative to the clock rising edge are verified. See figure 23.

9/ Test conditions P1 and P2, pattern FPAT 8, perform the same functional tests as FPAT 1 and FPAT 11 in note 1 except that the data inputs and Y outputs are inverted. The Y outputs are the only outputs checked. The A-address (source), B-address (destination), and Data (I = X37) minimum setup times relative to the clock rising edge are verified. See figure 18.

10/ Test conditions P1 and P2, pattern FPAT 9 performs the same functional tests as FPAT 2 and FPAT 12 in note 2 except that the data inputs and Y outputs are inverted. The Y outputs are the only outputs checked. The A-address (source), B-address (source), Data (I = X37) and I0 through I5 minimum setup time relative to the clock rising edge are verified. See figure 19.

11/ Test conditions P1 and P2, pattern FPAT 10 performs the same functional tests as FPAT 3 and FPAT 13 in note 3 except that the data inputs and the Y outputs are inverted. The Y outputs are the only outputs checked. The A-address (source), B-address (source), Data (I = X37), and I6 through I8 minimum setup time relative to the clock rising edge are verified. See figure 20.

12/ Test conditions P1 and P2, all patterns, verifies the minimum time clock low parameter. See figure 20.

13/ Test conditions P3 and P4, all patterns are tests in which the A and B-addresses are incremented from 0 to 15. A string of data is shifted into the appropriate shift inputs for the RAM. OP codes used are 434 and 533 for left shift, 634 and 733 for right shift. The A-address (source), B-address (source), RAM 0, and RAM 3 minimum setup times relative to the rising clock edge are verified. The minimum time clock high parameter is verified. See figure 25.

14/ Test conditions P5 and P6 are tests in which a read-modify-write test is performed on the RAM. OP codes 324 and 323 are used for the inversion and write operations while OP codes 134 and 133 are used for the read operations. The I5 through I0, A-address (source), B-address (source) minimum setup times, relative to the clock rising edge are verified. See figure 25.

15/ Test conditions P7 and P8 perform the same functional tests as P1 and P2, patterns FPAT 7 and FPAT 17. During this test, however, the maximum shift frequency for the Q register is verified. The Q0 and Q3 minimum setup times relative to the clock rising edge are verified. See figures 26 and 27.

16/ Test conditions P9 and P10 are tests which check various functions of the module which had not already been checked in previous patterns. There are no setup times verified in this pattern. See figure 28.

17/ Test conditions P1-P10 apply the functional test vectors (see 6.5). Each test condition tests mainly for the test parameters as described in the test descriptions, but also for the following operating conditions and parameters:

a. Loaded output voltage levels, V_{OH} = 2.4 minimum, V_{OL} = 0.5 V4 maximum.

b. Worst case input threshold voltage levels, V_{IL} = 0.5 V maximum, V_{IH} = 2.4 V minimum.

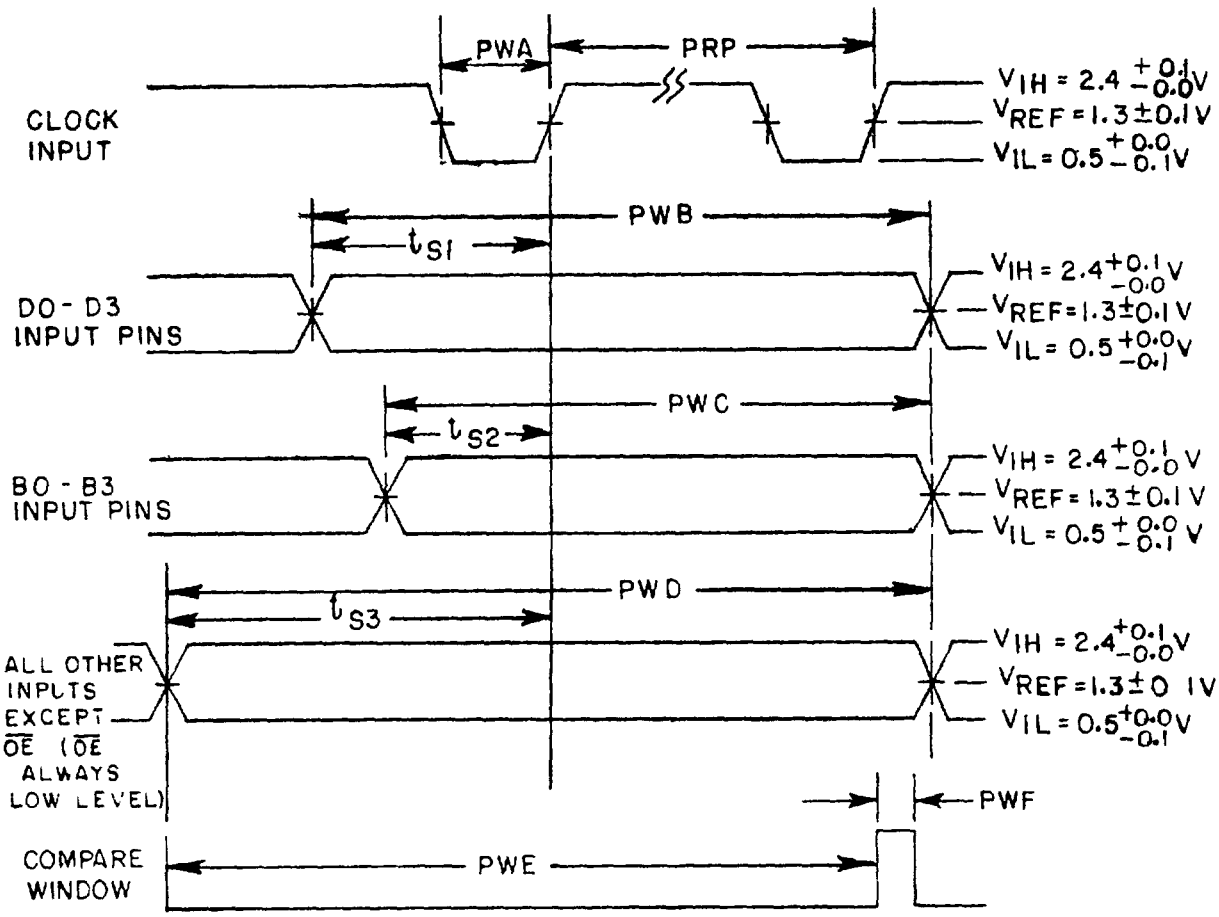
c. Worst case setup times as specified on figures 18 through 28

d. For test conditions P1, P3, P5, P7 and P9, V_{CC} = 4.5 V ± 1%.

For test conditions P2, P4, P6, P8 and P10, V_{CC} = 5.5 V ± 1%.

18/ Load all outputs with the load as specified on figure 14.

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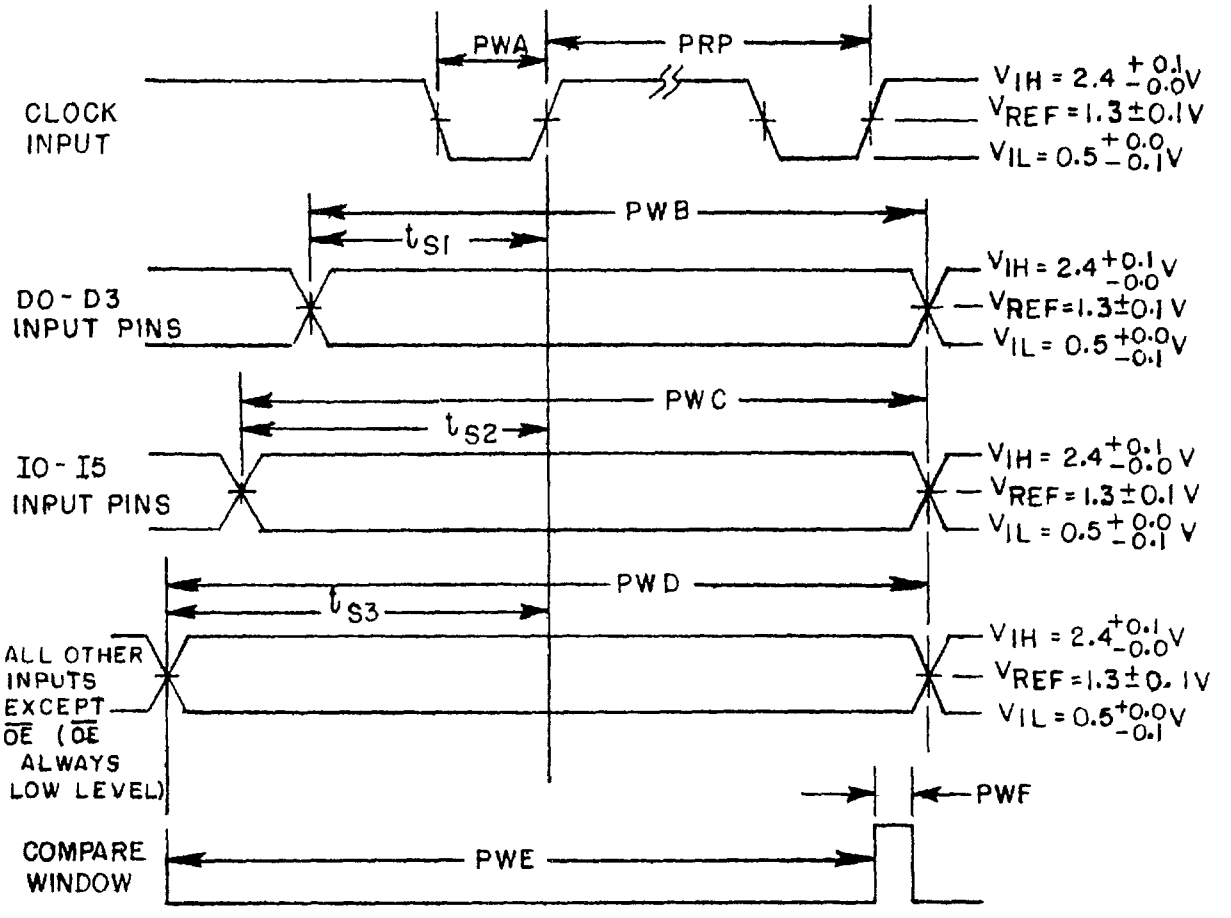


NOTES:

1. All inputs: $t_{THL} = t_{TLH} = 10 \pm 2$ ns measured at the 10% and 90% PA points.
2. Input pulse parameters:
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 170 ±1 ns.
   PWC = 150 ±1 ns.
   PWD = 215 ±1 ns.
   PWE = 200 ±1 ns.
   PWF = 10 +1 ns.
   t_{S1} = 65 ·2 ns.
   t_{S2} 45 ·2 ns.
   t_{S3} = 110 ·2 ns.
3. During compare window check, verify that high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

FIGURE 18. Dynamic functional test timing waveforms for test conditions P1, P2, FPAT 1, FPAT 8, and FPAT 11.

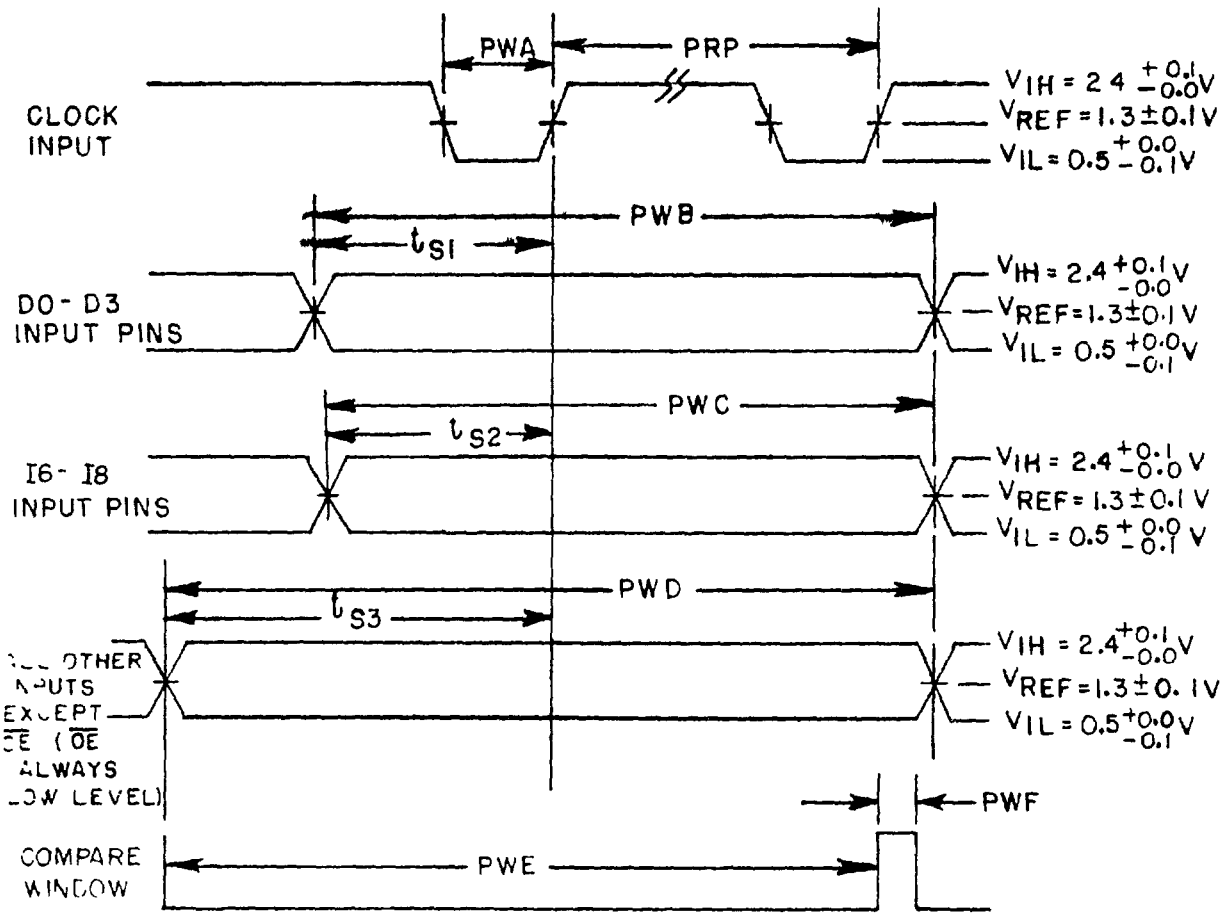
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NOTES
1. All inputs. $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points.
2. Input pulse parameters:
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 170 ±1 ns
   PWC = 190 ±1 ns.
   PWD = 215 ±1 ns.
   PWE = 200 ±1 ns.
   PWF = 10 ±1 ns.
   t_{S1} = 65 ±2 ns.
   t_{S2} = 85 ±2 ns
   t_{S3} = 110 ±2 ns.
3. During compare window check, verify that high level outputs are ≥ 2 4 V and low level outputs are ≤ 0 5 V.

FIGURE 19. Dynamic functional test timing waveforms for test conditions P1, P2, FPAT 2, FPAT 9 and FPAT 12.

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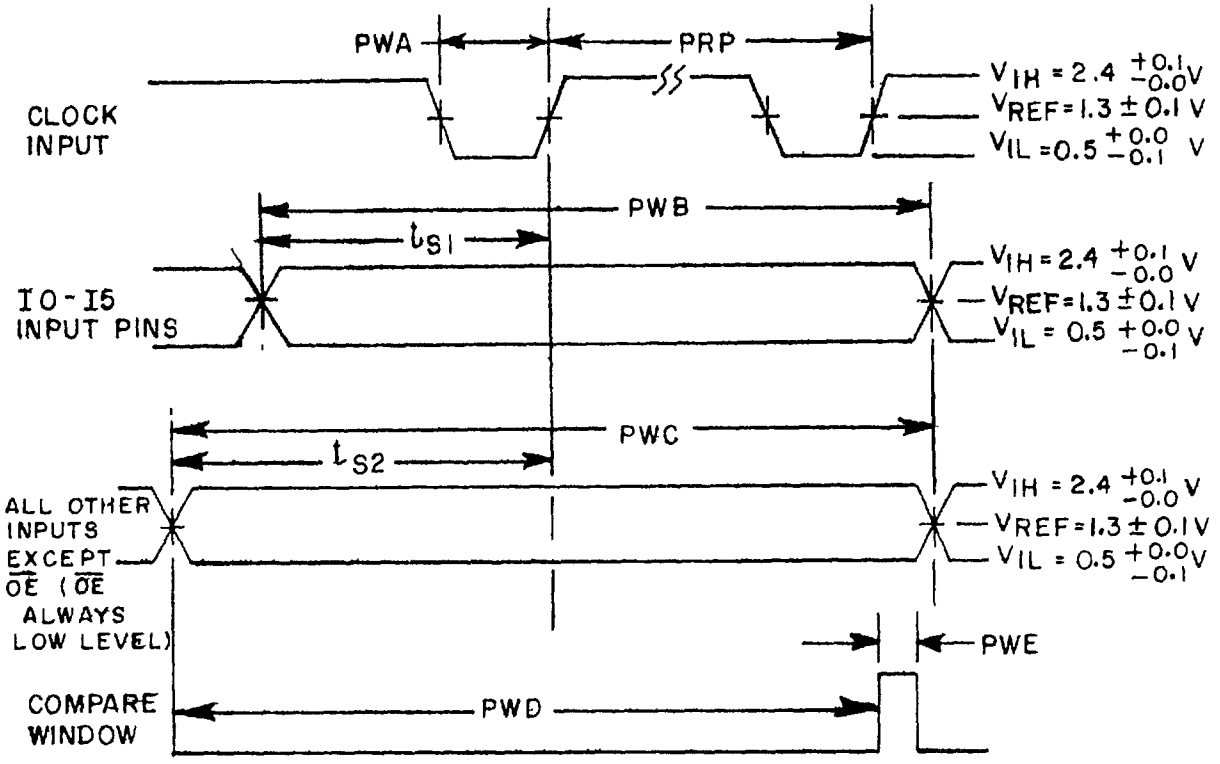


NOTES:

1. All inputs $t_{THL} = t_{TLH} = 10 \pm 2$ ns measured at the 10% and 90% PA points.
2. Input pulse parameters.
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 170 ±1 ns
   PWC = 165 ±1 ns.
   PWD = 215 ±1 ns.
   PWE = 200 ±1 ns.
   PWF = 10 ±1 ns.
   t_{S1} = 65 ±2 ns.
   t_{S2} = 60 ±2 ns.
   t_{S3} = 110 ±2 ns
3. During compare window check, verify that high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

FIGURE 20. Dynamic functional test timing waveforms for test conditions P1, P2, FPAT 3, FPAT 10, and FPAT 13.

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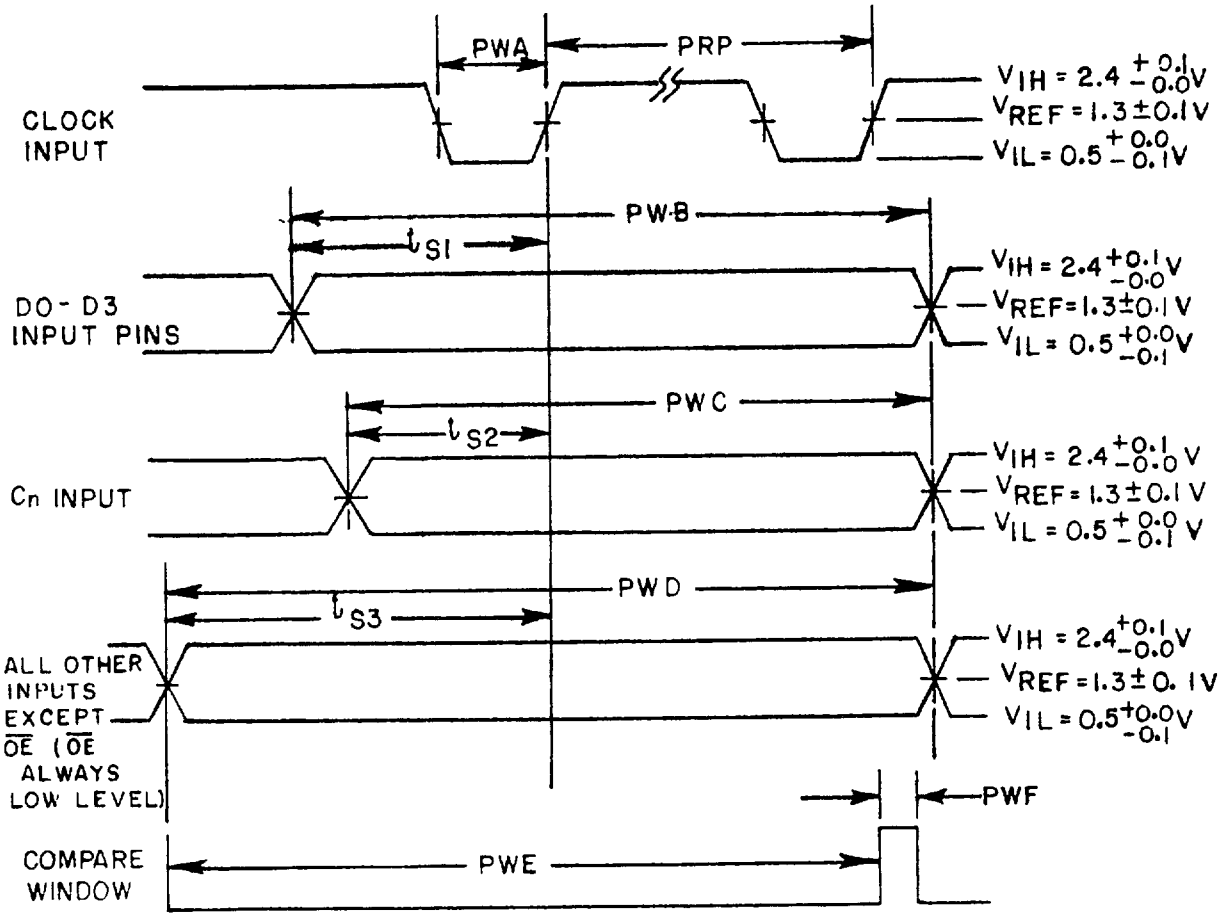


NOTES.

1. All inputs $t_{THL} = t_{TLH} = 10 \pm 2$ ns measured at the 10% and 90% PA points
2. Input pulse parameters.
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns
   PWB = 190 ±1 ns.
   PWC = 215 ±1 ns.
   PWD = 200 ±1 ns.
   PWE = 10 ±1 ns.
   t_{S1} = 85 ±2 ns.
   t_{S2} = 110 ±2 ns.
3. During compare window check, verify that the high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

FIGURE 21. Dynamic function test timing waveforms for test conditions P1, P2, FPAT 4, and FPAT 14.

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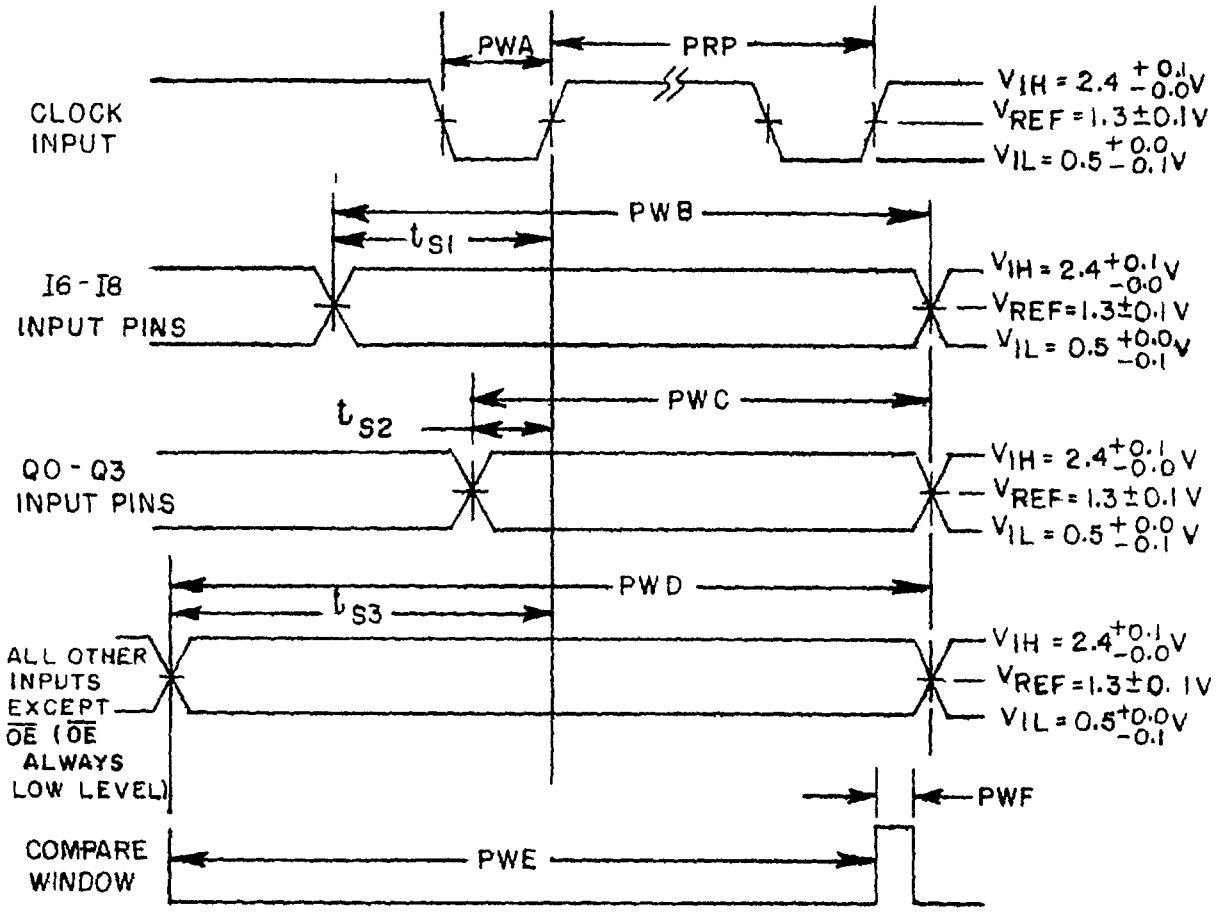


NOTES

1. All inputs: $t_{THL} = t_{TLH} = 10 \pm 2$ ns measured at the 10% and 90% PA points.
2. Input pulse parameters:
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 180 ±1 ns.
   PWC = 165 ±1 ns.
   PWD = 215 ±1 ns.
   PWE = 200 ±1 ns.
   PWF = 10 ±1 ns.
   t_{S1} = 75 ±2 ns.
   t_{S2} = 60 ±2 ns.
   t_{S3} = 110 ±2 ns.
3. During compare window check, verify that high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V

FIGURE 22. Dynamic functional test timing waveforms for test conditions P1, P2, FPAT 5, FPAT 6, FPAT 15, and FPAT 16.

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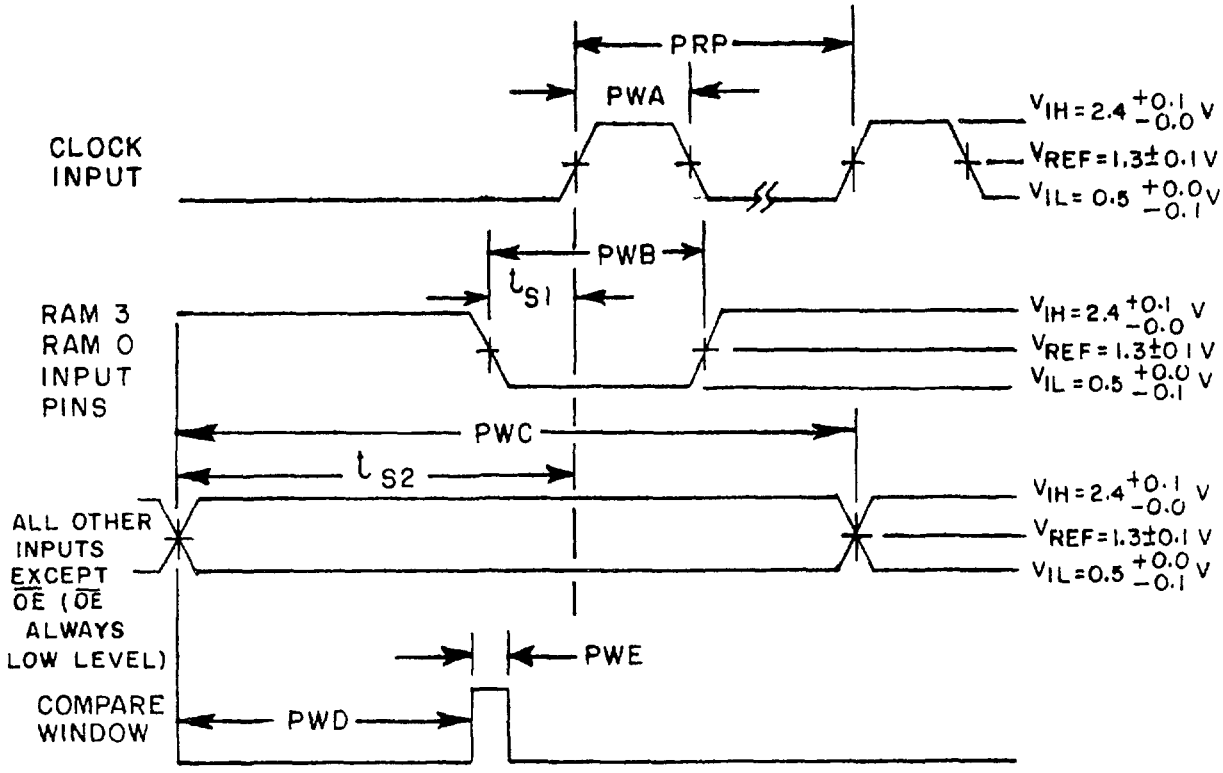


NOTES:

1. All inputs. $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points
2. Input pulse parameters:
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 165 ±1 ns.
   PWC = 130 ±1 ns.
   PWD = 215 ±1 ns.
   PWE = 200 ±1 ns.
   PWF = 10 ±1 ns.
   t_{S1} = 60 ±2 ns
   t_{S2} = 25 ±2 ns.
   t_{S3} = 110 ±2 ns.
3. During compare window check, verify that high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

FIGURE 23. Dynamic functional test timing waveforms for test conditions P1, P2, FPAT 7, and FPAT 17.

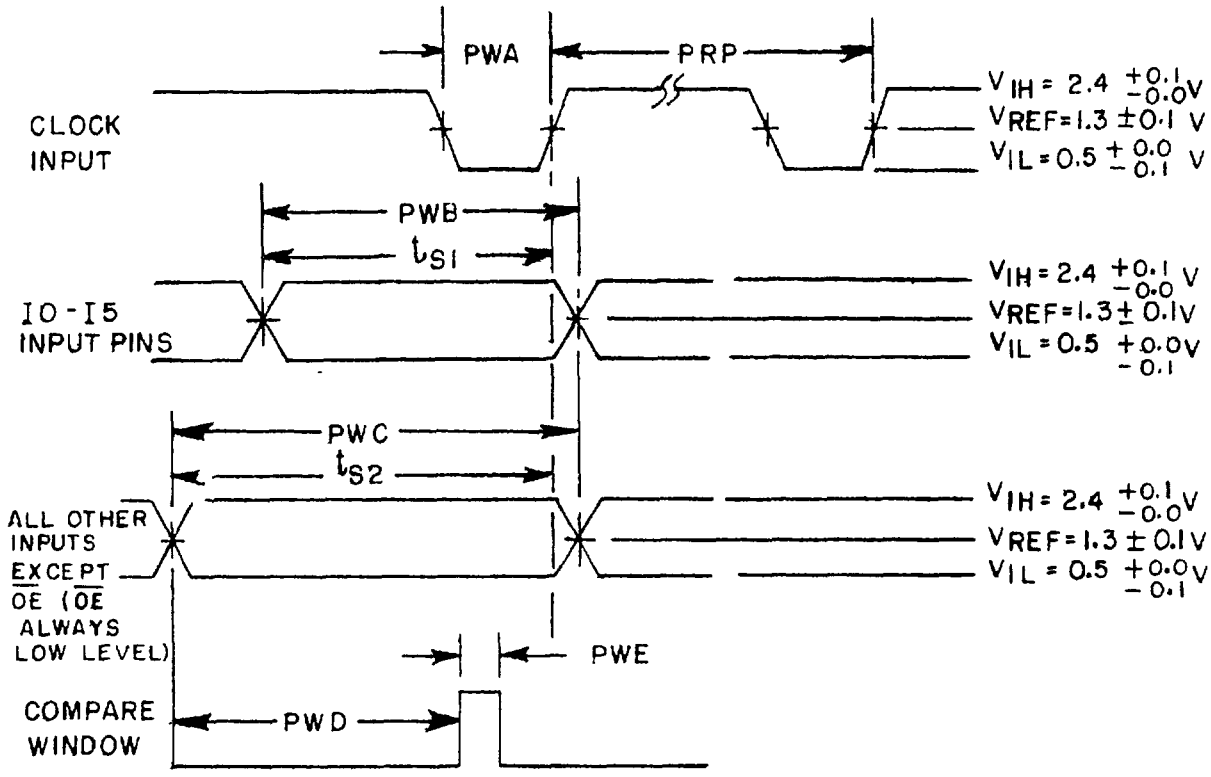
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NOTES:
1. All inputs: $t_{THL} = t_{TLH} = 10 \pm 2$ ns measured at the 10% and 90% PA points.
2. Input pulse parameters:
   PRP = 240 ±1 ns.
   PWA = 30 ±1 ns.
   PWB = 60 ±1 ns.
   PWC = 200 ±1 ns.
   PWD = 90 ±1 ns.
   PWE = 10 ±1 ns.
   t_{S1} = 25 ±2 ns.
   t_{S2} = 110 ±2 ns.
3. During compare window check, verify that high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

* FIGURE 24 Dynamic functional test timing waveforms for test conditions P3, P4, FPAT 18, FPAT 19, FPAT 20, and FPAT 21.

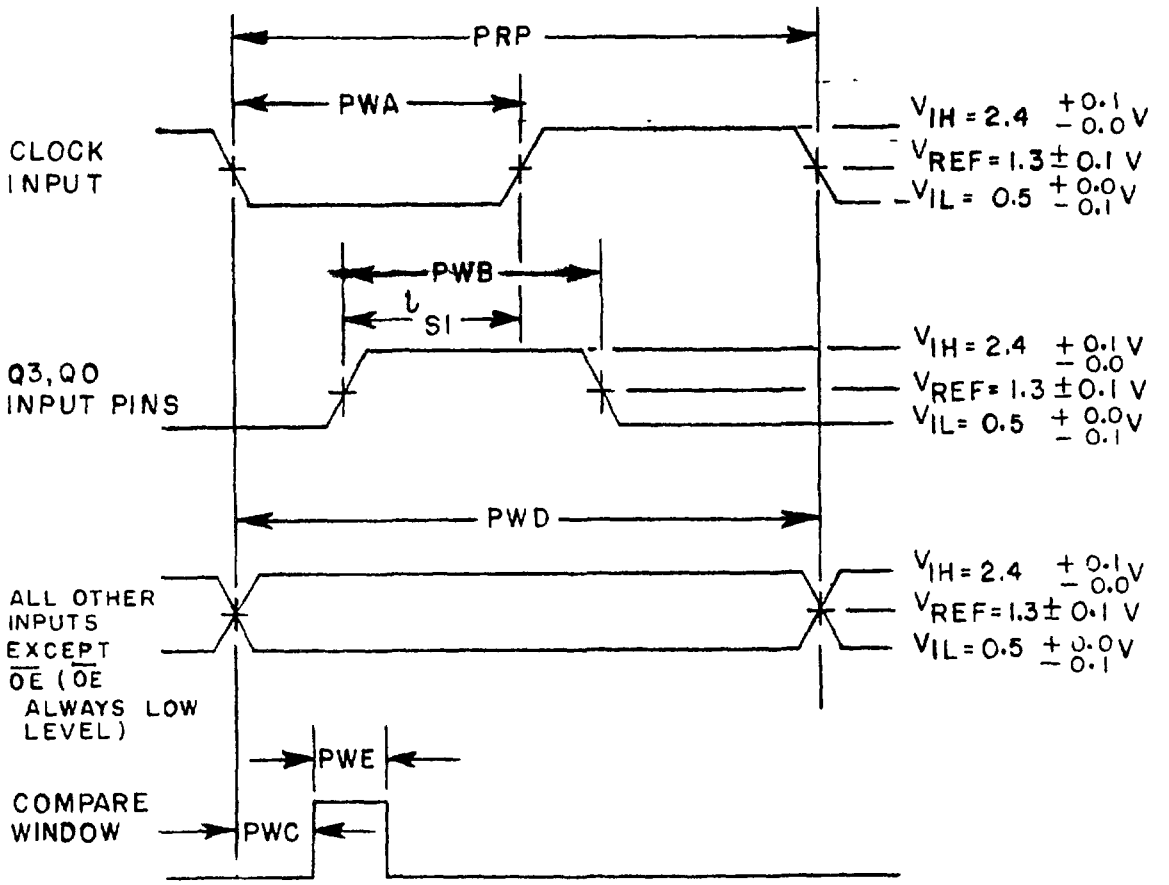
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NOTES
1 All inputs $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points
2 Input pulse parameters:
PRP = 112 ±1 ns.
PWA = 30 ±1 ns.
PWB = 87 ±1 ns.
PWC = 112 ±1 ns.
PWD = 95 ±1 ns
PWE = 10 ±1 ns.
t_{S1} = 85 ±2 ns.
t_{S2} = 110 ±2 ns.
3 During compare window check, verify that the high level outputs are ≥ 2.4 V and low level outputs are ≤ 0.5 V.

* FIGURE 25. Dynamic functional test timing waveforms for test conditions P5, P6, and FPAT 22

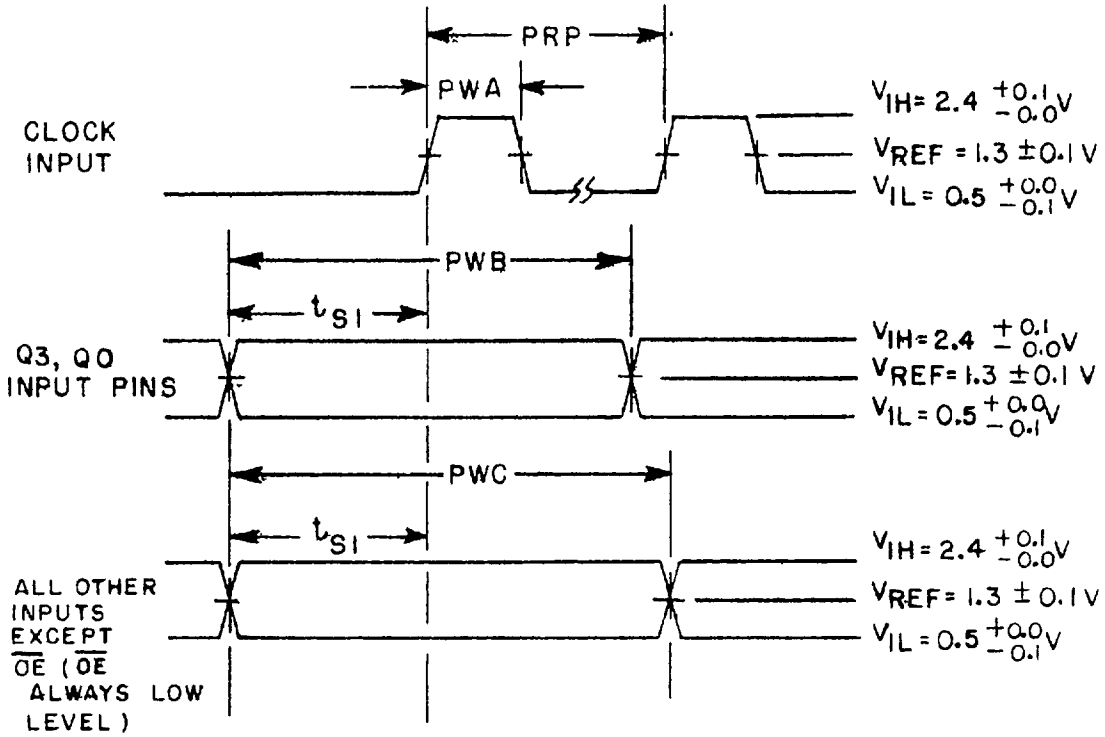
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
Check the source to verify that this is the current version before use.



NOTES
1. All inputs $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points
2. Input pulse parameters.
   PRP = 80 ±1 ns.
   PWA = 40 ±1 ns
   PWB = 35 ±1 ns.
   PWC = 20 ±1 ns.
   PWD = 80 ±1 ns.
   PWE = 10 ±1 ns.
   t_{SI} = 25 ±2 ns.
3. During compare window check, verify that the high level outputs are $\geq$ 2.4 V and low level outputs are $\leq$ 0.5 V.

* FIGURE 26 Dynamic functional test timing waveforms for test conditions P7, P8, FPAT 23, vectors 3-23, 26-46, 49-69, and 72-92.

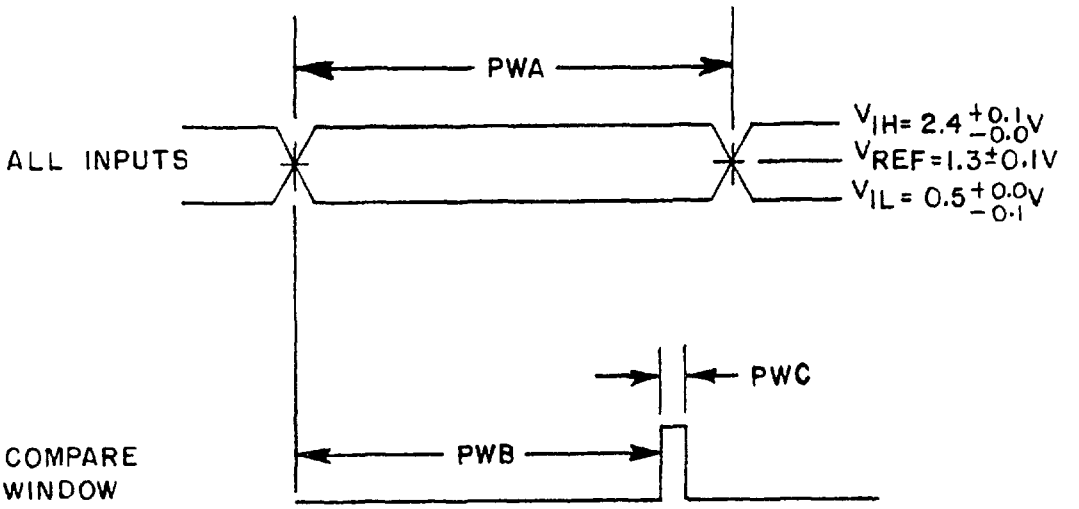
Source: https://assist.dla.mil --Downloaded: 2026-10-20T17:31Z
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NOTES:
1. All inputs: $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points
2. Input pulse parameters:
   PRP = 400 ±1 ns.
   PWA = 100 ±1 ns.
   PWB = 380 ±1 ns.
   PWC = 400 ±1 ns.
   t_{SI} = 180 ±2 ns.
3. These waveforms are required for setup of device prior to actual functional test
4. There are no outputs checked during these vectors.

FIGURE 27 Dynamic functional test timing waveforms for test conditions P7, P8, FPAT 23, vectors 1, 2, 24, 25, 47, 48, 70, and 71.

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NOTES:
1. All inputs: $t_{THL} = t_{TLH}$ = 10 ±2 ns measured at the 10% and 90% PA points.
2. Input pulse parameters:
   PWA = 240 ±1 ns.
   PWB = 200 ±1 ns.
   PWC = 10 ±1 ns.
3. During compare window check, verify that the high level outputs are ≥ 2.4 V and low level outputs are ≤ 0.5 V.

FIGURE 28 Dynamic functional test timing waveforms for test conditions P9, P10, and FPAT 24.

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TABLE XVII Transition time (t_{TLH}) test

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
Q1	Y0 Transition time, t_{TLH}	27,38,39,40	7,8,9,28,29,30, 34,35,36	37	Open	16	1	15	ns	29,30
Q2	Y1					17				
Q3	Y2					18				
Q4	Y3					19				
Q5	$\overline{P}$					15				
Q6	F3					11				
Q7	$\overline{G}$	7,9,27,38,39,40	8,28,29,30,34,35,36			12				
Q8	C_{n+4}	7,9,27,38,39,40	8,28,29,30,34,35,36			13				
Q9	OVR	7,9,27,38,39,40	8,28,29,30,34,35,36			14				
Q10	RAM 0	27,28,29,35	7,8,9,30,34,36	3,4,5 6		32				
Q11	RAM 3	27,34,35,36,38, 39,40	7,8,9,28,29,30	37		33				
Q12	Q3	27,28,29,34,35	7,8,9,30,36	2,3,4, 5,6		25				
Q13	Q0	27,28,29,35	7,8,9,30,34,36	3,4,5, 6,25		2				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
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Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
R1	Y0 Transition time, t_{THL}	27,38,39,40	7,8,9,28,29,30 34,35,36	37	Open	16	1	15	ns	29,30
R2	Y1					17				
R3	Y2					18				
R4	Y3					19				
R5	$\overline{P}$					15				
R6	F3					11				
R7	F=0					31				
R8	$\overline{G}$	7,9,27,38,39,40	8,28,29,30,34,35,36			12				
R9	C_{n+4}	7,9,27,38,39,40	8,28,29,30,34,35,36			13				
R10	OVR	7,9,27,38,39,40	8,28,29,30,34,35,36			14				
R11	RAM 0	27,28,29,35	7,8,9,30,34,36	3,4, 5,6		32				
R12	RAM 3	27,34,35,36,38, 39,40	7,8,9,28,29,30	37		33				
R13	Q3	27,28,29,34,35	7,8,9,30,36	2,3, 4,5, 6		25				
R14	Q0	27,28,29,35	7,8,9,30,34,36	3,4, 5,6, 25		2				

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TABLE XIX Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S1	A0 to Y0 Propagation delay time, t_{PLH},	21,22,24,27,38, 39,40	7,8,9,28,29,23, 30,34,35,36	37	37	16	7	75	ns	29,31
S2	A0 to F3	21,22,24,27,38, 39,40	7,8,9,28,29,23, 30,34,35,36			11	7	75		
S3	A0 to C_{n+4}	21,22,24,27,30, 38,39,40	7,8,9,28,29,34, 35,36,23			13	7	70		
S4	A0 to $\overline{P}$	21,22,24,27,30, 38,39,40	7,8,9,28,29,34, 35,36,23			15	5	59		
S5	A0 to F=0	21,22,24,27,30, 38,39,40	7,8,9,28,29,34, 35,36,23			31	8	85		
S6	A0 to OVR	7,9,21,22,24,27, 38,39,40	8,28,29,30,34, 35,36,23			14	7	76		
S7	A0 to $\overline{G}$	7,9,21,22,24,27, 38,39,40	8,28,29,30,34, 35,36,23			12	5	59		
S8	A0 to RAM 3	21,22,24,27,34, 35,36,38,39,40	7,8,9,28,29,30, 23			33	9	90		
S9	A0 to RAM 0	21,22,24,27,35, 36,38,39,40	7,8,9,28,29,30, 23,34			32	9	90		
S10	B0 to Y0	21,22,23,28,29, 37,39,40	7,8,9,27,30,34, 35,36,38	24	24	16	7	75		
S11	B0 to F3	21,22,23,28,29, 37,39,40	7,8,9,27,30,34, 35,36,38			11	7	75		
S12	B0 to C_{n+4}	21,22,23,28,29, 37,39,40	7,8,9,27,34,35, 36,38			13	7	70		
S13	B0 to $\overline{P}$	21,22,23,28,29, 30,37,39,40	7,8,9,27,34,35, 36,38			15	5	59		

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* TABLE XIX Propagation delay time (t_{PLH}) tests - Continued

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S14	BO to F=0 Propagation delay time, t_{PLH}	21,22,23,28,29 30,37,39,40	7,8,9,27,34,35, 36,38	24	24	31	8	85	ns	29,31
S15	BO to OVR	7,9,21,22,23,28, 29,37,39,40	8,27,30,34,35, 36,38			14	7	76		
S16	BO to $\overline{G}$	7,9,21,22,23,28, 29,37,39,40	8,27,30,34,35, 36,38			12	5	62		
S17	BO to RAM 0	7,9,21,22,23,28, 29,35,36,37,39,40	8,27,30,34,38,			32	9	90		
S18	BO to RAM 3	21,22,23,28,29, 34,35,36,37,39, 40	7,8,9,27,30,38,			33	9	90		
S19	DO to $\overline{G}$	21,22,23,24,37, 38,39,40,9,27, 28,29	7,8,30,34,35,36	3,4,5,6	6	12	3	31		
S20	DO to YO	21,22,23,24,37, 38,39,40,27,28, 29	7,8,9,30,34,35, 36			16	3	39		
S21	DO to F3	21,22,23,24,37, 38,39,40,27,28, 29	7,8,9,30,34,35, 36			11	3	37		
S22	DO to C_{n+4}	21,22,23,24,37, 38,39,40,27,28, 29,30	7,8,9,34,35,36			13	4	41		
S23	DO to $\overline{P}$	21,22,23,24,37, 38,39,40,27,28, 29,30	7,8,9,34,35,36			15	3	31		
S24	DO to F=0	21,22,23,24,37, 38,39,40,27,28, 29,30	7,8,9,34,35,36			31	5	55		
S25	DO to OVR	21,22,23,24,37, 38,39,40,27,28, 29,30	3,7,8,9,34,35,36	4,5,6		14	4	45		

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TABLE XIX. Propagation delay time (t_{PLH}) test - Continued

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S26	DO to RAM 3 Propagation delay time, t_{PLH}	21,22,23,24,27,28,29, 34,35,36,37,38,39,40	7,8,9,30	3,4,5,6	6	23	5	59	ns	29,31
S27	DO to RAM 0	27,28,29,35,21,22,23 24,36,37,38,39,40	7,8,9,30,34	3,4,5,6		32	5	59		
S28	DO to Y0	21,22,23,24,37,38,39, 40,7,9,27,28,29	8,30,34,35, 36	3,4,5,6		16	3	36		
S29	DO to F3	21,22,23,24,37,38, 39,40,7,9,27,28,29	8,30,34,35, 36	3,4,5,6		11	3	34		
S30	DO to F=0	21,22,23,24,37,38 39,40,7,9,27,28,29	8,30,34,35, 36	3,4,5,6		31	5	51		
S31	DO to RAM 3	21,22,23,24,37,38, 7,9,27,28,29,34,35, 36,39,40	8,30	3,4,5,6		33	5	53		
S32	DO to RAM 0	7,9,27,28,29,21,22, 23,24,35,36,37,38, 39,40	8,30,34	3,4,5,6		32	5	53		
S33	C_n to Y0	3,4,5,6,21,22,23, 24,27,28,29,37,38, 39,40	7,8,9,34, 35,36	30	30	16	2	27		
S34	C_n to F3					11	2	27		
S35	C_n to C_{n+4}					13	2	20		
S36	C_n to F=0					31	4	46		
S37	C_n to OVR	3,4,5,6,7,9,21,22, 23,24,27,28,29,37, 38,39,40	8,34,35,36			14	2	26		
S38	C_n to RAM 3	3,4,5,6,21,22,23, 24,27,28,29,34,35, 36,37,38,39,40	7,8,9			33	4	45		

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TABLE XIX Propagation delay time (t_{PLH}) test - Continued

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S39	C_n to RAM 0 Propagation delay time, t_{PLH}	3,4,5,6,21,22,23,24,27, 28,29,35,36,37,38,39,40	7,8,9,34	30	30	32	4	45	ns	29,31
S40	I1 to Y0	3,4,5,6,27,29,21,22,23, 24,37,38,39,40	7,8,9,30, 34,35,36	28	28	16	5	50		
S41	I1 to F3	3,4,5,6,21,22,23,24,27, 29,30,37,38,39,40	7,8,9,34, 35,36			11	5	50		
S42	I1 to C_{n+4}	3,4,5,6,21,22,23,24,27, 29,37,38,39,40	7,8,9,30, 34,35,36			13	4	46		
S43	I1 to $\overline{P}$	3,4,5,21,22,23,24,27, 29,37,38,39,40	6,7,8,9, 30,34,35, 36			15	4	41		
S44	I1 to F=0	3,4,5,6,21,22,23,24, 27,29,30,37,38,39,40	7,8,9,34, 35,36			31	6	65		
S45	I1 to OVR	4,5,6,21,22,23,24,27, 29,30,37,38,39,40	3,7,8,9, 34,35,36			14	5	57		
S46	I1 to RAM 3	3,4,5,6,21,22,23,24, 27,29,30,34,35,36,37, 38,39,40	7,8,9			33	7	70		
S47	I1 to RAM 0	3,4,5,6,21,22,23,24,27, 29,30,35,36,37,38,39, 40	7,8,9,34			32	7	70		
S48	I1 to $\overline{G}$	3,4,5,6,21,22,23,24,27, 29,30,35,36,37,38,39, 40	7,8,9,34			12	4	41		
S49	I3 to Y0	3,4,5,6,21,22,23,24,27, 28,29,30,37,38,39,40	8,9,34,35, 36	7	7	16	5	50		
S50	I3 to F3	3,4,5,6,21,22,23,24,27, 28,29,37,38,39,40	8,9,30, 34,35,36	7	7	11	5	50		
S51	I3 to C_{n+4}	3,4,5,6,21,22,23,24,27, 28,29,30,37,38,39,40	8,9,34, 35,36	7	7	13	5	50		

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TABLE XIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S52	I3 to $\overline{P}$ Propagation delay time, t_{PLH}	3,4,5,6,21,22,23,24,27,28,29,30,37,38,39,40	8,9,34,35,36	7	7	15	4	42	ns	29,31
S53	I3 to F=Q	3,4,5,6,21,22,23,24,27,28,29,30,37,38,39,40	8,9,34,35,36			31	6	65		
S54	I3 to $\overline{G}$	3,4,5,6,9,21,22,23,24,27,28,29,37,38,39,40	8,30,34,35,36			12	4	42		
S55	I3 to OVR	4,5,6,21,22,23,24,27,28,29,30,37,38,39,40	3,8,9,34,35,36			14	5	59		
S56	I3 to RAM 3	3,4,5,6,21,22,23,24,27,28,29,34,35,36,37,38,39,40	8,9,30			33	7	70		
S57	I3 to RAM 0	3,4,5,6,21,22,23,24,27,28,29,35,36,37,38,39,40	8,9,30,34			32	7	70		
S58	I6 to YO	3,4,5,6,9,21,22,23,24,27,28,29,34,37,38,39,40	7,8,30,35	36	36	16	2	26		
S59	AO to YO	21,22,23,24,27,34,39,40,38	7,8,9,28,29,30,35,36	37	37	16	3	35		
S60	CP to YO	21,22,23,24,28,34,35,37,38,39,40	7,8,9,27,29,30,36	2	26	16	5	52		
S61	CP to F3	21,22,23,24,28,34,35,37,38,39,40	7,8,9,27,29,30,36			11	5	52		
S62	CP to C_{n+4}	4,5,6,21,22,23,24,27,28,34,35,37,38,39,40	3,7,8,9,29,30,36			13	5	52		
S63	CP to $\overline{P}$	4,5,6,21,22,23,24,27,28,34,35,37,38,39,40	3,7,8,9,29,30,36			15	4	41		
S64	CP to $\overline{G}$	4,5,6,21,22,23,24,27,28,34,35,37,38,39,40	3,7,8,9,29,30,36			12	4	41		

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TABLE XIX Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S65	CP to F=0 Propagation delay time, t_{PLH}	4,5,21,22,23,24,27,28, 34,35,37,38,39,40	3,6,7,8,9, 29,30,36	2	26	31	7	70	ns	29, 31
S66	CP to OVR	4,5,21,22,23,24,27,28, 34,35,37,38,39,40	3,6,7,8,9, 29,30,36	2	26	14	5	57		
S67	CP to Q0	21,22,23,24,28,35,37, 38,39,40	7,8,9,27, 29,30,34, 36	25	26	2	3	30		
S68	CP to RAM 0	21,22,23,24,28,35,37, 38,39,40	7,8,9,27, 29,30,34, 36	25	26	32	7	71		
S69	CP to Q3	21,22,23,24,28,34, 35,37,38,39,40	7,8,9,27, 29,30,36	2	26	25	3	30		
S70	CP to RAM 3	21,22,23,24,28,34,35, 37,38,39,40	7,8,9,27, 29,30,36	2	26	33	7	71		

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* TABLE XX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	A0 to Y0 Propagation delay time, t_{PHL}	21,22,24,27,38,39, 40	7,8,9,28, 29,23,30, 34,35,36	37	37	16	7	75	ns	29,31
T2	A0 to F3	21,22,24,27,38,39, 40	7,8,9,28, 29,23,30, 34,35,36			11	7	75		
T3	A0 to C_{n+4}	21,22,24,27,30,38, 39,40	7,8,9,28, 29,34,35, 36,23			13	7	70		
T4	A0 to $\overline{P}$	21,22,24,27,30,38, 39,40	7,8,9,28,29, 34,35,36,23			15	5	59		
T5	A0 to F=0	21,22,24,27,30,38, 39,40	7,8,9,28,29, 34,35,36,23			31	8	85		
T6	A0 to OVR	7,9,21,22,24,27,38, 39,40	8,28,29,30, 34,35,36,23			14	7	76		
T7	A0 to $\overline{G}$	7,9,21,22,24,27,38, 39,40	8,28,29,30, 34,35,36,23			12	5	59		
T8	A0 to RAM 3	21,22,24,27,34,35, 36,38,39,40	7,8,9,28,29, 30,23			33	9	90		
T9	A0 to RAM 0	21,22,24,27,35, 36,38,39,40	7,8,9,28,29, 30,23,34			32	9	90		
T10	B0 to Y0	21,22,23,28,29,37, 39,40	7,8,9,27,30, 34,35,36,38	24	24	16	7	75		
T11	B0 to F3	21,22,23,28,29,37, 39,40	7,8,9,27,30, 34,35,36,38			11	7	75		
T12	B0 to C_{n+4}	21,22,23,28,29,30, 37,39,40	7,8,9,27,34, 35,36,38			13	7	70		
T13	B0 to $\overline{P}$	21,22,23,28,29,30, 37,39,40	7,8,9,27,34, 35,36,38			15	5	59		

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TABLE ... Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
T14	BO to F=0 Propagation delay time, t_{PHL}	21,22,23,28,29, 30,37,39,40	7,8,9,27,34, 35,36,38	24	24	31	8	85	ns	29,31
T15	BO to OVR	7,9,21,22,23,28, 29,37,39,40	8,27,30,34, 35,36,38			14	7	76		
T16	BO to $\overline{G}$	7,9,21,22,23,28, 29,37,39,40	8,27,30,34, 35,36,38			12	5	59		
T17	BO to RAM 0	7,9,21,22,23,28, 29,35,36,37,39,40	8,27,30,34, 38			32	9	90		
T18	BO to RAM 3	21,22,23,28,29,34, 35,36,37,39,40	7,8,9,27, 30,38			33	9	90		
T19	DO to $\overline{G}$	21,22,23,24,37,38, 39,40,9,27,28,29	7,8,30,34, 35,36	3,4,5,6	6	12	3	31		
T20	DO to YO	21,22,23,24,37,38, 39,40,27,28,29	7,8,9,30, 34,35,36			16	3	39		
T21	DO to F3	21,22,23,24,37,38, 39,40,27,28,29	7,8,9,30, 34,35,36			11	3	37		
T22	DO to C_{n+4}	21,22,23,24,37,38, 39,40,27,28,29,30	7,8,9,34, 35,36			13	4	41		
T23	DO to $\overline{P}$	21,22,23,24,37,38, 39,40,27,28,29,30	7,8,9,34, 35,36			15	3	31		
T24	DO to F=0	21,22,23,24,37,38, 39,40,27,28,29,30	7,8,9,34, 35,36			31	5	55		
T25	DO to OVR	21,22,23,24,37,38, 39,40,27,28,29,30	3,7,8,9, 34,35,36	4,5,6		14	4	45		

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* TABLE XX Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T26	DO to RAM 3 Propagation delay time, t_{PHL}	21,22,23,24,27,28,29, 34,35,36,37,38,39,40	7,8,9,30	3,4,5,6	6	33	5	59	ns	29,31
T27	DO to RAM 0	27,28,29,35,21,22,23, 24,36,37,38,39,40	7,8,9, 30,34			32	5	59		
T28	DO to YO	21,22,23,24,37,38,39, 40,7,9,27,28,29	8,30,34, 35,36			16	3	36		
T29	DO to F3	21,22,23,24,37,38,39, 40,7,9,27,28,29	8,30,34, 35,36			11	3	34		
T30	DO to F=0	21,22,23,24,37,38,39, 40,7,9,27,28,29	8,30,34, 35,36			31	5	51		
T31	DO to RAM 3	21,22,23,24,37,38,7,9, 27,28,29,34,35,36,39, 40	8,30			33	5	53		
T32	DO to RAM 0	7,9,27,28,29,21,22,23, 24,35,36,37,38,39,40	8,30,34			32	5	53		
T33	C_n to YO	3,4,5,6,21,22,23,24,27, 28,29,37,38,39,40	7,8,9,34, 35,36	30	30	16	2	27		
T34	C_n to F3					11	2	29		
T35	C_n to C_{n+4}					13	2	22		
T36	C_n to F=0					31	4	46		
T37	C_n to OVR	3,4,5,6,7,9,21,22,23, 24,27,28,29,37,38,39, 40	8,34,35,36			14	2	26		
T38	C_n to RAM 3	3,4,5,6,21,22,23,24, 27,28,29,37,35,36,37, 38,39,40	7,8,9			33	4	45		

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TABLE XX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T39	C_n to RAM 0 Propagation delay time, t_{PHL}	3,4,5,6,21,22,23,24,27, 28,29,35,36,37,38,39,40	7,8,9,34	30	30	32	4	45	ns	29,31
T40	I1 to Y0	3,4,5,6,27,29,21,22,23 24,37,38,39,40	7,8,9,30, 34,35,36	28	28	16	5	50		
T41	I1 to F3	3,4,5,6,21,22,23,24,27, 29,30,37,38,39,40	7,8,9,34, 35,36			11	5	50		
T42	I1 to C_{n+4}	3,4,5,6,21,22,23,24,27, 29,37,38,39,40	7,8,9,30, 34,35,36			13	4	46		
T43	I1 to $\overline{P}$	3,4,5,21,22,23,24,27, 29,37,38,39,40	6,7,8,9,30, 34,35,36			15	4	41		
T44	I1 to F=0	3,4,5,6,21,22,23,24, 27,29,30,37,38,39,40	7,8,9,34, 35,36			31	6	65		
T45	I1 to OVR	4,5,6,21,22,23,24,27,29, 30,37,38,39,40	3,7,8,9, 34,35,36			14	5	57		
T46	I1 to RAM 3	3,4,5,6,21,22,23,24,27, 29,30,34,35,36,37,38, 39,40	7,8,9			33	7	70		
T47	I1 to RAM 0	3,4,5,6,21,22,23,24,27, 29,30,35,36,37,38,39, 40	7,8,9,34			32	7	70		
T48	I1 to $\overline{G}$	3,4,5,6,21,22,23,24, 27,29,30,35,36,37,38, 39,40	7,8,9,34			12	4	41		
T49	I3 to Y0	3,4,5,6,21,22,23,24, 27,28,29,30,37,38, 39,40	8,9,34, 35,36	7	7	16	5	50		
T50	I3 to F3	3,4,5,6,21,22,23,24, 27,28,29,37,38, 39,40	8,9,30, 34,35,36	7	7	11	5	50		
T51	I3 to C_{n+4}	3,4,5,6,21,22,23,24, 27,28,29,30,37,38,39, 40	8,9,34, 35,36	7	7	13	5	50		

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* TABLE XX Propagation delay time (t_{PHL}) test - Continued.

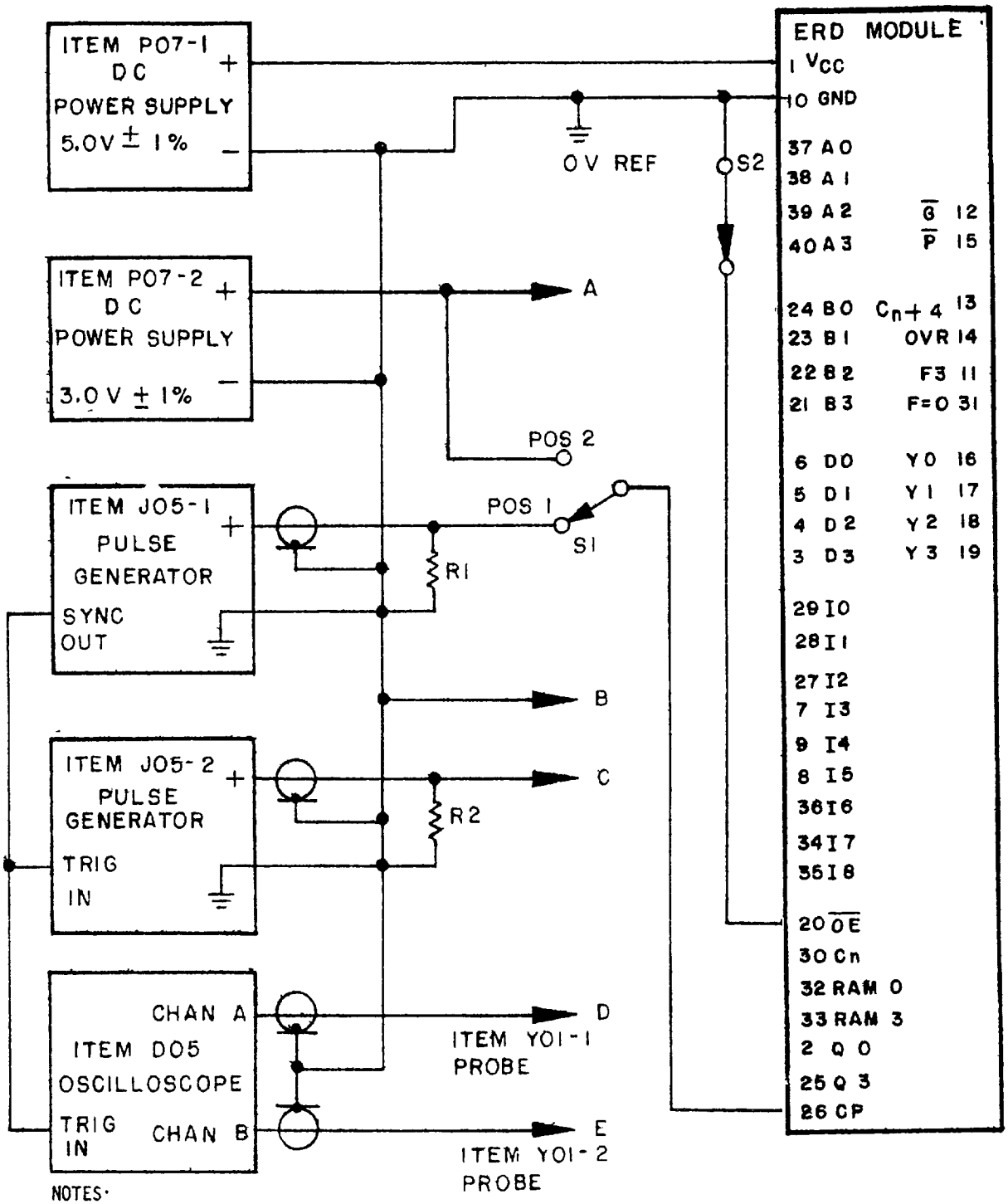
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T52	I3 to $\overline{P}$ Propagation delay time, t_{PHL}	3,4,5,6,21,22,23,24,27, 28,29,30,37,38,39,40	8,9,34, 35,36	7	7	15	4	42	ns	29,31
T53	I3 to F=0	3,4,5,6,21,22,23,24,27, 28,29,30,37,38,39,40	8,9,34, 35,36			31	6	65		
T54	I3 to $\overline{G}$	3,4,5,6,9,21,22,23,24, 27,28,29,37,38,39,40	8,30,34, 35,36			12	4	42		
T55	I3 to OVR	4,5,6,21,22,23,24,27, 28,29,30,37,38,39,40	3,8,9,34, 35,36			14	5	59		
T56	I3 to RAM 3	3,4,5,6,21,22,23,24, 27,28,29,34,35,36,37, 38,39,40	8,9,30			33	7	70		
T57	I3 to RAM 0	3,4,5,6,21,22,23,24,27, 28,29,35,36,37,38,39,40	8,9,30,34			32	7	70		
T58	I6 to YO	3,4,5,6,9,21,22,23,24, 27,28,29,34,37,38,39, 40	7,8,30,35	36	36	16	2	26		
T59	A0 to YO	21,22,23,24,27,34,39, 40,38	7,8,9,28, 29,30,35, 36	37	37	16	3	35		
T60	CP to YO	21,22,23,24,28,34,35, 37,38,39,40	7,8,9,27, 29,30,36	2	26	16	5	52		
T61	CP to F3	21,22,23,24,28,34,35, 37,38,39,40	7,8,9,27, 29,30,36			11	5	52		
T62	CP to C_{n+4}	4,5,6,21,22,23,24,27, 28,34,35,37,38,39,40	3,7,8,9, 29,30,36			13	5	52		
T63	CP to $\overline{P}$	4,5,6,21,22,23,24,27, 28,34,35,37,38,39,40	3,7,8,9, 29,30,36			15	4	45		
T64	CP to $\overline{G}$	4,5,6,21,22,23,24,27, 28,34,35,37,38,39,40	3,7,8,9, 29,30,36			12	4	43		
T65	CP to F=0	4,5,21,22,23,24,27,28, 34,35,37,38,39,40	3,6,7,8,9, 29,30,36			31	7	70		

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TABLE XX Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T66	CP to OVR Propagation delay time, t_{PHL}	4,5,21,22,23,24,27,28, 34,35,37,38,39,40	3,6,7,8,9, 29,30,36	2	26	14	5	63	ns	29,31
T67	CP to Q0	21,22,23,24,28,35,37, 38,39,40	7,8,9,27, 29,30,34, 36	25	26	2	3	33		
T68	CP to RAM 0	21,22,23,24,28,35,37, 38,39,40	7,8,9,27, 29,30,34, 36	25	26	32	7	71		
T69	CP to Q3	21,22,23,24,28,34,35, 37,38,39,40	7,8,9,27, 29,30,36	2	26	25	3	30		
T70	CP to RAM 3	21,22,23,24,28,34,35, 37,38,39,40	7,8,9,27, 29,30,36	2	26	33	7	71		

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. S1 = SPDT switch.
3. S2 = SPST switch.
4. Load all outputs per figure 14.
5. The module is to be tested in sequential order and the program should be verified and/or re-entered at the beginning of each test sequence (Q1, R1, S1, T1).
6. Switch S2 open for test conditions Q8 and Q9 only; closed at all other times.
7. Switch S1 in position 1 for test conditions S60-S70 and T60-T70 and in position 2 for all other tests.

FIGURE 29 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVII, XVIII, XIX, and XX.

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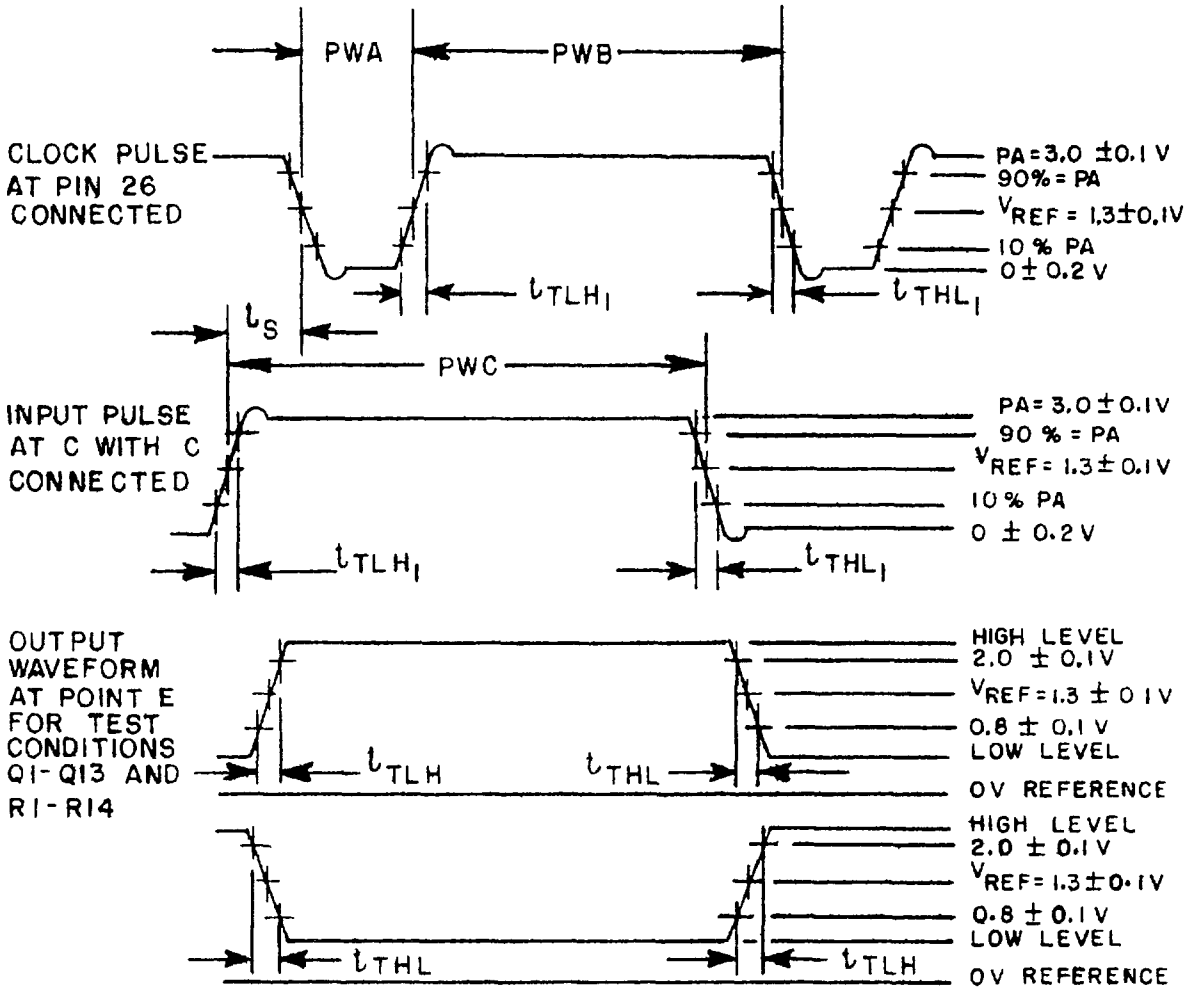
6. The following table is to be used for programming the RAM:

Step	A to pins	B to pins	C to pins	Switch S1	Output pins 16 17 18 19	Figure
A	7,9,21,22,23 27,28,29,34, 36,38,39,40	8,30,35	3,4,5,6,24, 37	Position 1	X X X X	29, 30
B	7,9,21,22,23, 28,29,36,38, 39,40,26	8,27,30,34, 35,24	Open	Position 2	L L L L	29, 30
C	21,22,23,24, 28,29,36,26, 7,9	8,27,30,34, 35	Open	Position 2	H H H H	29, 30
D	27,36,38,39, 40,26,7,9	8,28,29,30, 34,35,37	Open	Position 2	L L L L	29, 30
E	27,36,37,38, 39,40,26,7, 9	8,28,29,30, 34,35	Open	Position 2	H H H H	29, 30

X = Don't care.
L = Low level $\leq$ 0.4 V.
H = High level $\geq$ 2.4 V.

FIGURE 29. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVII, XVIII, XIX, and XX - Continued.

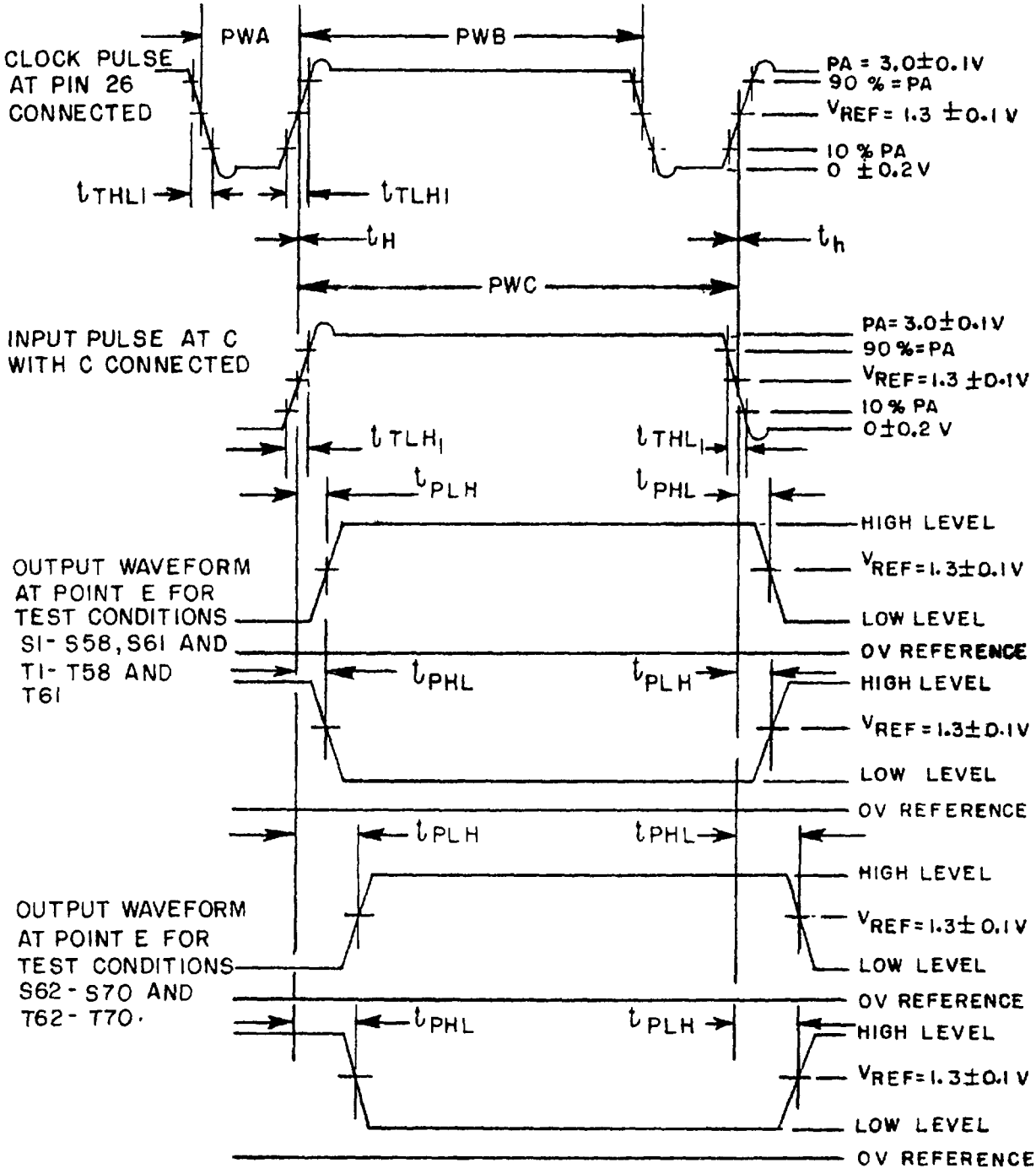
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
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NOTES
1. Input pulse parameters:
   PWA = 50 ±5 ns.
   PWB = 400 ±25 ns.
   PWC = 450 ±30 ns.
   t_S = 200 ±50 ns.
   $t_{TLH1} = t_{THL1}$ = 3.5 ±1.5 ns.
2. Input pulse C has a 50% duty cycle.
3 The clock pulse is in the double pulse mode and is used only for programming the RAM and test conditions Q12, Q13, R13, and R14.

FIGURE 30 Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XVII and XVIII.

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NOTES.
1. Input pulse parameters·
   PWA = 50 ±5 ns
   PWB = 400 ±25 ns.
   PWC = 450 ±30 ns.
   $t_{TLH1} = t_{THL1} = 3.5 \pm 1.5$ ns.
   $t_H = 0 \pm 1$ ns.
2. Input pulse C has a 50% duty cycle.
3. Before taking the specified measurement, verify that the high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.

FIGURE 31. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIX and XX.

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TABLE XXI. Disable/enable recovery time test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	YO Disable recovery time, t_{PLZ}	36	3,4,5,6, 34,35	20	16	6	66	ns	32,33
U2	Y1				17				
U3	Y2				18				
U4	Y3				19				
U5	YO Enable recovery time, t_{PZL}				16		60		
U6	Y1				17				
U7	Y2				18				
U8	Y3				19				
U9	YO Disable recovery time, t_{PHZ}	3,4,5,6, 36	34,35		16		66		
U10	Y1				17				
U11	Y2				18				
U12	Y3				19				
U13	YO Enable recovery time, t_{PZH}				16		60		
U14	Y1				17				
U15	Y2				18				
U16	Y3				19				

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TABLE XXII Disable/enable recovery time tests

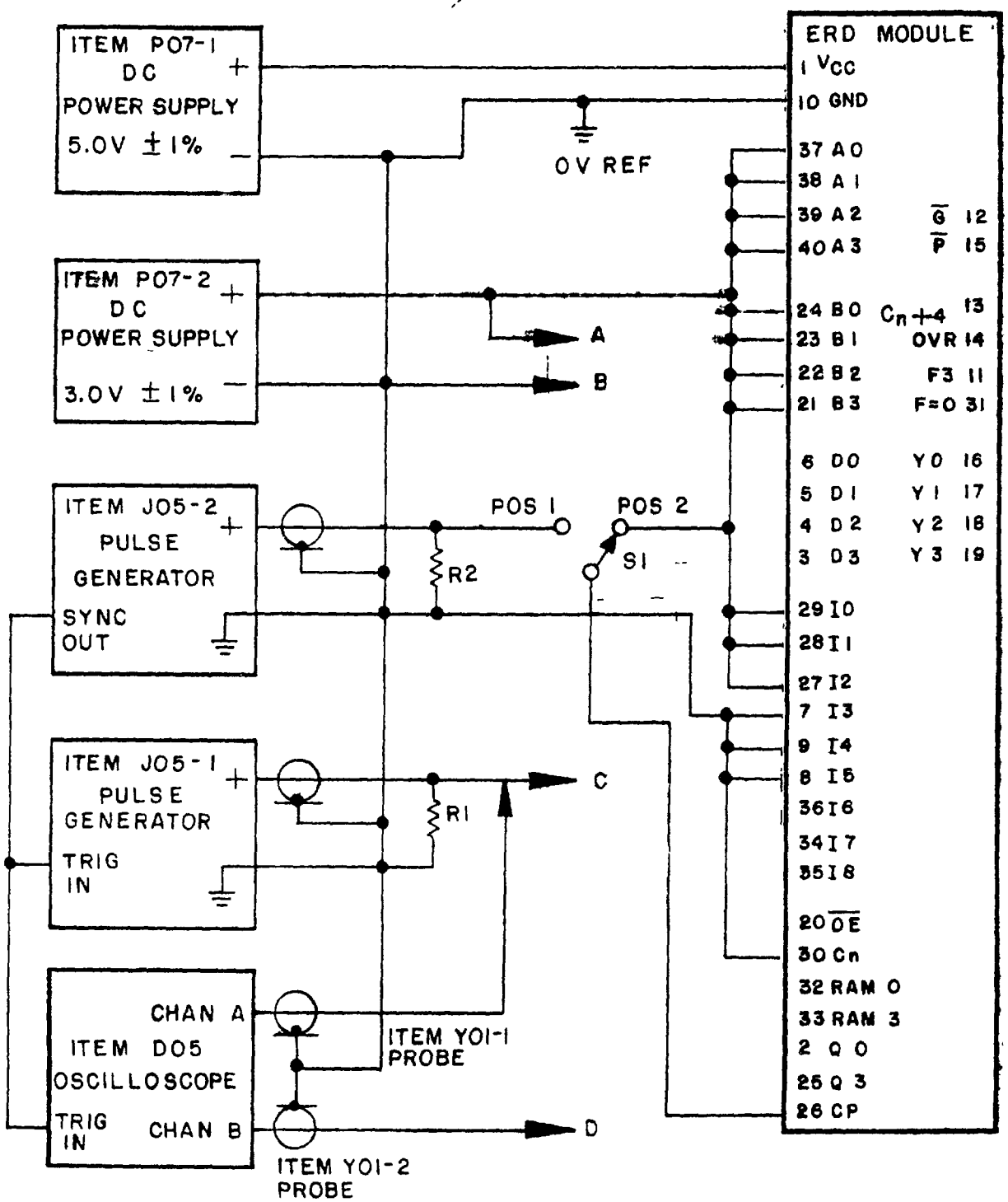
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	RAM 0 Disable recovery time, t_{PLZ}	3,4,5, 33	6,20,34	35,36	32	9	90	ns	32,34
V2	RAM 3	Open	3,4,5,6, 20	34,35, 36	33				
V3	Q0	25	3,4,5,6, 20,34,36	35	2				
V4	Q3	Open	3,4,5,6, 20	34,35, 36	25				
V5	RAM 0 Enable recovery time, t_{PZL}	3,4,5 33	6,20,34	35,36	32	5	50		
V6	RAM 3	Open	3,4,5,6, 20	34,35, 36	33				
V7	Q0	25	3,4,5,6, 20,34,36	35	2				
V8	Q3	Open	3,4,5,6, 20	34,35, 36	25				
V9	RAM 0 Disable recovery time, t_{PHZ}	3,4,5, 6,33	20,34	35,36	32	9	90		
V10	RAM 3	3	4,5,6,20	34,35, 36	33		90		
V11	Q0	3,4,5, 6,25	20,34 36	35	2		105		
V12	Q3	3	4,5,6,20	34,35, 36	25		90		

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TABLE XXII. Disable/enable recovery time tests - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V13	RAM 0 Enable recovery time, t_{PZH}	3,4,5,6, 33	20,34	35,36	32	5	50	ns	32,34
V14	RAM 3	3	4,5,6, 20	34,35, 36	33				
V15	Q0	3,4,5,6, 25	20,34, 36	35	2				
V16	Q3	3	4,5,6, 20	34,35, 36	25				

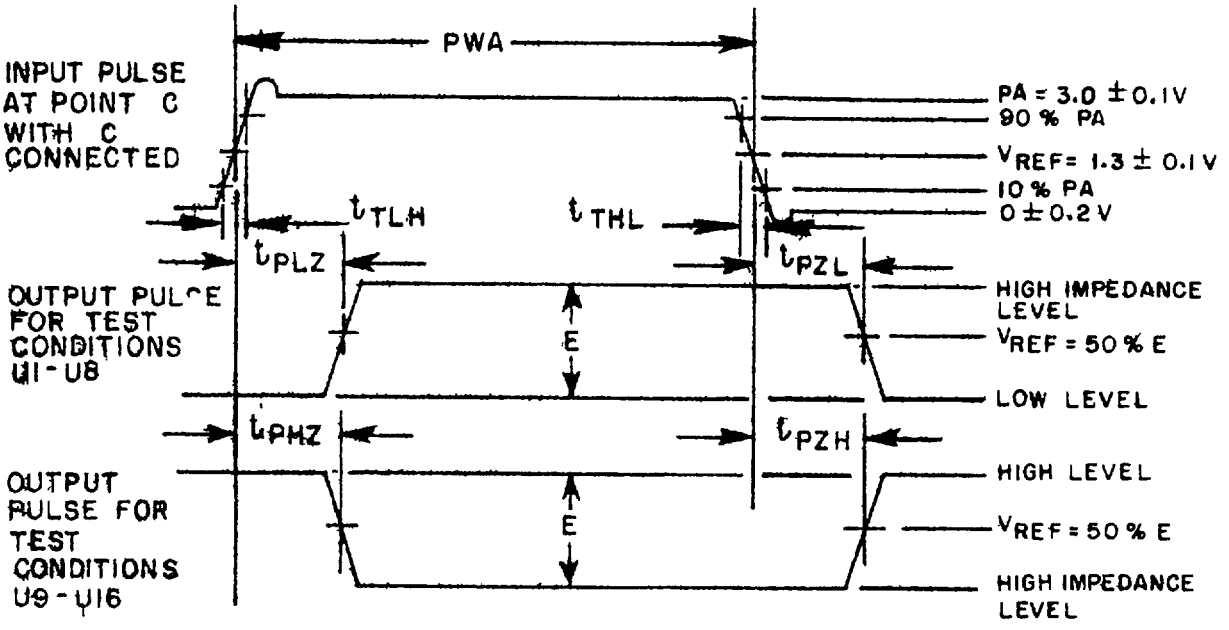
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. S1 = SPDT switch.
3. Load all outputs per figure 14.
4 Switch S1 to position 1 for test conditions V3, V4, V7, V8, V11, V12, V15, and V16 and position 2 for all other test conditions.

FIGURE 32. Disable/enable recovery time test circuit for tables XXI and XXII.

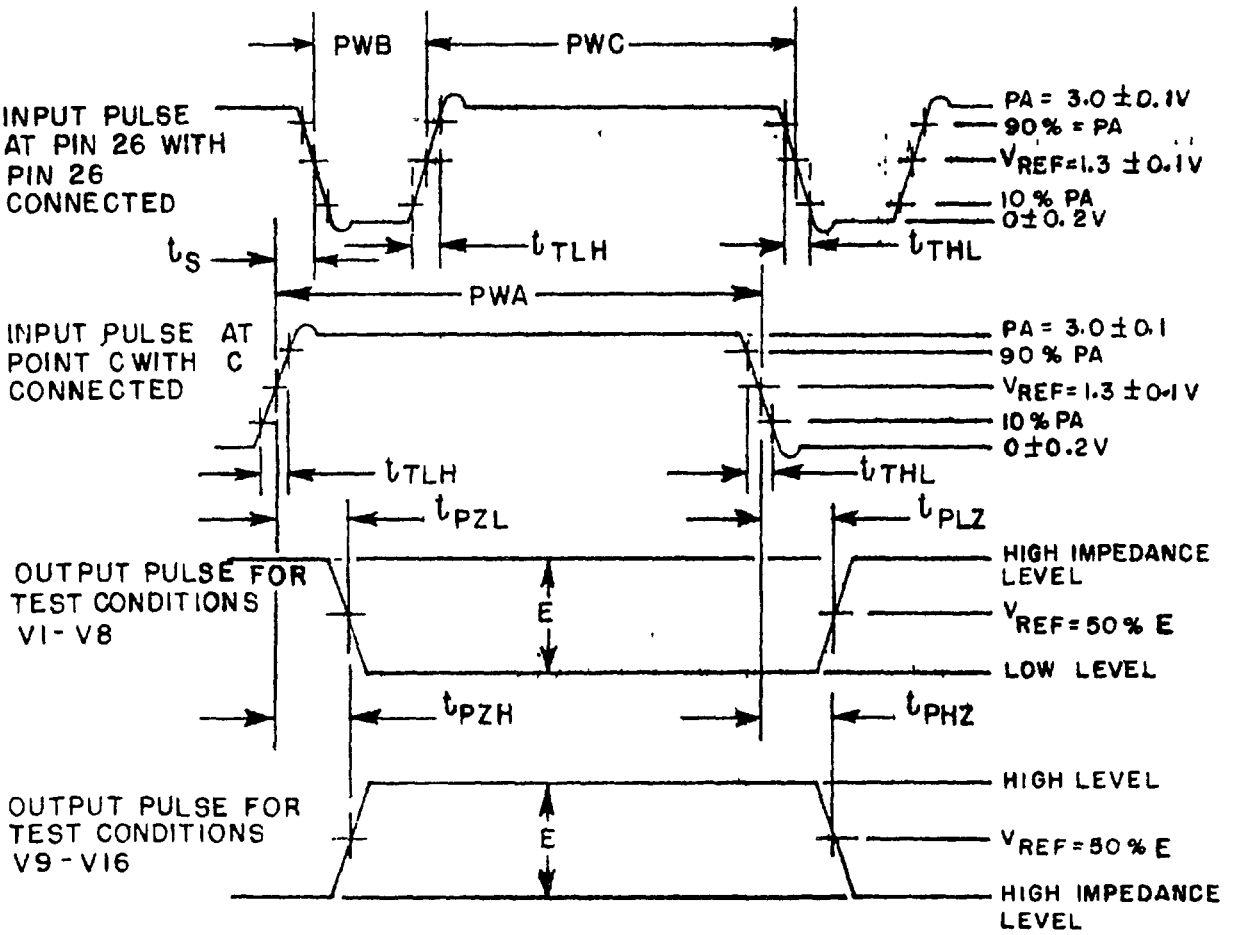
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NOTES:
1. Input pulse parameters:
   PWA = 450 ±30 ns at 50% duty cycle.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. Before taking the specified measurements, verify that the low level is $\leq$ 0.5 V and the high level is $\geq$ 2.4 V.

FIGURE 33. Disable/enable recovery time test circuit waveforms for table XXI.

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NOTES

1. Input pulse parameters:
   PWA = 450 ±30 ns at 50% duty cycle.
   PWB = 50 ±5 ns.
   PWC = 400 ±25 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
   $t_S = 200 \pm 20$ ns.
2. Before taking the specified measurements, verify that the low level is $\leq$ 0.5 V and the high level is $\geq$ 2.4 V.

FIGURE 34. Disable/enable recovery time test circuit waveforms for table XXII.

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TABLE XXIII. Output crosstalk (low-level) test.

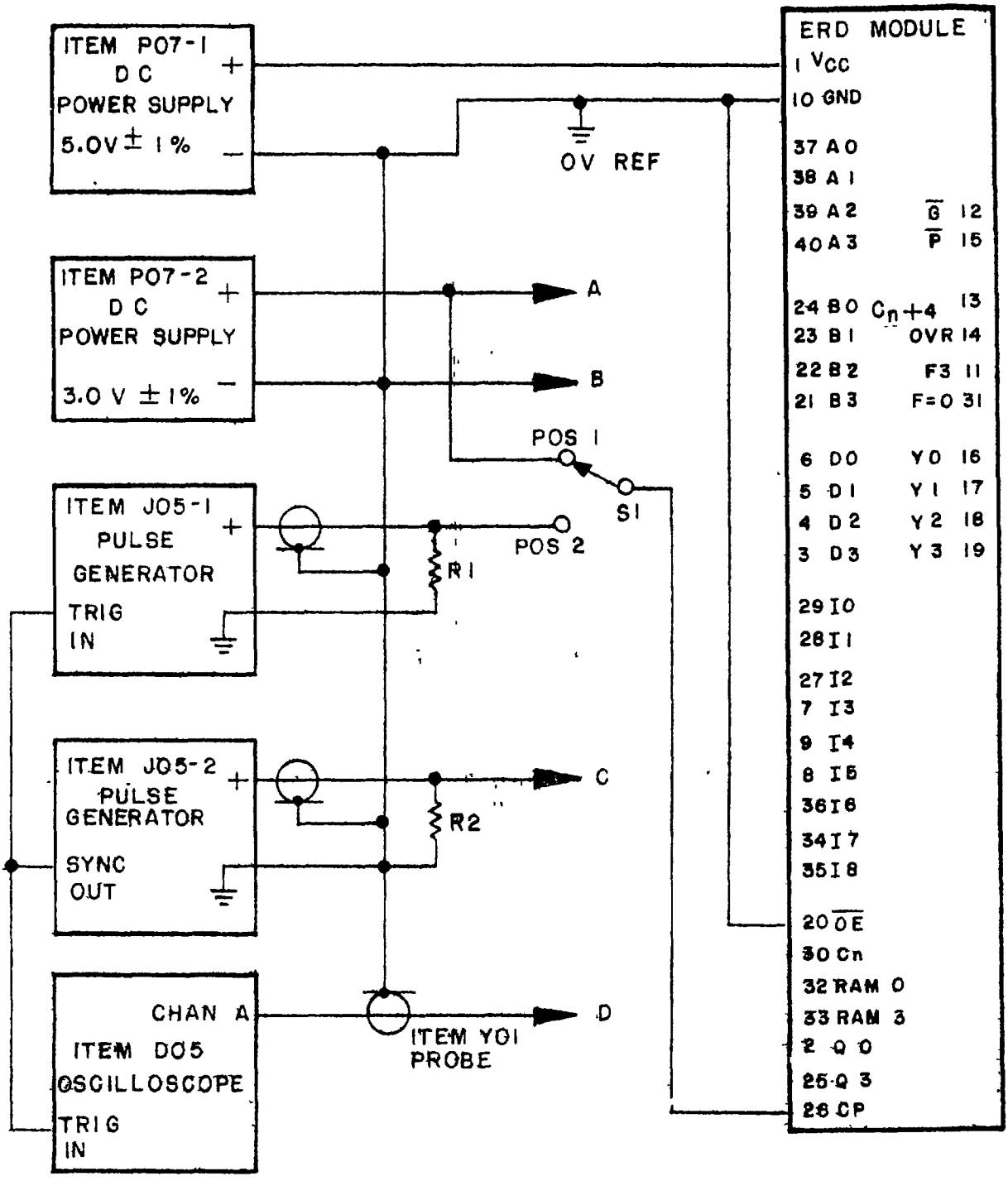
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	$\overline{P}$ Output crosstalk, low level	7,9,27,28, 29,36	8,30,34, 35	3,4,5, 6	15	0	600	mV	35,36
W2	C_{n+4}	7,27,28, 29,36	8,9,30, 34,35	3,4,5 6	13				
W3	OVR	7,27,28, 29,36	8,9,30, 34,35	3,4,5, 6	14				
W4	F=0	6,7,9,27, 28,29,36	8,30, 34,35	3,4,5	31				
W5	Y0	8,9,27,28, 29,36	6,7,30, 34,35	3,4,5	16				
W6	Y1		5,7,30, 34,35	3,4,6	17				
W7	Y2		4,7,30, 34,35	3,5,6	18				
W8	Y3		3,7,30, 34,35	4,5,6	19				
W9	F3		3,7,30, 34,35	4,5,6	11				
W10	$\overline{G}$	3,8,9,27, 28,29,36	7,30,34, 35	4,5,6	12				

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TABLE XXIV. Output crosstalk (high-level) test

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	$\overline{G}$ Output crosstalk, high level	7,27,28,29, 36	8,9,30,34, 35	3,4,5, 6	12	0	400	mV	35,36
X2	F3	3,7,9,27, 28,29,36	8,30,34,35	4,5,6	11				
X3	Y3	3,7,9,27, 28,29,36		4,5,6	19				
X4	Y2	4,7,9,27, 28,29,36		3,5,6	18				
X5	Y1	5,7,9,27, 28,29,36		3,4,6	17				
X6	Y0	6,7,9,27, 28,29,36		3,4,5	16				
X7	C_{n+4}	7,9,27,28, 29,30,36	8,34,35	3,4,5, 6	13				
X8	OVR	7,9,27,28, 29,30,36	8,34,35	3,4,5, 6	14				
X9	$\overline{P}$	27,28,29, 30,36	3,7,8,9, 34,35	4,5,6	15				

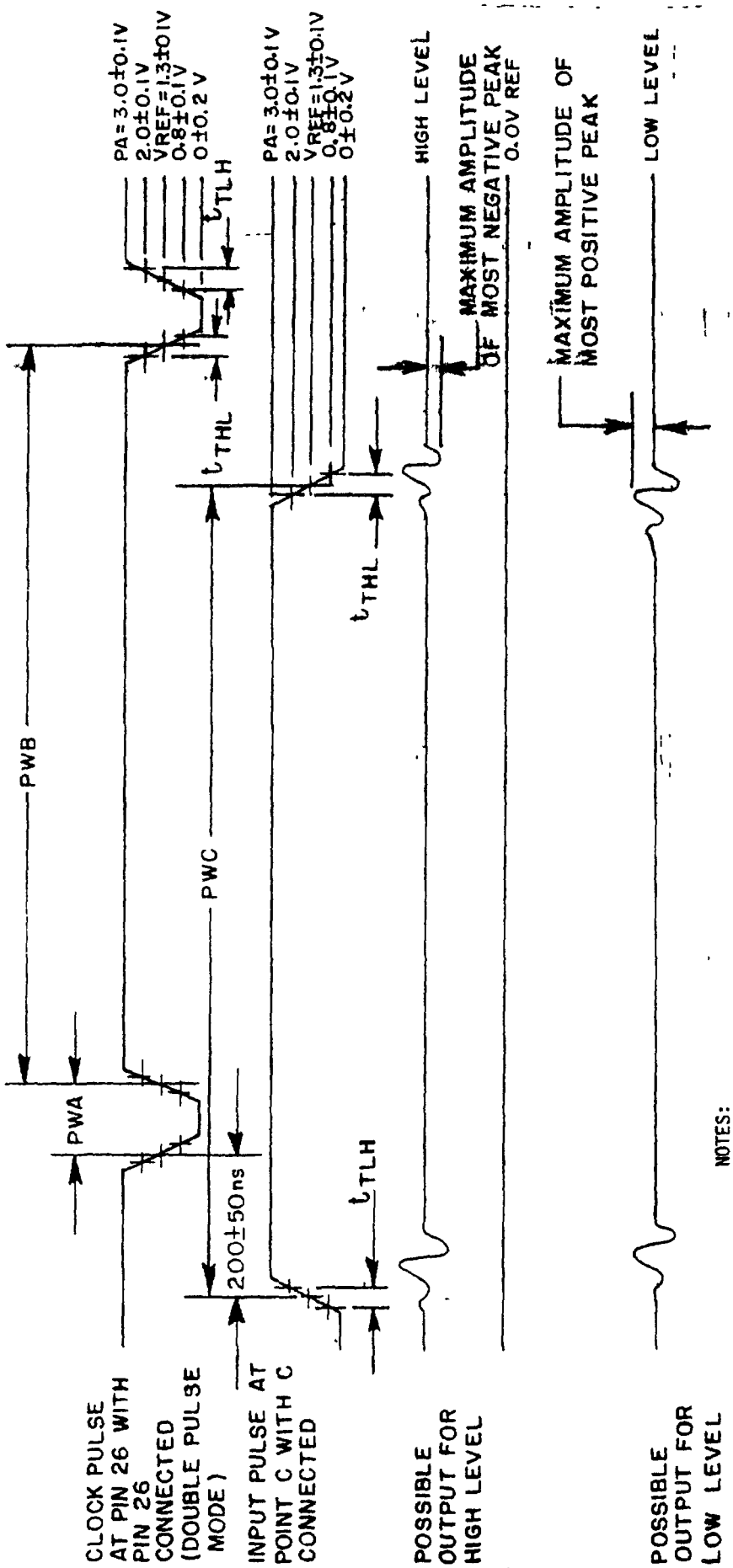
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NOTES:
1. R1 = R2 = 51Ω ±5%.
2. S1 = SPDT switch.
3. Load all outputs per figure 14.

FIGURE 35. Output crosstalk (low-level and high-level) test circuit for tables XXIII and XXIV

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NOTES:
1. Input pulse parameters:
   PWA = 50 ±5 ns.
   PWB = 400 ±25 ns.
   PWC = 450 ±30 ns at 50% duty cycle.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured between the 0.8 and 2.0 V points).
2. Setup for programming the RAM:
   a. Ground pin 30 (C_n).
   b. Switch S1 to position 2.
   c. Apply input pulse from point C to D0-D3, A0 and B0.
   d. Apply 3.5 V to A1-A3, and B1-B3.
   e. Apply microcode I_{337} for a minimum of 2 seconds.
   f. Switch S1 to position 1 and verify B RAM is programmed microcode I_{133}.
   g. Verify A RAM is programmed microcode I_{134}
   h. Begin testing per test conditions.

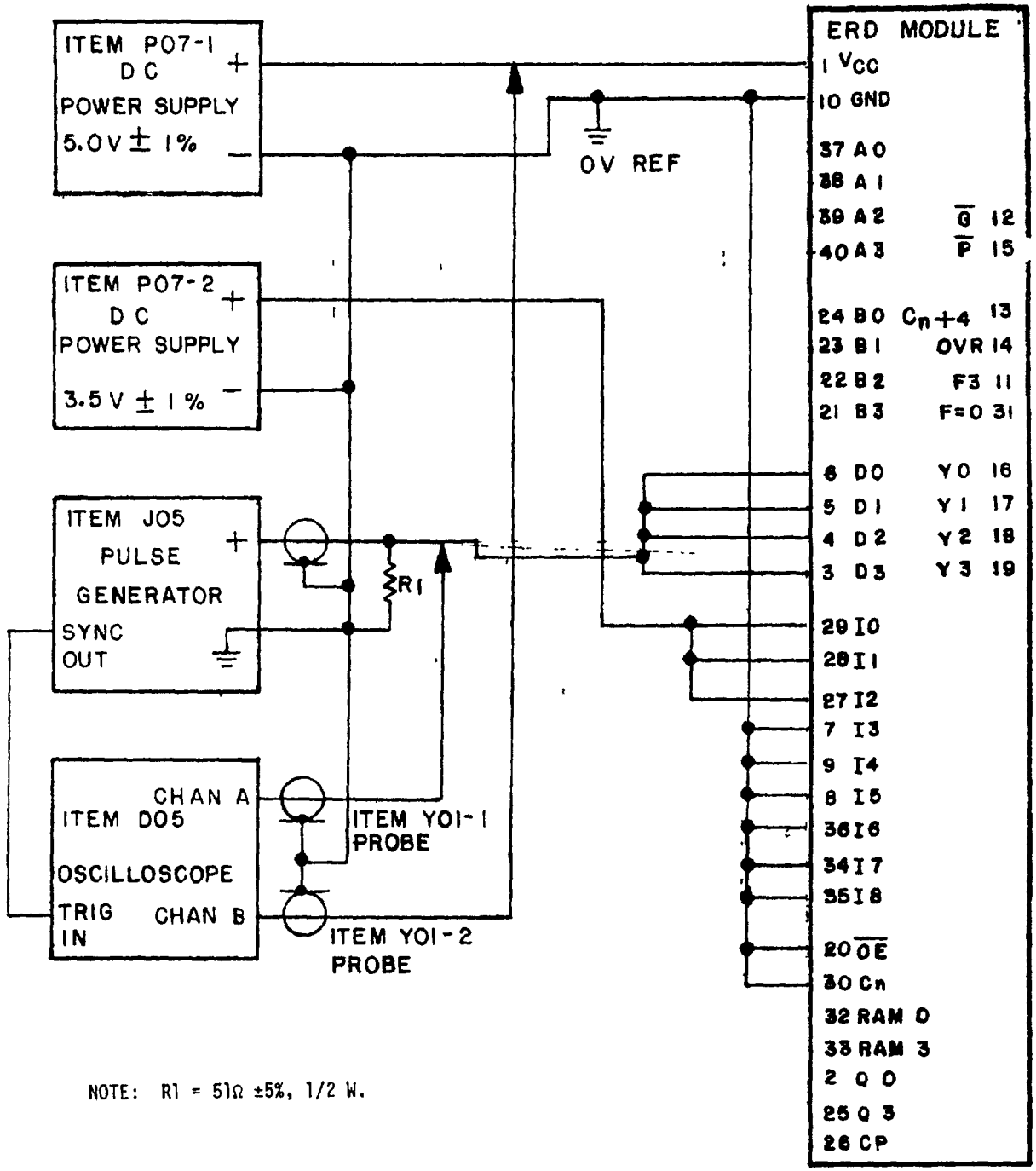
FIGURE 36. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIII and XXIV.

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TABLE XXV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	37,38

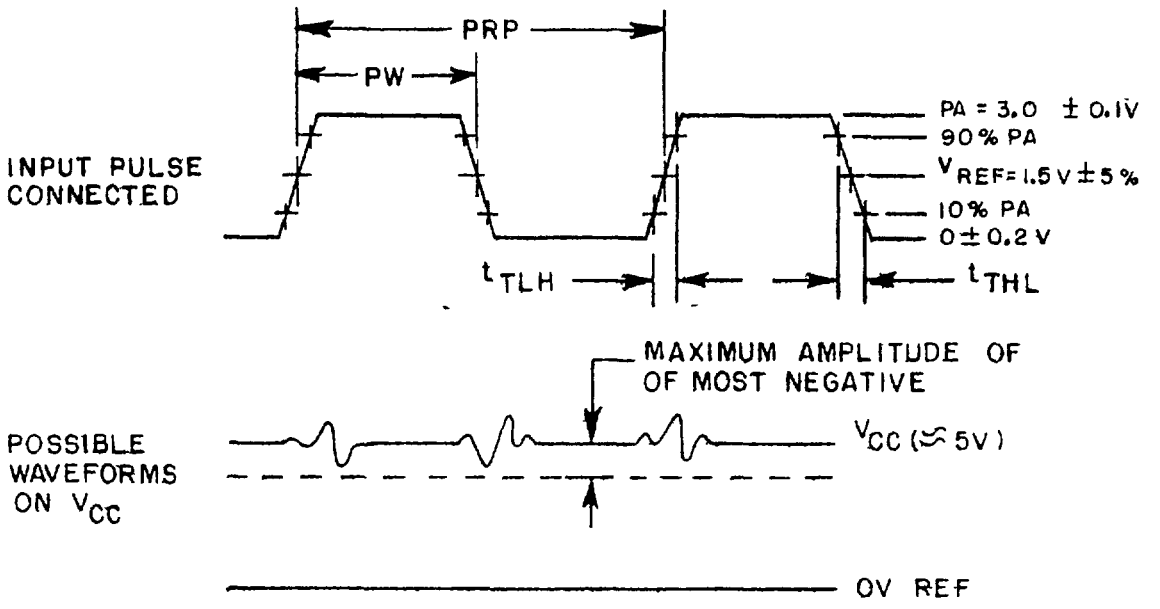
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:31Z
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 37. Power supply transient amplitude test circuit for table XXV.

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NOTE: Input pulse parameters
PW = 430 ±30 ns.
PRP = 860 ±60 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 38. Power supply transient amplitude test circuit waveforms for table XXV.

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* TABLE XXVI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
Z	Power supply filter capacitance	1	10	3.30	5.40	µF	39

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* TABLE XXVII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	A0 Input capacitance	37	10	2	30	pF	39
AA2	A1	38					
AA3	A2	39					
AA4	A3	40					
AA5	B0	24					
AA6	B1	23					
AA7	B2	22					
AA8	B3	21					
AA9	D0	6					
AA10	D1	5					
AA11	D2	4					
AA12	D3	3					
AA13	I0	29					
AA14	I1	28					
AA15	I2	27					
AA16	I3	7					
AA17	I4	9					
AA18	I5	8					

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* TABLE XXVII Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA19	I6 Input capacitance	36	10	2	30	pF	39
AA20	I7	34					
AA21	I8	35					
AA22	$\overline{OE}$	20					
AA23	C_n	30					
AA24	CP	26					

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* TABLE XXVIII. Input/output capacitance test.

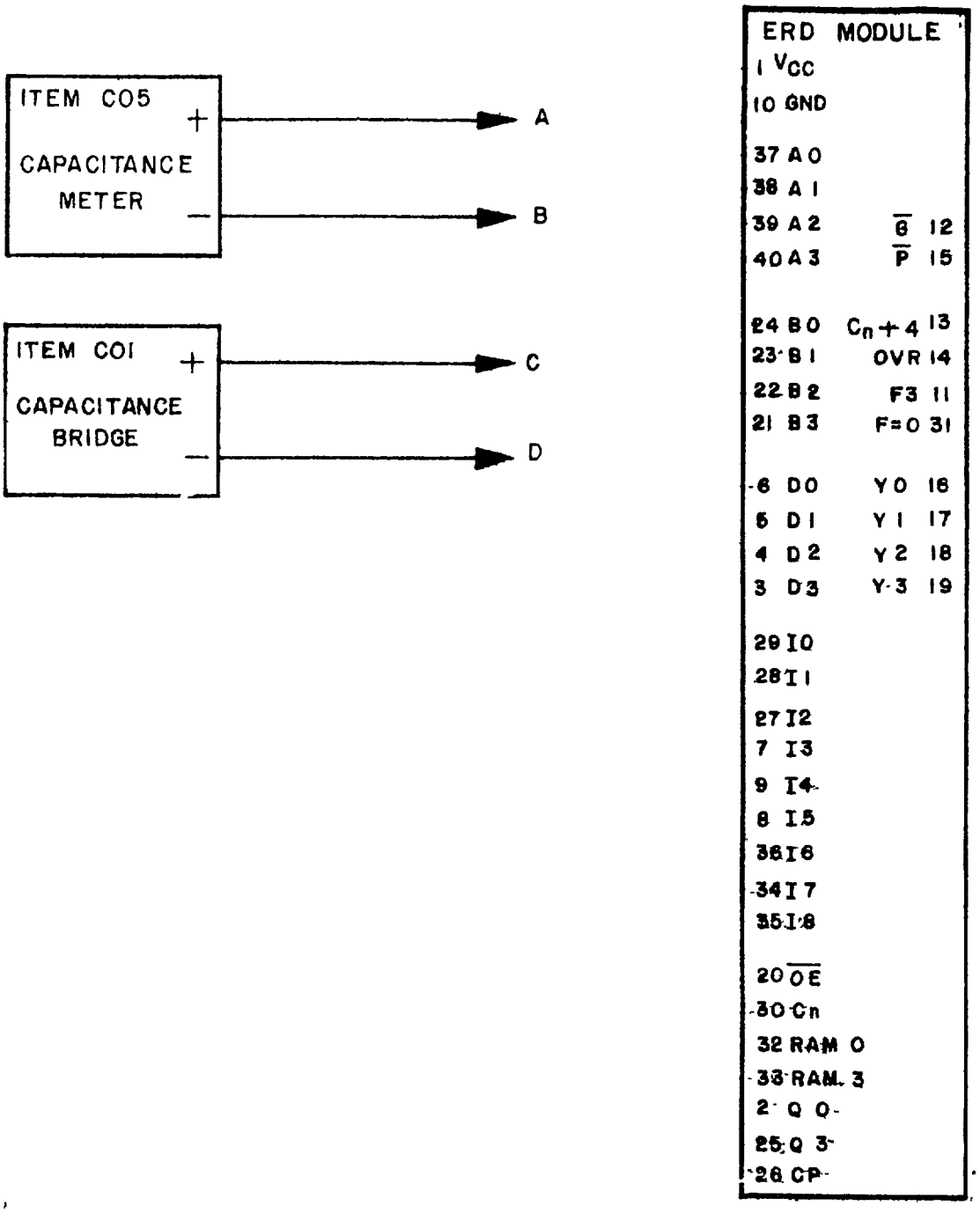
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	RAM 0 Input/output capacitance	32	10	2	30	pF	39
BB2	RAM 3	33					
BB3	Q0	2					
BB4	Q3	25					

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* TABLE XXIX. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°, ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	$\overline{E}$ Output capacitance	12	10	2	30	pF	39
CC2	$\overline{P}$	15					
CC3	C_{n+4}	13					
CC4	OVR	14					
CC5	F3	11			35		
CC6	F=0	31			30		
CC7	Y0	16					
CC8	Y1	17					
CC9	Y2	18					
CC10	Y3	19					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (COI) setup:
   Frequency = 120 Hz
   V_{pk} = 0.5 V maximum

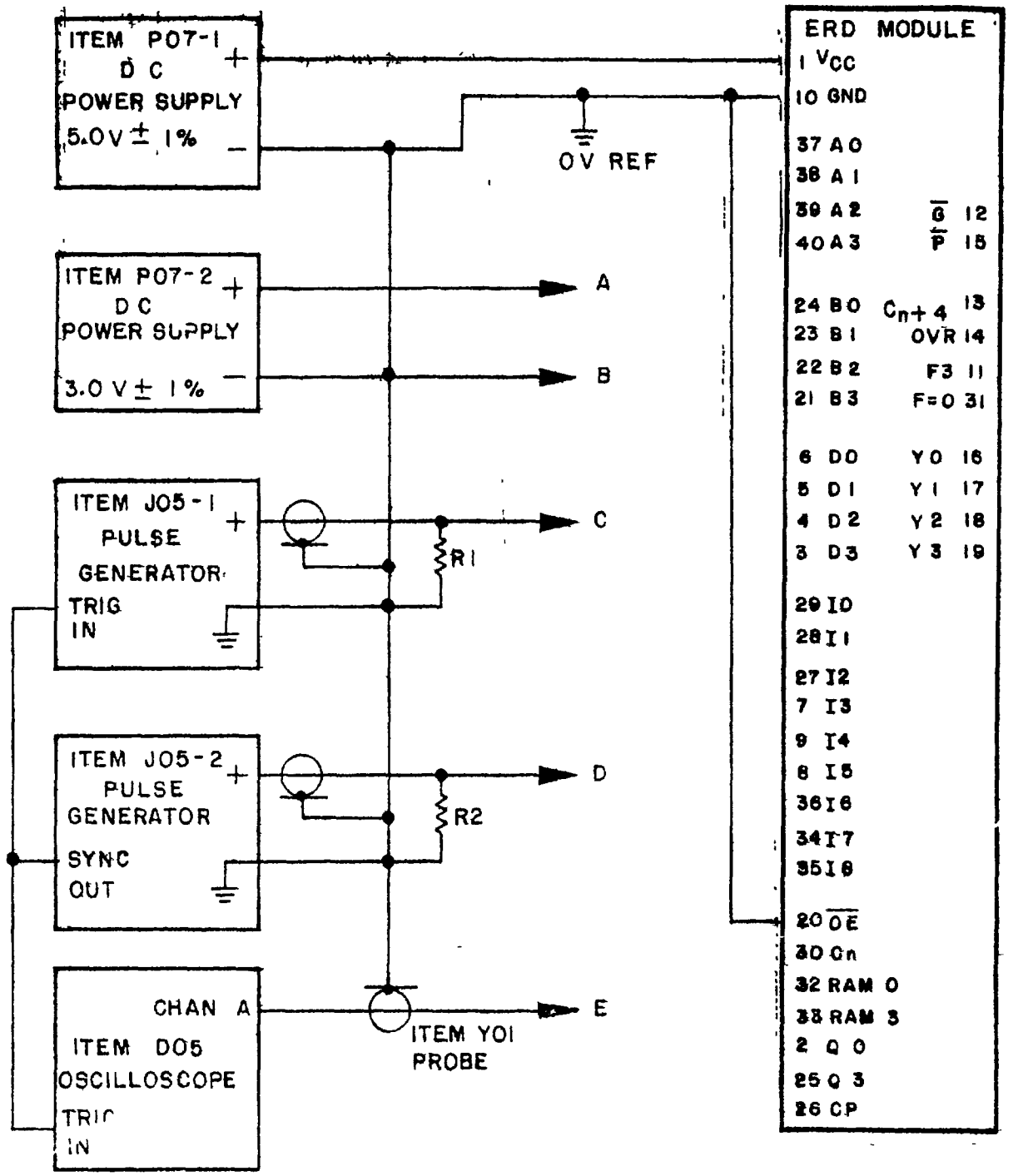
FIGURE 39. Power supply filter capacitance, input, input/output and output capacitance test circuit for tables XXVI, XXVII, XXVIII, and XXIX

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TABLE XXX. Holdtime (t_{HH} and t_{HL}) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
DD1	Q0 Holdtime, t_{HH}	3,4,5,6,21,22,23, 24,28,37,38,39, 40,34,35	7,8,9,27,29,36	26	2	16	900	1100	ns	40,41
DD2	Q0 Holdtime, t_{HL}	3,4,5,6,21,22,23, 24,28,37,38,39,40 34,35	7,8,9,27,29,36		2	16				40,42
DD3	Q3 Holdtime, t_{HH}	3,4,5,6,21,22,23, 24,28,35,37,38, 39,40	7,8,9,27,29,34, 36		25	19				40,41
DD4	Q3 Holdtime, t_{HL}	3,4,5,6,21,22,23, 24,28,35,37,38, 39,40	7,8,9,27,29,34, 36		25	19				40,42
DD5	RAM 0 Holdtime, t_{HH}	3,4,5,6,21,22,23, 24,28,29,34,35,36, 37,38,39,40	7,8,9,27		32	16				40,41
DD6	RAM 0 Holdtime, t_{HL}	3,4,5,6,21,22,23, 24,28,29,34,35, 36,37,38,39,40	7,8,9,27		32	16				40,42
DD7	RAM 3 Holdtime, t_{HH}	3,4,5,6,21,22,23, 24,28,29,35,36, 37,38,39,40	7,8,9,27,34		33	19				40,41
DD8	RAM 3 Holdtime, t_{HL}	3,4,5,6,21,22,23, 24,28,29,35,36, 37,38,39,40	7,8,9,27,34		33	19				40,42

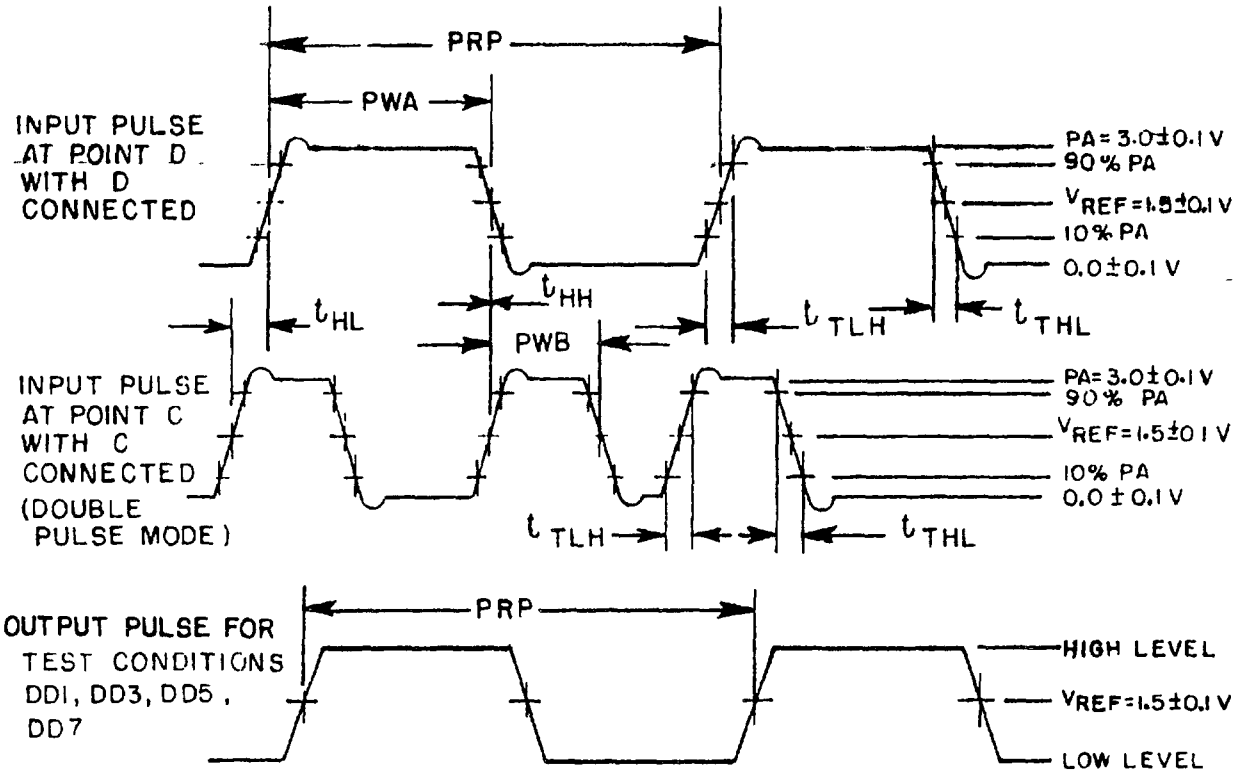
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 40. Hold time (t_{HH} and t_{HL}) test circuit for table XXX.

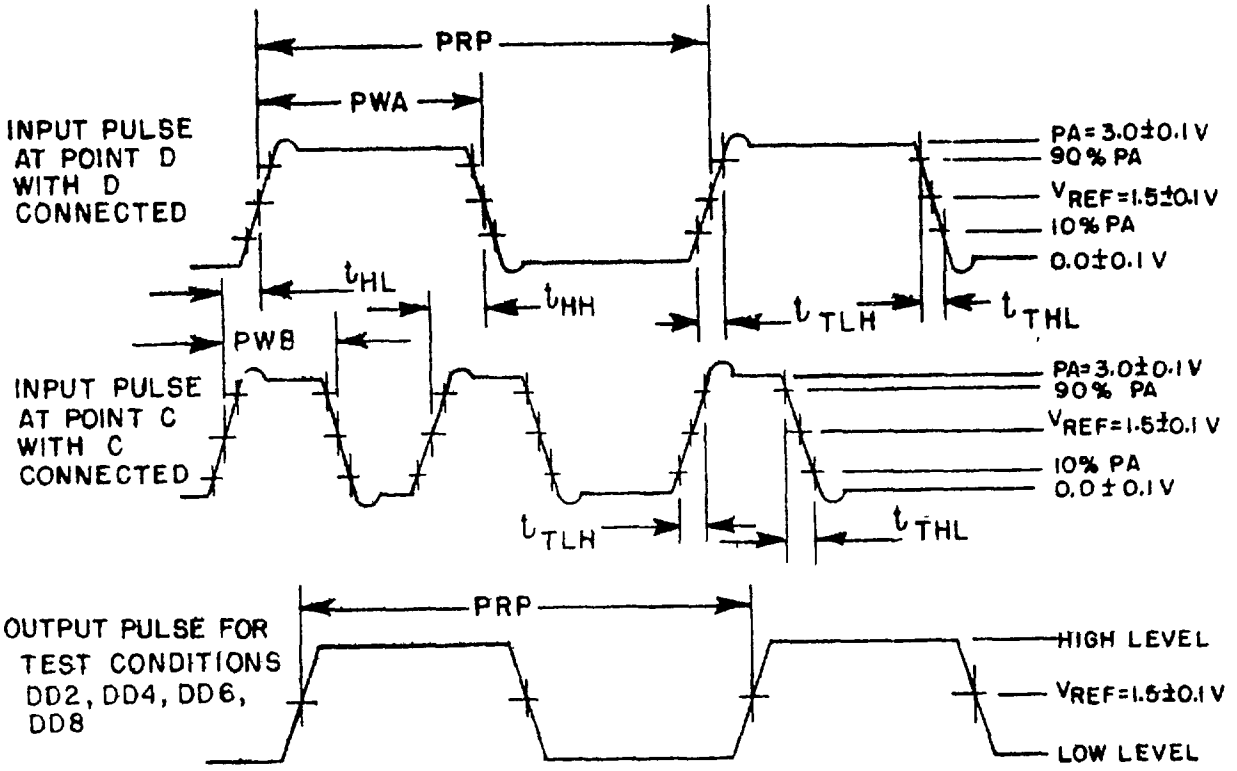
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NOTES
1. Input pulse parameters:
   PWA = 500 +50 ns.
   PWB = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_{HL} = 10 ±2 ns.
   $t_{HH} = 0\ ^{+2}_{-0}$ ns.
2. Before taking the specified measurements, verify that the low level is $\leq$ 0.5 V and the high level is $\geq$ 2.4 V

FIGURE 41 Hold time (t_{HH}) test circuit waveforms for table XXX.

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NOTES
1. Input pulse parameters:
   PWA = 500 ±50 ns
   PWB = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{HL} = 0\ ^{+2}_{-0}$ ns.
   $t_{HH} = 10 \pm 2$ ns.
2. Before taking the specified measurements, verify that the low level is ≤ 0.5 V and the high level is ≥ 2.4 V.

FIGURE 42. Hold time (t_{HL}) test circuit waveforms for table XXX

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